

ALGEBRA 2

Unit 8

RATIONAL EXPRESSION

SIMPLIFYING Rational Expressions

COMPLEX FRACTIONS

Steps to SIMPLIFY

KEY FEATURES of Rational Functions

VARIATION Models

Identifying EQUATIONS

RATIONAL FUNCTIONS

NOTES • HOMEWORK • QUIZZES • TEST

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Unit 8 – Rational Functions: Sample Unit Outline

	TOPIC	HOMEWORK
DAY 1	Simplifying Rational Expressions; Multiplying & Dividing Rational Expressions	HW #1
DAY 2	Adding and Subtracting Rational Expressions	HW #2
DAY 3	Quiz 8-1	None
DAY 4	Complex Fractions	HW #3
DAY 5	Review all Operations	HW #4
DAY 6	Applications	HW #5
DAY 7	Quiz 8-2	None
DAY 8	Graphing Basic Rational Functions (Reciprocal Function)	HW #6
DAY 9	Key Features of Rational Functions (x-Intercepts, Vertical & Horizontal Asymptotes, Holes)	HW #7
DAY 10	Graphing Rational Functions	HW #8
DAY 11	Graphing Review	HW #9
DAY 12	Solving Rational Equations	HW #10
DAY 13	Quiz 8-3	None
DAY 14	Direct, Inverse, Joint, and Constant Variation	HW #11
DAY 15	Unit 8 Review	Study for Test
DAY 16	UNIT 8 TEST	None

MULTIPLYING

Rational Expressions

If the problem contains-

- **Monomials Only:** Multiply together, then simplify.
- **Binomials/Trinomials:** Factor everything you can FIRST, then simplify.

9. $\frac{6x^2y^3}{2x^2y^2} \cdot \frac{10x^3y^4}{18y^2}$

10. $\frac{4a^2b^2}{15ab^3} \cdot \frac{5a^3b^6}{12a^4b^7}$

11. $\frac{4x^2 - 4x}{2x^2 + 4x - 6} \cdot \frac{x^2 + x - 6}{4x^2 + 8x}$

12. $\frac{10v - 5v^2}{v^2 - 11v + 18} \cdot \frac{v^2 - 8v - 9}{15v}$

DIVIDING

Rational Expressions

To divide rational expressions, multiply by the _____!

13. $\frac{5pq}{16p^3} \div \frac{35p^2q^2}{8q^5}$

14. $\frac{10}{4x - 8} \div \frac{2x^2 + 6x}{x^2 + x - 6}$

15. $\frac{2a^3 - 12a^2}{a^2 - 4a - 12} \div \frac{24a^2 - 8a^3}{a^2 - 8a + 15}$

16. $\frac{k + 3}{k} \div (4k + 1) \cdot \frac{16k^2 - 1}{k + 3}$

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Unit 8: Rational Functions

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Homework 1: Simplify, Multiply, & Divide
Rational Expressions**** This is a 2-page document! ******Simplify each expression below.**

1. $\frac{16m^2}{24m^7}$

2. $\frac{n^2 + 7n}{4n^2 + 28n}$

3. $\frac{x^2 - 10x - 24}{x + 2}$

4. $\frac{1 - 9w^2}{12w - 4}$

5. $\frac{4a^2 - 36a}{2a^4 - 24a^3 + 54a^2}$

6. $\frac{y^2 - 36}{5y^2 - 26y - 24}$

Find the product. Give your answer in simplest form.

7. $\frac{32x^3y}{5xy^2} \cdot \frac{15y}{8x^2y^4}$

8. $\frac{m^2 - 6m + 8}{2m - 2} \cdot \frac{10}{m - 4}$

9. $\frac{28n + 40}{35n + 50} \cdot \frac{12n + 24}{8n + 16}$

10. $\frac{p + 10}{9 - p} \cdot \frac{p^2 - 5p - 36}{4p^2 + 16p}$

11. $\frac{v^2 - 49}{20v^3} \cdot \frac{4v^2 - 24v}{v^2 + v - 42}$	12. $\frac{2n - 3}{n + 1} \cdot \frac{2n^2 + 5n + 3}{9 - 4n^2}$
13. $\frac{12k^2 - 54k}{6k} \cdot \frac{6}{81 - 18k}$	14. $\frac{6c^2 + 13c - 63}{6c^2 - 17c + 7} \cdot \frac{2c^2 - 9c + 4}{12c + 54}$
Find the quotient. Give your answer in simplest form.	
15. $\frac{14m^4}{3m} \div \frac{7m^2}{18m^5}$	16. $\frac{x^2 - 3x - 28}{6x} \div \frac{2x - 14}{2}$
17. $\frac{2a^2 + 14a}{8a^2} \div (10a + 70)$	18. $\frac{1 - h^2}{2h^2 - 10h - 12} \div \frac{2h - 2}{6}$
19. $\frac{2r + 2}{r + 2} \div \frac{4r^2 + 8r + 4}{12r + 12}$	20. $\frac{20x - 4}{8x^2} \div (5x - 1) \cdot \frac{6x}{5}$

Name:	Date:
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Main Ideas/Questions	Notes/Examples	
ADDING & SUBTRACTING Rational Expressions	❶	Rewrite the problem using a common denominator.
	❷	Combine the numerators and keep the common denominator.
	❸	Simplify the remaining expression.
Examples with LIKE BASES	1. $\frac{13a}{4a^2} + \frac{5a}{4a^2}$	2. $\frac{9x+12}{16} - \frac{7x+2}{16}$
	3. $\frac{y^2+8y}{y^2-9} + \frac{2y+21}{y^2-9}$	4. $\frac{1}{12a-3} - \frac{16a^2}{12a-3}$
	5. $\frac{n^2+3n-7}{n^2+n-6} - \frac{n+1}{n^2+n-6}$	6. $\frac{13k^2-9k}{6k^2-5k+1} + \frac{k^2+2k}{6k^2-5k+1}$
Examples with UNLIKE BASES	7. $\frac{15}{2x} - \frac{16}{3x}$	8. $\frac{4a}{a^2} - \frac{7}{8a}$

$$9. \frac{k^2 + 16}{5k - 10} - \frac{4}{k - 2}$$

$$10. \frac{7}{y + 8} + \frac{35}{y^2 + 11y + 24}$$

$$11. \frac{x^2 - 5}{x^2 + 5x + 6} + \frac{1}{x + 2}$$

$$12. \frac{m^2 + 5m + 6}{m^2 - 4} + \frac{2}{m - 2}$$

$$13. \frac{5}{w + 1} + \frac{8}{w - 7}$$

$$14. \frac{b}{b + 3} - \frac{4}{b}$$

$$15. \frac{2}{4p - 20} + \frac{p - 6}{p^2 - 8p + 15}$$

$$16. \frac{r}{2r + 8} - \frac{8}{r^2 + 4r}$$

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Homework 2: Adding & Subtracting
Rational Expressions**** This is a 2-page document! ******Find each sum/difference. Give your answer in simplest form.**

1. $\frac{9w^3}{2w^5} + \frac{3w^3}{2w^5}$

2. $\frac{8k+9}{18} - \frac{2k+1}{18}$

3. $\frac{x^2+5x}{x+7} - \frac{14}{x+7}$

4. $\frac{3y^2-19y}{y^2-25} + \frac{y^2-y}{y^2-25}$

5. $\frac{8m^2}{10m^3+60m^2} - \frac{6m^2-12m}{10m^3+60m^2}$

6. $\frac{v^2-4v}{5v^2+9v-2} - \frac{2v+16}{5v^2+9v-2}$

7. $\frac{5x}{4} + \frac{x}{6}$

8. $\frac{a}{12b} - \frac{b}{8a}$

9. $\frac{7}{n+9} + \frac{12}{4n+36}$

10. $\frac{p^2-9p}{p^2-4p-45} - \frac{2}{p+5}$

$$11. \frac{k+5}{k+1} + \frac{8}{k^2-1}$$

$$12. \frac{22w+11}{8w^2-6w} - \frac{3}{2w}$$

$$13. \frac{x}{2x-5} - \frac{4x+15}{4x^2-25}$$

$$14. \frac{8a-18}{3a^2+14a+8} + \frac{7}{3a+2}$$

$$15. 4 - \frac{5}{4m+8}$$

$$16. \frac{3c}{c-4} - \frac{2}{5}$$

$$17. \frac{6}{v+4} + \frac{2}{v-6}$$

$$18. \frac{r}{r^2-4} + \frac{1}{r^2-6r+8}$$

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Unit 8: Rational Functions

Quiz 8-1: Simplifying & Operations with Rational Expressions

Simplify each expression.

1. $\frac{20x^2 + 30x}{4x^4 + 6x^3}$

2. $\frac{k^2 + 13k + 40}{k^2 - 2k - 35}$

3. $\frac{7n - 42}{36 - n^2}$

4. $\frac{6a^2 + 5a - 6}{3a^2 - 5a + 2}$

1. _____

2. _____

3. _____

4. _____

Find each product or quotient.

5. $\frac{5w + 20}{3w + 15} \cdot \frac{7w + 35}{10w + 40}$

6. $\frac{x - 2}{x^2 + 4x + 4} \cdot \frac{x + 2}{2x - 4}$

7. $\frac{3m^3}{m^2 - 25} \cdot \frac{m^2 + 6m + 5}{m^2}$

8. $\frac{7x^4}{24y^5} \div \frac{21x}{12y^4}$

5. _____

6. _____

7. _____

8. _____

$$9. \frac{p^2 - 2p}{p^2 + 7p - 18} \div \frac{p^2 - 11p + 18}{p^2 - 81}$$

$$10. \frac{r^2 - 3r - 10}{2r^2 - 11r + 5} \div \frac{r^2 - 5r + 6}{2r^2 - 7r + 3}$$

9. _____

10. _____

Find each sum or difference.

$$11. \frac{4}{15v} + \frac{2}{15v}$$

$$12. \frac{2x^2}{x^2 - 7x + 12} - \frac{x^2 + 9}{x^2 - 7x + 12}$$

11. _____

12. _____

13. _____

14. _____

15. _____

16. _____

$$13. \frac{5x}{3} - \frac{7x}{8}$$

$$14. \frac{n^2 + 11n + 10}{4n + 36} + \frac{2}{n + 9}$$

$$15. \frac{a}{a+2} - \frac{a+10}{a^2-4}$$

$$16. \frac{3}{x+1} + \frac{x}{x-1}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
COMPLEX FRACTIONS	- A complex fraction is a fraction in which the numerator and denominator both contain fractions. - To simplify a complex fraction means to rewrite as a single fraction with no fractions in the numerator or denominator.	
<i>Steps to</i> SIMPLIFY	①	Create a single fraction in the numerator and denominator.
	②	Set up as a division problem, then multiply by the reciprocal.
	③	Simplify! (Factor first if possible.)
EXAMPLES	1. $\frac{\frac{a}{2}}{\frac{a^2}{4}}$	2. $\frac{\frac{6}{xy}}{\frac{3y}{8x^3}}$
	3. $\frac{\frac{18p}{8p-2}}{\frac{4p^2}{4p-1}}$	4. $\frac{\frac{k-1}{k}}{\frac{k^2-1}{k^2}}$
	5. $\frac{\frac{7c+35}{c}}{\frac{c+5}{c^2}}$	6. $\frac{\frac{b+2}{b^2+6b+8}}{\frac{1}{b^2-b-20}}$
	7. $\frac{\left(\frac{15-15r}{15r^2}\right)}{(12r-12)}$	8. $\frac{\frac{2m^2-2m-40}{3m+12}}{\frac{m^3-5m^2}{9m^2+36m}}$

9. $\frac{\frac{x-3}{3-\frac{x}{x-3}}}{x}$	10. $\frac{\frac{1}{pq}}{\frac{1}{p}-\frac{1}{q}}$
11. $\frac{\frac{5}{4}}{\frac{20}{n-4}-\frac{5}{4n-16}}$	12. $\frac{\frac{w}{w+2}}{1-\frac{w}{w+2}}$
13. $\frac{\frac{1}{a^2}-\frac{1}{b^2}}{\frac{1}{b}+\frac{1}{a}}$	14. $\frac{\frac{m}{n}-1}{\frac{m}{n}+1}$
15. $\frac{\frac{1}{4}+\frac{1}{4v}}{\frac{1}{v}+\frac{1}{4}}$	16. $\frac{\frac{a^2}{a+3}+\frac{a^2}{5a+15}}{\frac{a}{a+3}+\frac{1}{5}}$

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Homework 3: Complex Fractions

**** This is a 2-page document! ******Simplify each expression.**

1.
$$\frac{\frac{6a^3}{5b}}{\frac{21a}{10b^2}}$$

2.
$$\frac{\frac{18x^2}{x-1}}{\frac{x}{2x-2}}$$

3.
$$\frac{\frac{4c^2-1}{3c}}{\frac{2c+1}{9c^2}}$$

4.
$$\frac{\frac{n^2-9n+14}{4n+28}}{\frac{n-7}{n+7}}$$

5.
$$\frac{\frac{8x^2+24x}{x^2-9}}{2x}$$
$$\frac{15-5x}{15-5x}$$

6.
$$\frac{\frac{2v^2+16v}{2v^2-5v-7}}{4v^2+28v-32}$$
$$\frac{10v^2-10}{10v^2-10}$$

7.
$$\frac{\frac{x^2}{4}}{\frac{x^2}{4}-\frac{4}{x}}$$

8.
$$\frac{1-\frac{a}{a-b}}{\frac{1}{a-b}}$$

$$9. \frac{\frac{1}{p} + \frac{2}{3}}{\frac{1}{p}}$$

$$10. \frac{\frac{5q}{q+1}}{1 - \frac{q}{q+1}}$$

$$11. \frac{\frac{8}{y} - 4}{12 - \frac{24}{y}}$$

$$12. \frac{\frac{1}{r} + \frac{1}{s}}{\frac{1}{r^2} - \frac{1}{s^2}}$$

$$13. \frac{\frac{k^2}{4} - 9}{\frac{k}{6} - 1}$$

$$14. \frac{\frac{m}{4} - \frac{1}{m}}{\frac{1}{3m} + \frac{1}{6}}$$

$$15. \frac{\frac{w+4}{4} + \frac{1}{2}}{\frac{3}{w} + \frac{1}{2}}$$

$$16. \frac{\frac{2}{x+3} - \frac{1}{3}}{\frac{x^2}{3} - 3}$$

REVIEW: Operations with Rational Expressions

SIMPLIFYING

Write each expression in simplest form.

1. $\frac{6x^3 + 6x^2}{2x^3 - 2x}$

2. $\frac{4n^2 - 12n + 9}{2n^2 + 15n - 27}$

MULTIPLYING & DIVIDING RATIONAL EXPRESSIONS

Find each product or quotient. Write each answer in simplest form.

3. $\frac{42a^2 - 24a}{21a - 12} \cdot \frac{3a - 9}{6a^2 + 24a}$

4. $\frac{49 - y^2}{6y^2 + 42y} \cdot \frac{48y^3}{y^2 - 10y + 21}$

5. $\frac{w^2 - 81}{4w^2 + 27w - 7} \div \frac{w^2 - 5w - 36}{w^2 + 11w + 28}$

6. $\frac{30c^3 - 60c^2}{3c} \div (2c^3 - 16c^2 + 24c)$

ADDING & SUBTRACTING RATIONAL EXPRESSIONS

Find each sum or difference. Write each answer in simplest form.

7. $\frac{17k + 3}{8k + 4} + \frac{3k + 7}{8k + 4}$

8. $\frac{9m - 12}{m^2 - 25} - \frac{m + 28}{m^2 - 25}$

$$9. \frac{3}{12y} + \frac{9}{24}$$

$$10. \frac{p^2 + 2p - 51}{p^2 - 6p - 27} - \frac{4}{p + 3}$$

$$11. \frac{r}{r+6} - \frac{1}{r}$$

$$12. \frac{3}{h-2} + \frac{5}{h-5}$$

COMPLEX FRACTIONS

Simplify each expression.

$$13. \frac{\frac{15x-10}{x-8}}{\frac{3x-2}{x^2-12x+32}}$$

$$14. \frac{\frac{m}{3} - \frac{1}{3}}{\frac{1-m^2}{21}}$$

$$15. \frac{\frac{p^2}{5} - 20}{\frac{p}{10} - 1}$$

$$16. \frac{\frac{4v}{v} - \frac{1}{v^2}}{1 - \frac{1}{2v}}$$

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Homework 4: Rational Expressions Review

**** This is a 2-page document! ******Simplify each expression.**

1. $\frac{y^2 - 14y + 48}{y^2 - 64}$

2. $\frac{12w^4 + 30w^3}{16w^2 + 40w}$

3. $\frac{16a^2b^4}{3ab^3} \cdot \frac{6a^2b^3}{8a^5b}$

4. $\frac{x^2 + x - 6}{x^2 - x - 12} \cdot \frac{x^2 + x - 20}{x^2 - 4x + 4}$

5. $\frac{2m^2 + 9m - 5}{3m^2 + 13m - 10} \div \frac{14m - 7}{4 - 9m^2}$

6. $\frac{4p^2 - 7p - 15}{2p^2 + 3p - 27} \div (4p + 5)$

7. $\frac{2k - 3}{3k + 24} - \frac{k - 11}{3k + 24}$

8. $\frac{2x - 13}{x^2 + 9x - 36} + \frac{3x - 2}{x^2 - 9x - 36}$

$$9. \frac{a}{3} - \frac{4}{5}$$

$$10. \frac{2}{3m+1} + \frac{19m+17}{3m^2-14m-5}$$

$$11. \frac{x}{x-2} + \frac{5}{x}$$

$$12. \frac{3}{n-6} - \frac{3}{n+1}$$

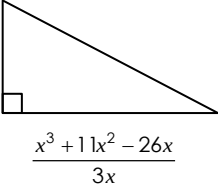
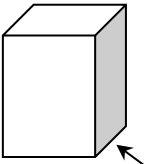
$$13. \frac{\frac{w^2-1}{8w-8}}{\frac{w^2+6w+5}{18w}}$$

$$14. \frac{1 - \frac{r}{r+2}}{\frac{r^2}{r+2}}$$

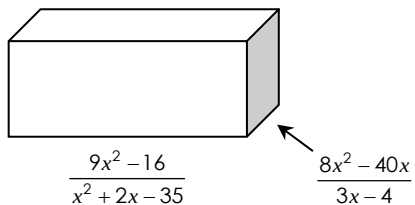
$$15. \frac{\frac{12}{c} - 6}{8 - \frac{16}{c}}$$

$$16. \frac{\frac{k^2-9k}{2} + 10}{\frac{k}{8} - \frac{k+5}{16}}$$

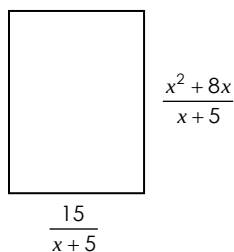
RATIONAL EXPRESSIONS: Applications

Example	Work and Solution
1. Write an expression in simplest form to represent the area of the rectangle. $\frac{2x}{x^2 - 12x + 27}$ $\frac{x^2 - 81}{4x^3 + 36x^2}$	
2. Write an expression in simplest form to represent the area of the rectangle. $\frac{x^2 + 3x - 10}{7x - 14}$ $\frac{3x^2 - 11x - 20}{25 - x^2}$	
3. If the area of a rectangular garden is $\frac{a^2}{b} - b$ and the length is $\frac{a}{b} + 1$, write an expression to represent the width of the garden in simplest form.	
4. Write an expression to represent the area of the triangle below in simplest form. $\frac{9x + 18}{6x^3 - 24x}$ 	
5. Write an expression to represent the volume of the rectangular prism shown below in simplest form. $\frac{5x^2 - 32x + 12}{10x^3}$  $\frac{18x^2}{45x^2 - 18x}$ $\frac{x^2 + 14x + 48}{36 - x^2}$	

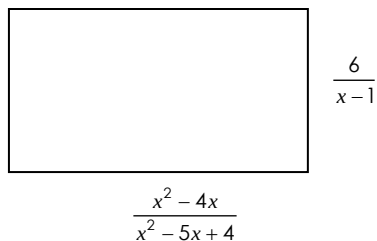
6. The **volume** of a rectangular prism below is $12x + 16$. Find the height of the prism.



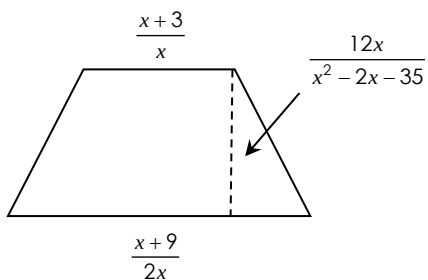
7. Write an expression to represent the **perimeter** of the rectangle in simplest form.



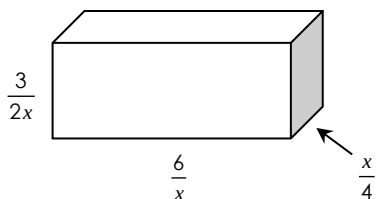
8. Write an expression to represent the **perimeter** of the rectangle in simplest form.



9. Write an expression to represent the **area** of the trapezoid below in simplest form.



10. Write an expression to represent the **surface area** of the rectangular prism below.





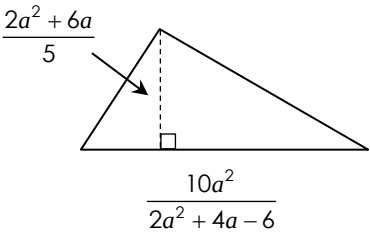
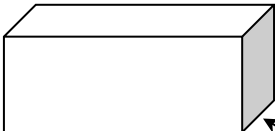
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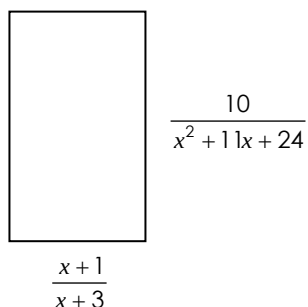
Homework 5: Rational Expressions Applications

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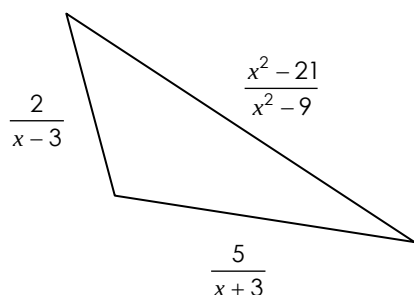
Example	Work and Solution
1. Find an expression to represent the area of the rectangle below. $\frac{6m^2 + 7m - 5}{10m^2 - 5m}$$\frac{10m^2}{9m^2 - 25}$	
2. The area of a rectangle is $x - \frac{16}{x}$. If the width of rectangle is $\frac{x}{4} + 1$, find an expression to represent the length.	
3. Find an expression to represent the area of the triangle below. $\frac{2a^2 + 6a}{5}$$\frac{10a^2}{2a^2 + 4a - 6}$	
4. Find an expression to represent the volume of the rectangular prism below. $\frac{x^2 - 1}{6x^3 + 30x^2}$$\frac{3x}{8}$	

5. The **volume** of a rectangular prism is $\frac{7}{x} - \frac{3}{5x}$. If it has a width of $\frac{2}{x}$ and a height of $\frac{4}{x}$, find the length of the prism.

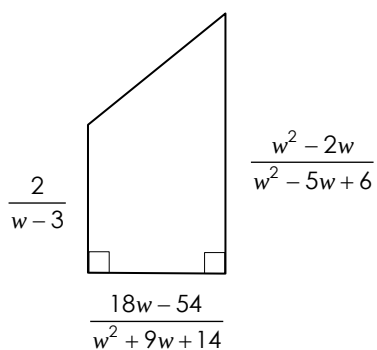
6. Find an expression to represent the **perimeter** of the rectangle below.



7. Find an expression to represent the **perimeter** of the triangle below.



8. Find an expression to represent the **area** of the trapezoid below.



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Unit 8: Rational Functions

Quiz 8-2: Rational Expressions & Applications

Simplify each expression.

1. $\frac{28m^2 - 42m}{12m^3 - 18m^2}$

2. $\frac{16k - 24}{6k^2 - 13k + 6}$

1. _____

2. _____

Find each product or quotient.

3. $\frac{15cd^3}{2d^2} \cdot \frac{4c^3d}{45c^4d}$

4. $\frac{x^2 + 2x - 3}{2x^2 - x - 21} \cdot \frac{4x^2 - 49}{x - 1}$

3. _____

4. _____

5. _____

5. $\frac{5w^2 + 20w}{16 - w^2} \div \frac{15w^2}{w^2 - 11w + 28}$

6. $\frac{8a^2 - 16a - 24}{5a^2 - a - 6} \div (4a - 12)$

6. _____

Find each sum or difference.

7. $\frac{6p}{p-1} - \frac{p+5}{p-1}$

8. $\frac{9x}{10} + \frac{7x}{8}$

7. _____

8. _____

9. $\frac{y^2 + y - 14}{y^2 - 49} + \frac{2}{y + 7}$

10. $\frac{n}{n - 4} - \frac{n}{n + 3}$

9. _____

10. _____

Simplify each complex fraction.

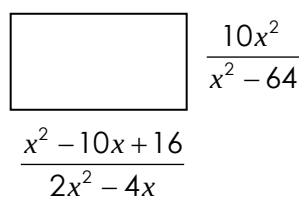
11. $\frac{2 + \frac{x - 8}{x - 2}}{\frac{x}{4} - 1}$

12. $\frac{\frac{r^2 - 2r}{4r} - \frac{2}{r}}{1 - \frac{4}{r}}$

11. _____

12. _____

13. Find the **area** of the rectangle below.

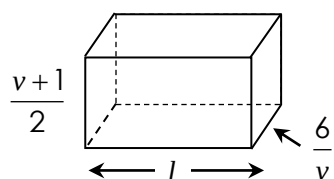

 $\frac{10x^2}{x^2 - 64}$
 $\frac{x^2 - 10x + 16}{2x^2 - 4x}$

13. _____

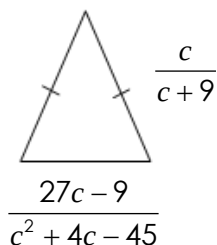
14. _____

15. _____

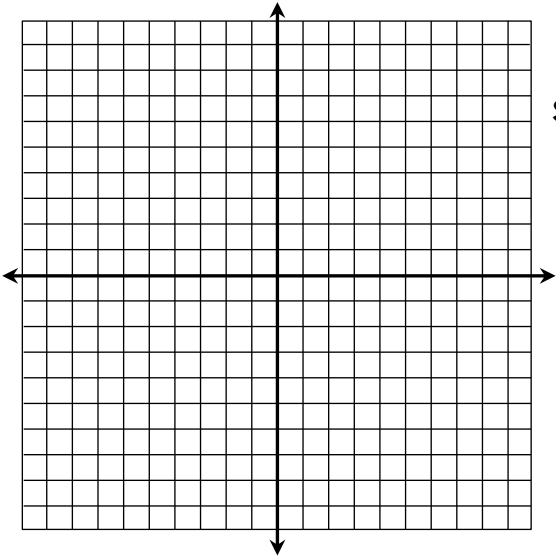
14. If the rectangular prism below has a **volume** of $\frac{2v^2 - 3v - 5}{8v - 20}$, find its **length, l**.



15. Find the **perimeter** of the triangle below.

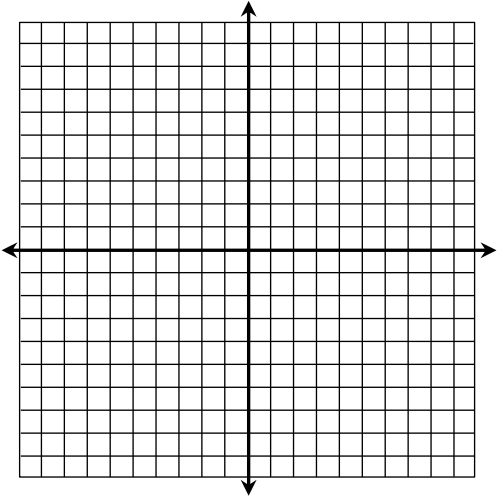
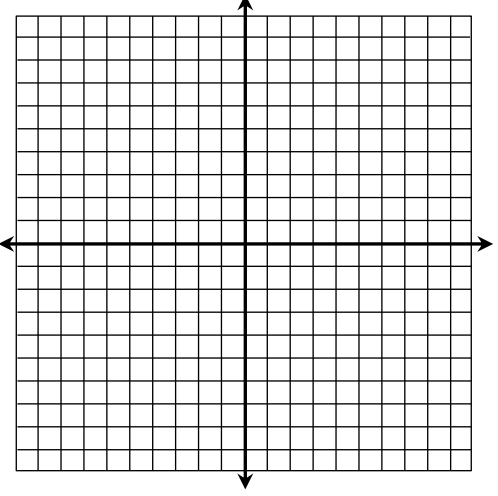
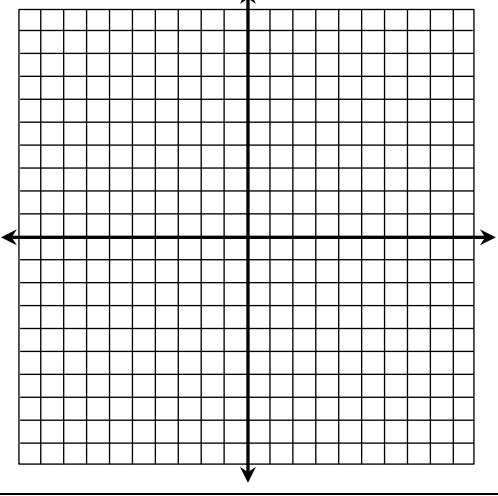

 $\frac{c}{c + 9}$
 $\frac{27c - 9}{c^2 + 4c - 45}$

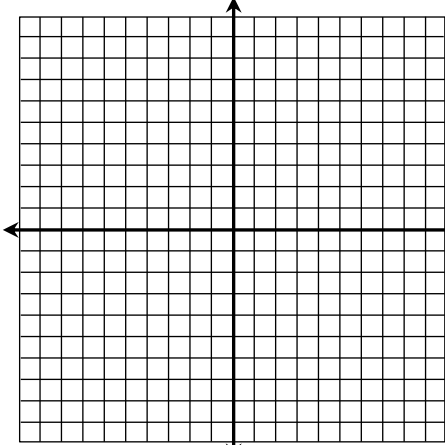
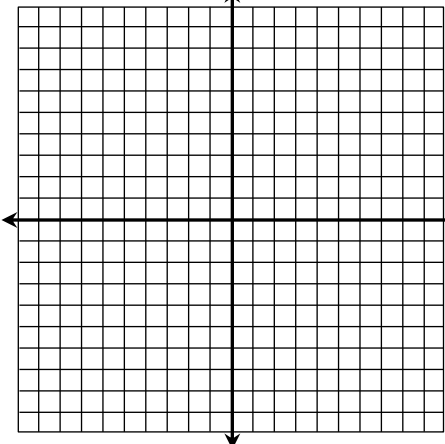
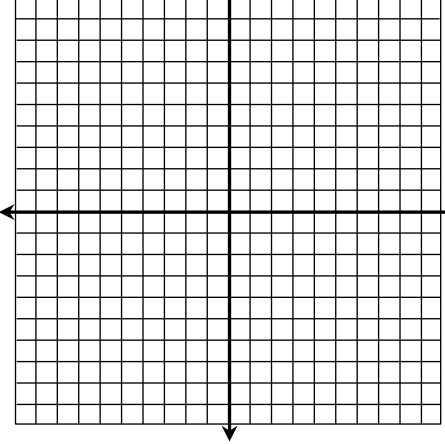
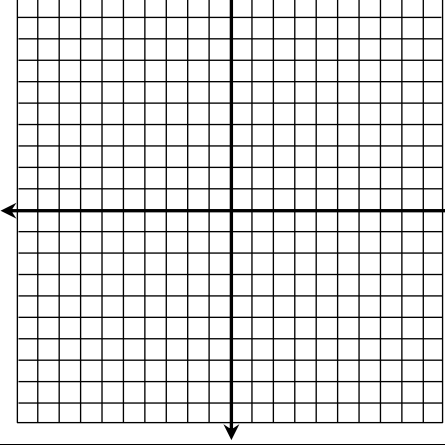
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
RATIONAL Function	A rational function is a function of the form , where $p(x)$ and $q(x)$ are _____.	
RECIPROCAL Function	One of the most basic rational functions is $f(x) = \frac{1}{x}$, which is also known as a reciprocal function. Graph this function below and identify the key characteristics.  Shape of Graph: _____ Domain: _____ Range: _____ Vertical Asymptote: _____ Horizontal Asymptote: _____	
TRANSFORMING Reciprocal Functions $f(x) = \frac{a}{x - h} + k$	TRANSFORMATIONS: h is the _____ shift. (+ shifts _____, - shifts _____) k is the _____ shift. (+ shifts _____, - shifts _____) - If a is negative, the function is _____ across the ____ - _____. $a > 1$ represents a vertical _____. $0 < a < 1$ represents a vertical _____. VERTICAL ASYMPTOTE: HORIZONTAL ASYMPTOTE:	

DESCRIBING Transformations	For each function, identify the transformations from $f(x) = \frac{1}{x}$, the vertical asymptote, and the horizontal asymptote.		
	1. $f(x) = \frac{1}{x} + 4$	Transformations:	
		<table border="1"> <tr> <td>VA:</td> <td>HA:</td> </tr> </table>	VA:
	VA:	HA:	
	2. $f(x) = \frac{-2}{x+1}$	Transformations:	
		<table border="1"> <tr> <td>VA:</td> <td>HA:</td> </tr> </table>	VA:
	VA:	HA:	
	3. $f(x) = \frac{1}{x-5} - 2$	Transformations:	
		<table border="1"> <tr> <td>VA:</td> <td>HA:</td> </tr> </table>	VA:
	VA:	HA:	
4. $f(x) = \frac{4}{x+3} - 2$	Transformations:		
	<table border="1"> <tr> <td>VA:</td> <td>HA:</td> </tr> </table>	VA:	HA:
VA:	HA:		
WRITING Functions	5. The reciprocal parent function is reflected across the x-axis, then translated 4 units right and 3 units down. Write an equation to represent the new function and identify the asymptotes.		
	6. The reciprocal parent function is vertically stretched by a factor of 2, then translated 7 units up and 1 unit left. Write an equation to represent the new function and identify the asymptotes.		
	7. A reciprocal function has a vertical asymptote located at $x = -2$ and a horizontal asymptote located at $y = 5$. Write an equation that could represent this function.		

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
Graphing RECIPROCAL FUNCTIONS	Graph each function. Identify the domain, range, and asymptotes.		
	1. $f(x) = \frac{1}{x} - 2$ 	Domain:	
		Range:	
		Vertical Asymptote:	
		Horizontal Asymptote:	
	2. $f(x) = \frac{1}{x+3}$ 	Domain:	
		Range:	
		Vertical Asymptote:	
		Horizontal Asymptote:	
	3. $f(x) = \frac{1}{x-2} - 4$ 	Domain:	
		Range:	
		Vertical Asymptote:	
		Horizontal Asymptote:	

	4. $f(x) = \frac{-4}{x} + 1$ 	Domain:
		Range:
		Vertical Asymptote:
		Horizontal Asymptote:
	5. $f(x) = \frac{3}{x-2} - 2$ 	Domain:
		Range:
		Vertical Asymptote:
		Horizontal Asymptote:
	6. $f(x) = \frac{2}{x+4} - 7$ 	Domain:
		Range:
		Vertical Asymptote:
		Horizontal Asymptote:
	7. $f(x) = \frac{-1}{x-5} + 3$ 	Domain:
		Range:
		Vertical Asymptote:
		Horizontal Asymptote:

Name: _____

Unit 8: Rational Functions

Date: _____ Per: _____

Homework 6: Graphing Reciprocal Functions

** This is a 2-page document! **

Transformations from the function $f(x) = \frac{1}{x}$ are described below. Write an equation to represent the new function, then identify the vertical asymptote (VA) and horizontal asymptote (HA).

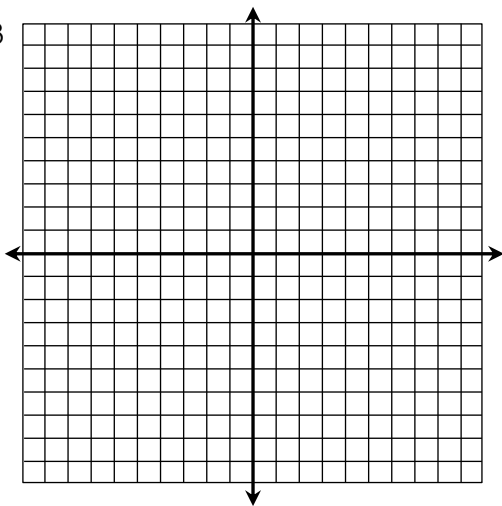
Description of Transformations	New Function	VA	HA
1. Translated 2 units left and 9 units down.			
2. Reflected across the x-axis, then translated 5 units up.			
3. Vertically stretched by a factor of 4, then translated 3 units right.			
4. Vertically stretched by a factor of 2, reflected across the x-axis, then translated 1 unit left and 8 units up.			

The function $f(x) = \frac{1}{x}$ is transformed so that it has the given asymptotes below. Write an equation that could represent the new function.

5. Asymptotes: $x = 3$ and $y = -2$	6. Asymptotes: $x = -7$ and $y = 0$
-------------------------------------	-------------------------------------

Graph each function. Identify the domain, range, and asymptotes.

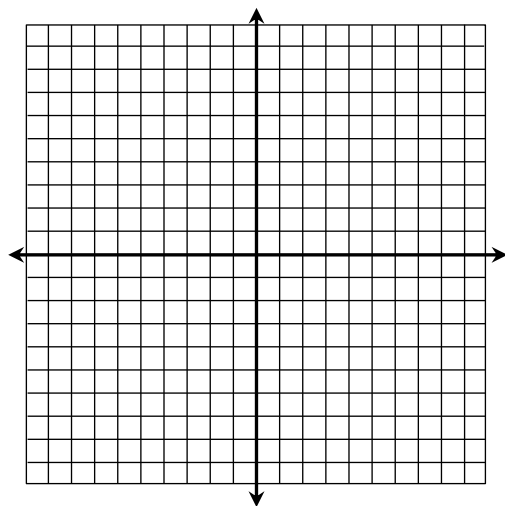
7. $f(x) = \frac{1}{x} + 3$



Domain: _____ VA: _____

Range: _____ HA: _____

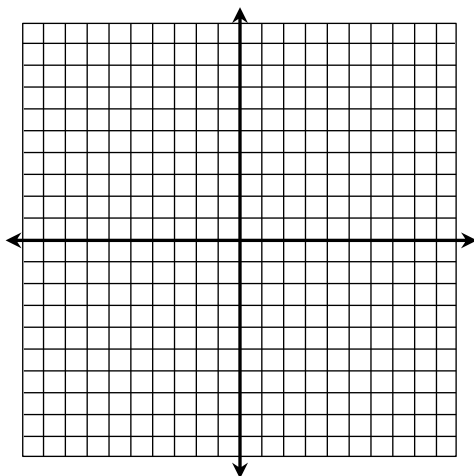
8. $f(x) = \frac{-1}{x}$



Domain: _____ VA: _____

Range: _____ HA: _____

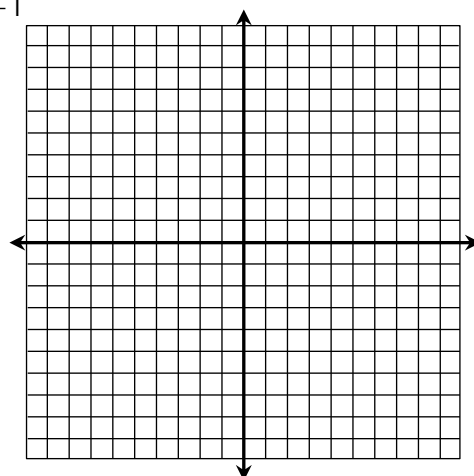
9. $f(x) = \frac{1}{x-4}$



Domain: _____ VA: _____

Range: _____ HA: _____

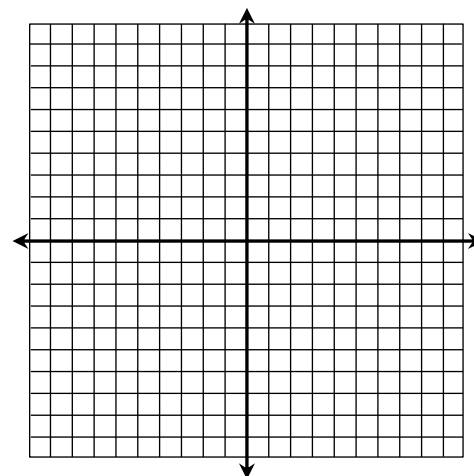
10. $f(x) = \frac{3}{x+2} - 1$



Domain: _____ VA: _____

Range: _____ HA: _____

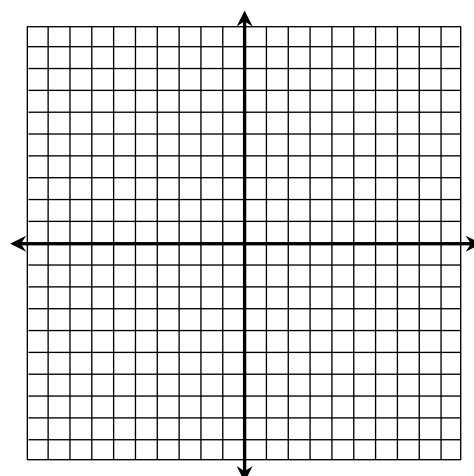
11. $f(x) = \frac{2}{x} - 5$



Domain: _____ VA: _____

Range: _____ HA: _____

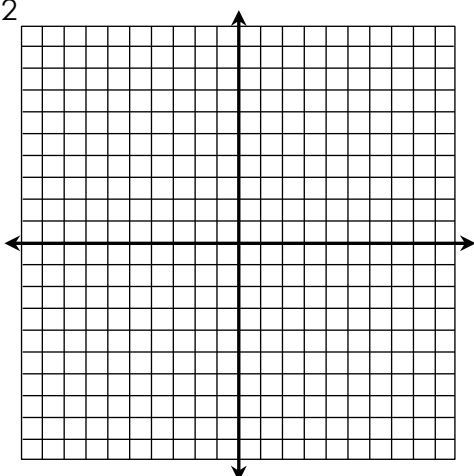
12. $f(x) = \frac{-1}{x-6}$



Domain: _____ VA: _____

Range: _____ HA: _____

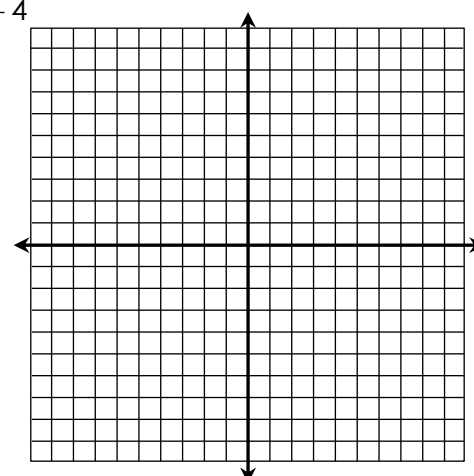
13. $f(x) = \frac{4}{x-1} - 2$



Domain: _____ VA: _____

Range: _____ HA: _____

14. $f(x) = \frac{-3}{x+5} + 4$



Domain: _____ VA: _____

Range: _____ HA: _____

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
KEY FEATURES of Rational Functions	A rational function is a function of the form $f(x) = \frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomial functions. So far, we have looked at the graph of a very basic rational function (a reciprocal function). Now, we will look at more complex rational functions and their key features: x-intercepts, vertical asymptotes, horizontal asymptotes, and holes. A hole is a point (x, y) at which there is a break in the graph.	
	Identify the key features of the rational function given each graph below.	
	1. $f(x) = \frac{x^2 - 4}{x^2 - 5x + 6}$	x-Intercept(s): Vertical Asymptote(s): Horizontal Asymptote: Hole(s):
	2. $f(x) = \frac{2x + 8}{x^2 + 6x + 8}$	x-Intercept(s): Vertical Asymptote(s): Horizontal Asymptote: Hole(s):
	3. $f(x) = \frac{x^3 - 9x}{2x^2 - 8}$	x-Intercept(s): Vertical Asymptote(s): Horizontal Asymptote: Hole(s):

Finding Features ALGEBRAICALLY (x-intercepts, vertical asymptotes, & holes)	Follow to find the x-intercepts, vertical asymptotes and holes of: $f(x) = \frac{x^2 - 2x - 3}{x^2 + 3x + 2}$	
	Write the function in <u>factored form</u>. Then, simplify by crossing out common factors in the numerator and denominator.	
	To find the x-intercept(s): Set the <u>numerator</u> of the simplified function equal to 0 and solve for x.	
	To find the vertical asymptote(s): Set the <u>denominator</u> of the simplified function equal to 0 and solve for x.	
	To find the hole(s): If there was a <u>common factor</u> , set it equal to 0 and solve for x. Substitute this x-value into the simplified function to find y.	
EXAMPLES	Find the x-intercepts, vertical asymptote(s), and hole(s) of each function.	
	4. $f(x) = \frac{3x^2 - 12}{x^2 + 2x - 8}$	x-Intercept(s):
		Vertical Asymptote(s):
		Hole(s):
	5. $f(x) = \frac{x + 3}{x^2 + 8x + 15}$	x-Intercept(s):
		Vertical Asymptote(s):
		Hole(s):
	6. $f(x) = \frac{6x^3 - 6x}{x^2 - 2x}$	x-Intercept(s):
		Vertical Asymptote(s):
		Hole(s):

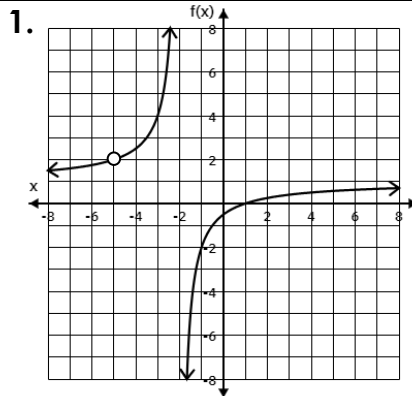
	7. $f(x) = \frac{x-6}{x^2+3x}$	x-Intercept(s):
		Vertical Asymptote(s):
		Hole(s):
	8. $f(x) = \frac{x^2+6x+5}{x^2+4x+3}$	x-Intercept(s):
		Vertical Asymptote(s):
		Hole(s):
HORIZONTAL ASYMPTOTES	To find the horizontal asymptote of a rational function, we look at the original function $f(x) = \frac{p(x)}{q(x)}$ and follow the rules below.	
	CASE	HORIZONTAL ASYMPTOTE
	degree of $p >$ degree of q	
	degree of $p <$ degree of q	
	degree of $p =$ degree of q	
	Give the horizontal asymptote of each function below.	
	9. $f(x) = \frac{x^2-9}{x^2+2x-15}$	10. $f(x) = \frac{x^2+4x}{2x-4}$
	11. $f(x) = \frac{3x+1}{x^2-2x+1}$	12. $f(x) = \frac{8x+6}{4x-2}$
	13. $f(x) = \frac{x^3-x^2+2x}{4x^2+4x}$	14. $f(x) = \frac{8}{4x-6}$
	15. $f(x) = \frac{-2x^2+8x}{6x^2-3x}$	16. $f(x) = \frac{3x^2-6x}{x+1}$

Name: _____

Unit 8: Rational Functions

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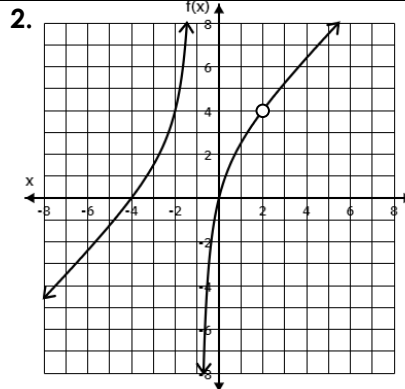
Homework 7: Key Features of Rational Functions

Identify the x -intercept(s), vertical asymptote(s), horizontal asymptote, and hole(s) given the graph. x -intercept(s):

Vertical Asymptote(s):

Horizontal Asymptote:

Hole(s):

 x -intercept(s):

Vertical Asymptote(s):

Horizontal Asymptote:

Hole(s):

Find the x -intercept(s), vertical asymptote(s), horizontal asymptote, and hole(s) of each function.

3. $f(x) = \frac{-6x+12}{3x-12}$

 x -intercept(s):

Vertical Asymptote(s):

Horizontal Asymptote:

Hole(s):

4. $f(x) = \frac{2x-4}{x^2+4x-12}$

 x -intercept(s):

Vertical Asymptote(s):

Horizontal Asymptote:

Hole(s):

5. $f(x) = \frac{x^3+3x^2-4x}{x^2-2x-24}$

 x -intercept(s):

Vertical Asymptote(s):

Horizontal Asymptote:

Hole(s):

6. $f(x) = \frac{x^2-9}{x^2+5x-14}$

 x -intercept(s):

Vertical Asymptote(s):

Horizontal Asymptote:

Hole(s):

7. $f(x) = \frac{2x^3+5x^2+2x}{x^2+5x+6}$

 x -intercept(s):

Vertical Asymptote(s):

Horizontal Asymptote:

Hole(s):

8. $f(x) = \frac{3x^2-6x-9}{x^3-2x^2-3x}$

 x -intercept(s):

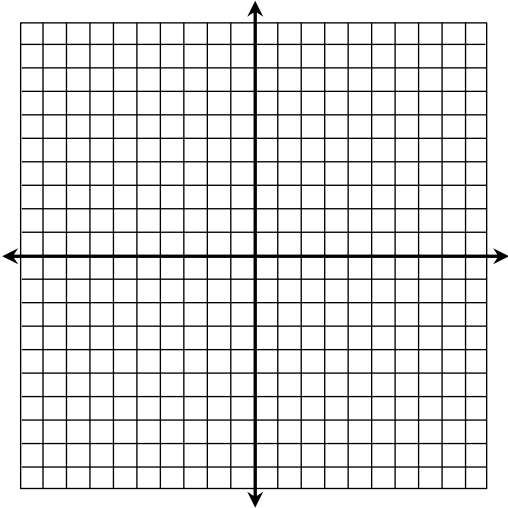
Vertical Asymptote(s):

Horizontal Asymptote:

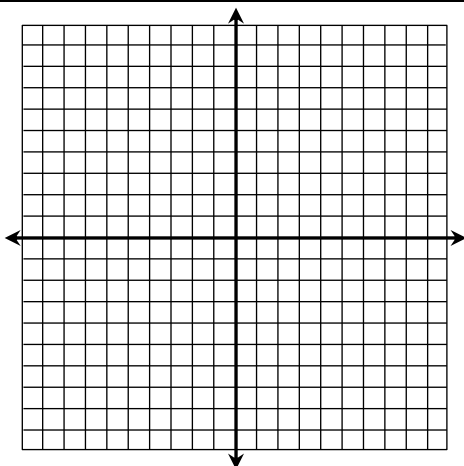
Hole(s):

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
GRAPHING Rational Functions	Use the key features of a rational function to aid in graphing:		
	❶	Write the function in FACTORED FORM , then SIMPLIFY .	
	❷	Find the x-intercept(s) by setting the numerator equal to 0.	
	❸	Find the vertical asymptote(s) by setting the denominator equal to 0.	
	❹	Find the horizontal asymptote using the rules below.	
		CASE	HORIZONTAL ASYMPTOTE
		degree of $p >$ degree of q	
		degree of $p <$ degree of q	
		degree of $p =$ degree of q	
	❺	Identify any holes in the function using common factors.	
A hole is a point (x, y) at which there is a break in the graph. A hole occurs when there is a common factor between the numerator and denominator. » To find the x-coodinate: Set the common factor equal to 0. » To find the y-coordinate: Substitute the x-coordinate into the simplified function.			

Directions: Graph each function and identify its key characteristics.		
1. $f(x) = \frac{x - 4}{x + 4}$		x-int: _____ VA: _____ HA: _____ Hole: _____ Domain: _____ Range: _____

2. $f(x) = \frac{2x-6}{x-1}$



x-int: _____

VA: _____

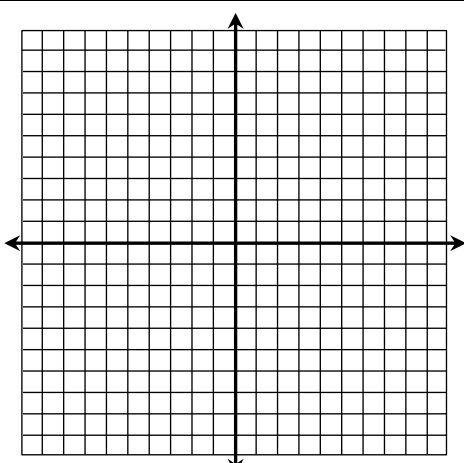
HA: _____

Hole: _____

Domain: _____

Range: _____

3. $f(x) = \frac{x^2-1}{x-1}$



x-int: _____

VA: _____

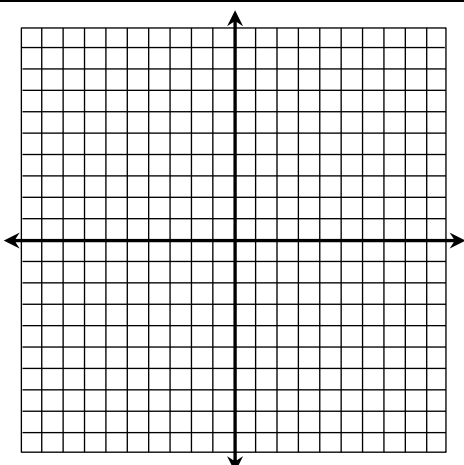
HA: _____

Hole: _____

Domain: _____

Range: _____

4. $f(x) = \frac{x+5}{x^2+9x+20}$



x-int: _____

VA: _____

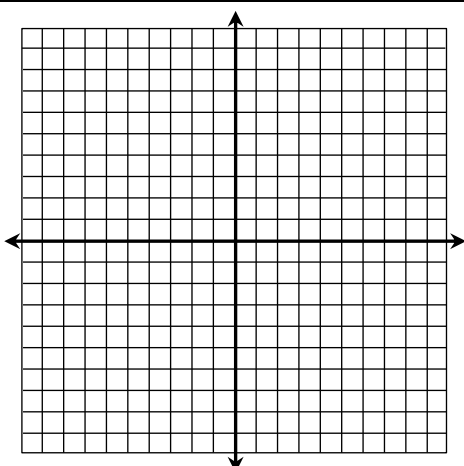
HA: _____

Hole: _____

Domain: _____

Range: _____

5. $f(x) = \frac{2x^2-x-1}{x^2-x-12}$



x-int: _____

VA: _____

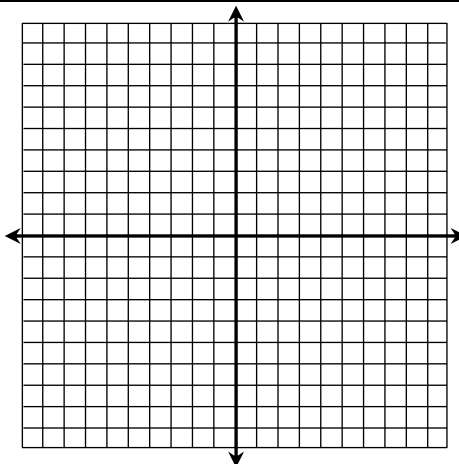
HA: _____

Hole: _____

Domain: _____

Range: _____

6. $f(x) = \frac{x^3 + 3x^2}{x^2 + 2x - 3}$



x-int: _____

VA: _____

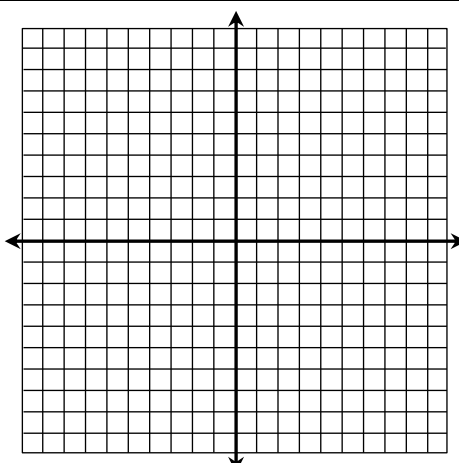
HA: _____

Hole: _____

Domain: _____

Range: _____

7. $f(x) = \frac{x-6}{x^2-6x}$



x-int: _____

VA: _____

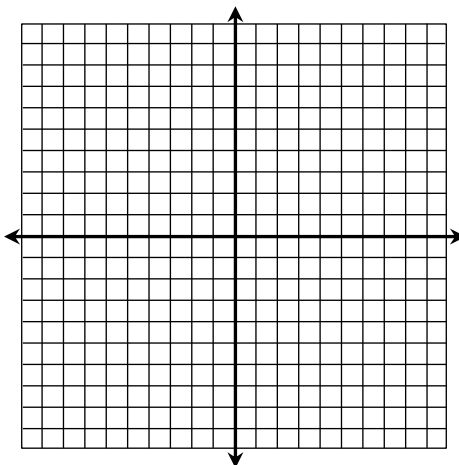
HA: _____

Hole: _____

Domain: _____

Range: _____

8. $f(x) = \frac{x^2 + 3x - 28}{x^2 + 12x + 35}$



x-int: _____

VA: _____

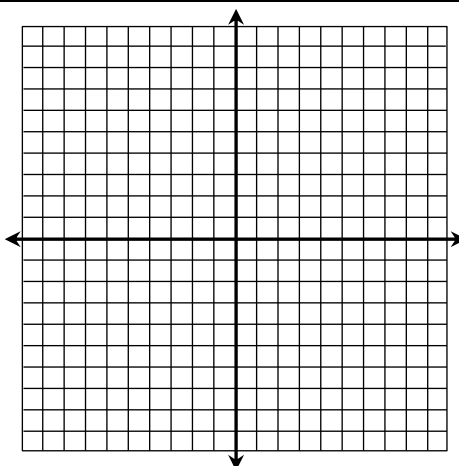
HA: _____

Hole: _____

Domain: _____

Range: _____

9. $f(x) = \frac{x^2 - 4x - 5}{2x + 2}$



x-int: _____

VA: _____

HA: _____

Hole: _____

Domain: _____

Range: _____

Name: _____

Unit 8: Rational Functions

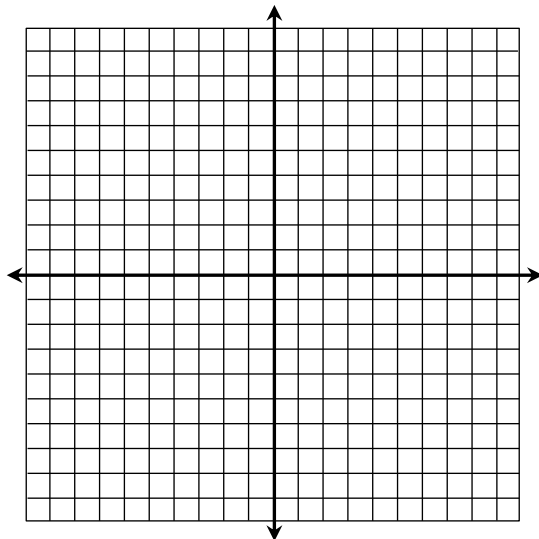
Date: _____ Per: _____

Homework 8: Graphing Rational Functions

**** This is a 2-page document! ****

Graph each function. Identify the domain, range, asymptotes, and holes.

1. $f(x) = \frac{x+3}{x-2}$



x-int: _____

VA: _____

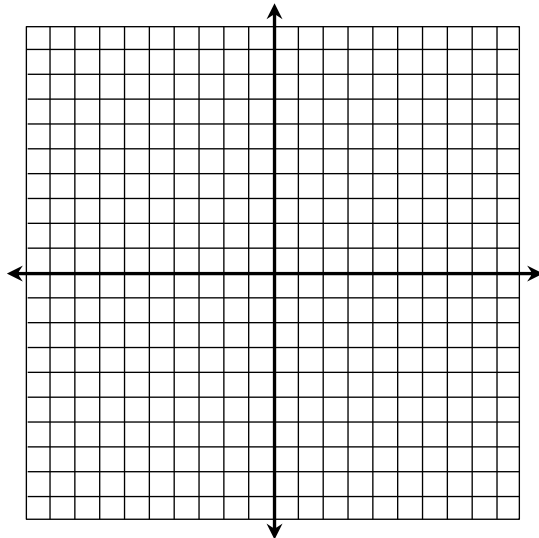
HA: _____

Hole: _____

Domain: _____

Range: _____

2. $f(x) = \frac{4x-24}{2x-4}$



x-int: _____

VA: _____

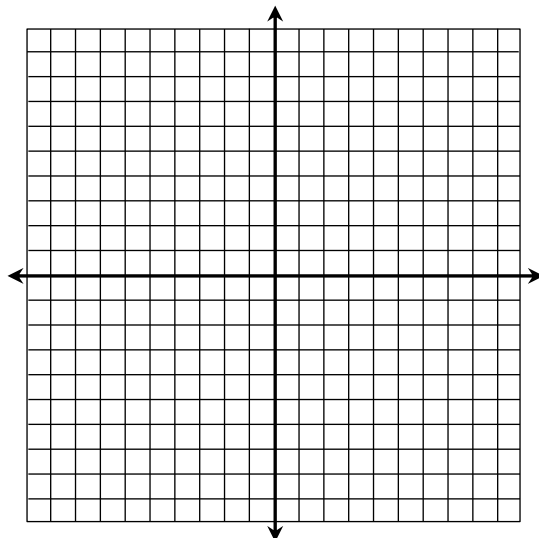
HA: _____

Hole: _____

Domain: _____

Range: _____

3. $f(x) = \frac{x^2+7x+10}{x+5}$



x-int: _____

VA: _____

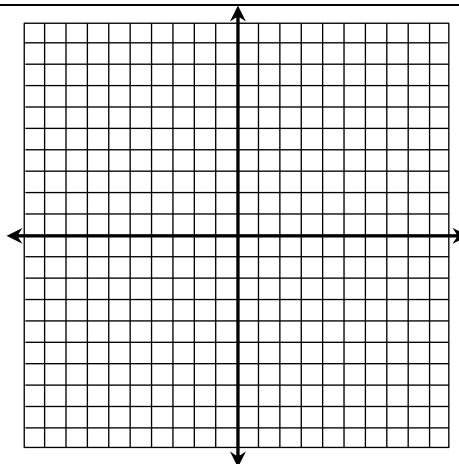
HA: _____

Hole: _____

Domain: _____

Range: _____

4. $f(x) = \frac{x+3}{x^2-9}$



x-int: _____

VA: _____

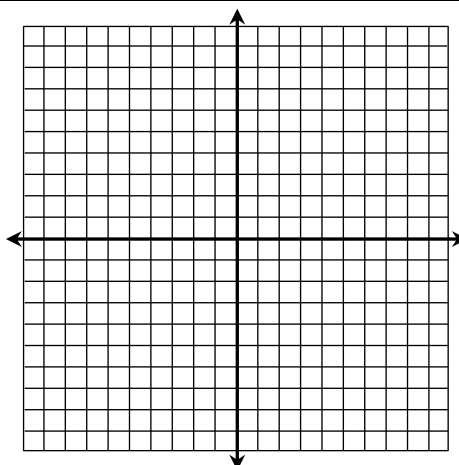
HA: _____

Hole: _____

Domain: _____

Range: _____

5. $f(x) = \frac{x^2+4x}{x^2+3x-4}$



x-int: _____

VA: _____

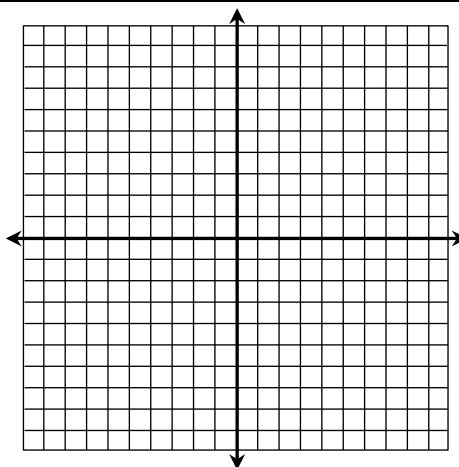
HA: _____

Hole: _____

Domain: _____

Range: _____

6. $f(x) = \frac{x^2-x-6}{x}$



x-int: _____

VA: _____

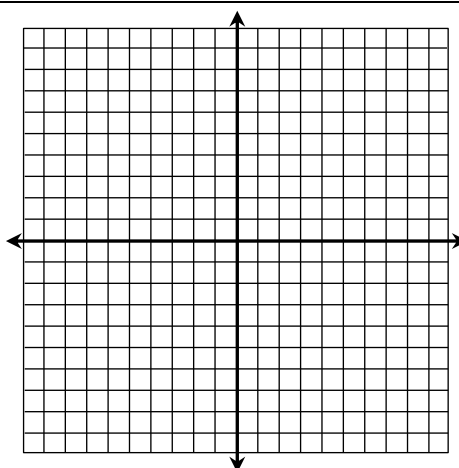
HA: _____

Hole: _____

Domain: _____

Range: _____

7. $f(x) = \frac{3x^2-5x-2}{x^2-3x+2}$



x-int: _____

VA: _____

HA: _____

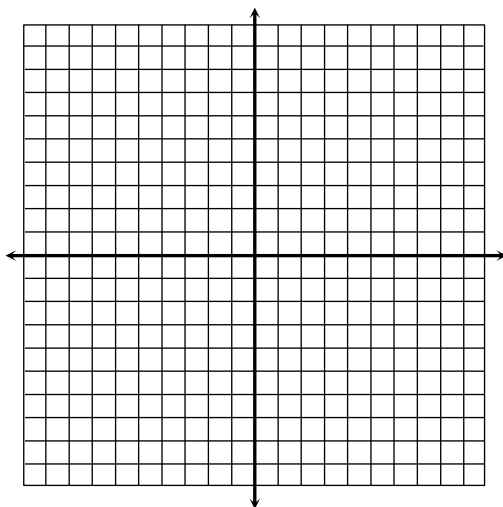
Hole: _____

Domain: _____

Range: _____

RECIPROCAL Functions

PARENT FUNCTION:



Vertical Asymptote: _____ **D:** _____

Horizontal Asymptote: _____ **R:** _____

..... Transformations

STANDARD FORM:

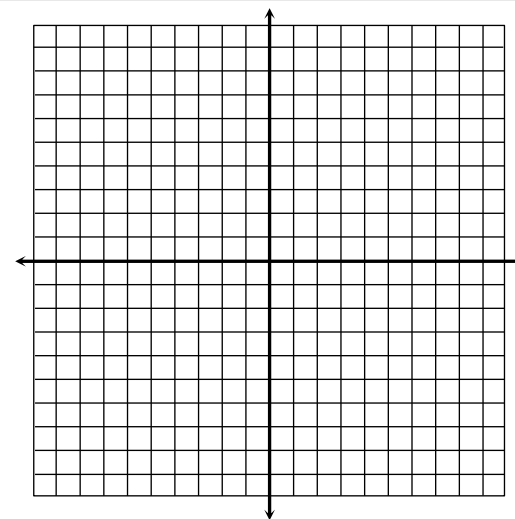
- h is the _____ shift. (+ shifts _____, - shifts _____)
- k is the _____ shift. (+ shifts _____, - shifts _____)
- If a is negative, the function is _____ across the x -axis.
- $|a| > 1$ represents a vertical _____.
- $0 < |a| < 1$ represents a vertical _____.

VERTICAL ASYMPTOTE:

HORIZONTAL ASYMPTOTE:

EXAMPLES

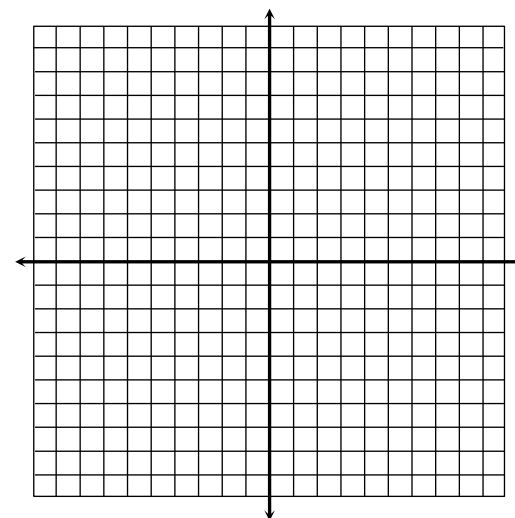
① $f(x) = \frac{1}{x+1} - 5$



Vertical Asymptote: _____ **D:** _____

Horizontal Asymptote: _____ **R:** _____

② $f(x) = \frac{-2}{x+3}$



Vertical Asymptote: _____ **D:** _____

Horizontal Asymptote: _____ **R:** _____

RATIONAL Functions

EQUATION FORM:

X-INTERCEPTS:

VERTICAL ASYMPTOTES:

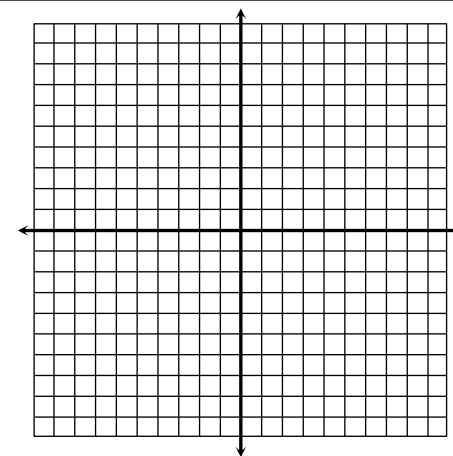
HORIZONTAL ASYMPTOTES:

- If degree of $p >$ degree of q :
- If degree of $p <$ degree of q :
- If degree of $p =$ degree of q :

HOLES:

EXAMPLES

1 $f(x) = \frac{x^2 - 9}{x + 2}$

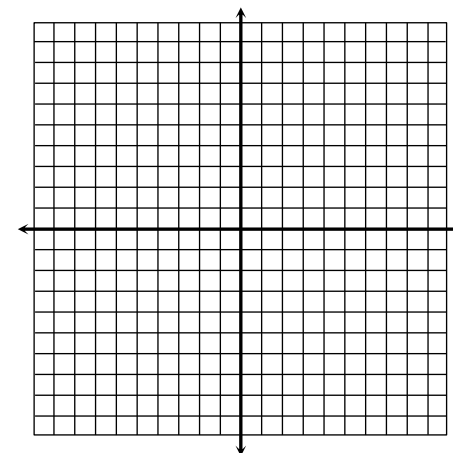


x-int: _____ Holes: _____

Vertical Asymptote: _____ D: _____

Horizontal Asymptote: _____ R: _____

2 $f(x) = \frac{x^2 - 4x - 5}{x^2 - 1}$



x-int: _____ Holes: _____

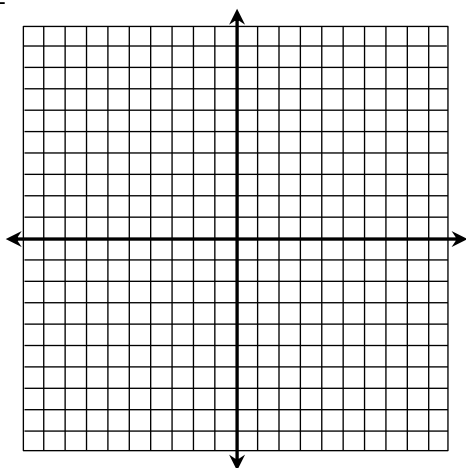
Vertical Asymptote: _____ D: _____

Horizontal Asymptote: _____ R: _____

GRAPHING RATIONAL FUNCTIONS *Practice!*

Directions: Graph each function and identify its key characteristics.

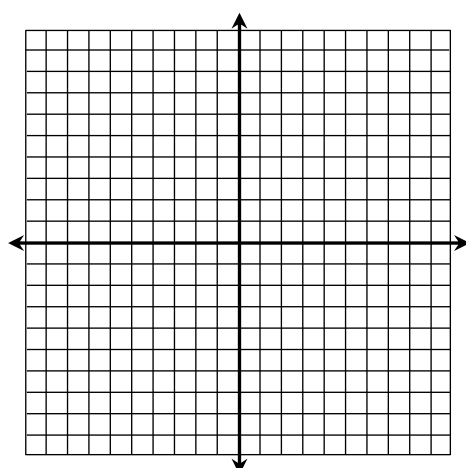
1. $f(x) = \frac{1}{x+5} - 2$



Domain: _____ VA: _____

Range: _____ HA: _____

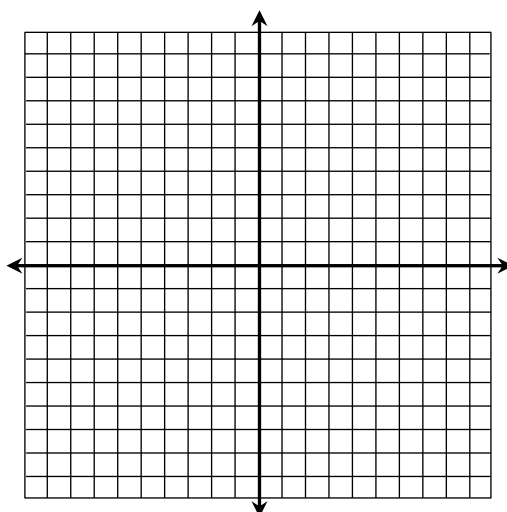
2. $f(x) = \frac{-2}{x} + 6$



Domain: _____ VA: _____

Range: _____ HA: _____

3. $f(x) = \frac{x+7}{x-2}$



x-int: _____

VA: _____

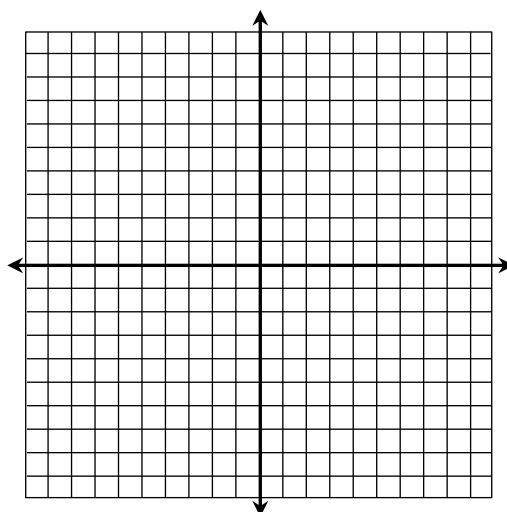
HA: _____

Hole: _____

Domain: _____

Range: _____

4. $f(x) = \frac{2x+2}{x^2-1}$



x-int: _____

VA: _____

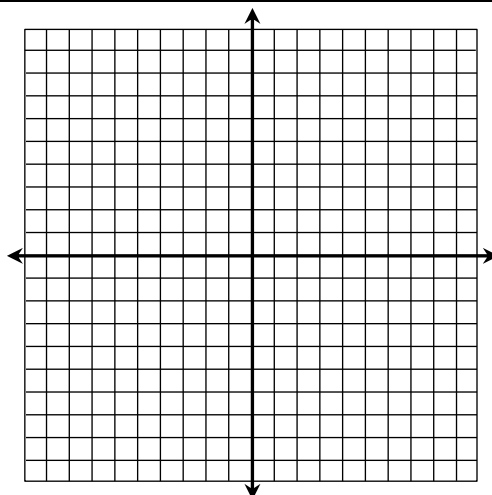
HA: _____

Hole: _____

Domain: _____

Range: _____

5. $f(x) = \frac{x^2 + x - 2}{3x + 6}$



x-int: _____

VA: _____

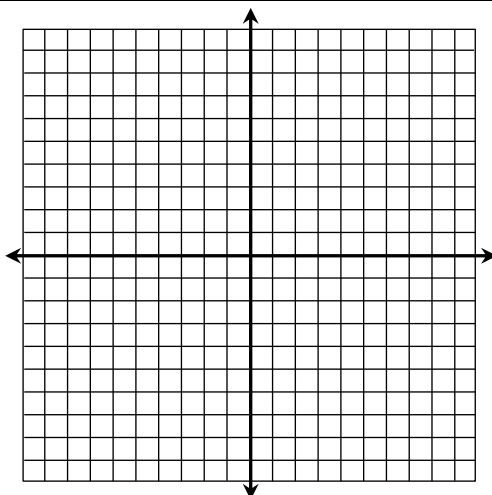
HA: _____

Hole: _____

Domain: _____

Range: _____

6. $f(x) = \frac{x^2 - 6x + 8}{x^2 - 13x + 36}$



x-int: _____

VA: _____

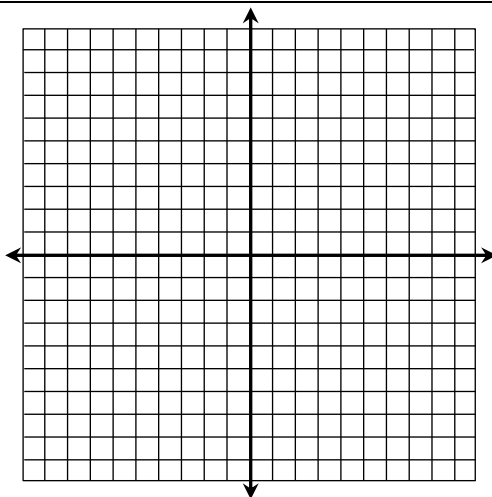
HA: _____

Hole: _____

Domain: _____

Range: _____

7. $f(x) = \frac{x^2 + 5x}{2x^2 + 7x - 15}$



x-int: _____

VA: _____

HA: _____

Hole: _____

Domain: _____

Range: _____

8. The **reciprocal parent function** is translated so that it has asymptotes at $x = 4$ and $y = -9$. Write an equation that could represent this new function.

9. Write a **rational function** with an x-intercept at $(1, 0)$ and vertical asymptotes at $x = -3$ and $x = -5$. Then give the horizontal asymptote of the function.

Name: _____

Unit 8: Rational Functions

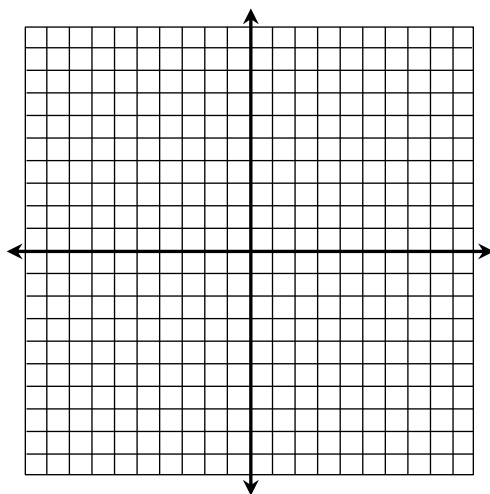
Date: _____ Per: _____

Homework 9: Graphing Review

**** This is a 2-page document! ****

Graph each function and identify its key characteristics.

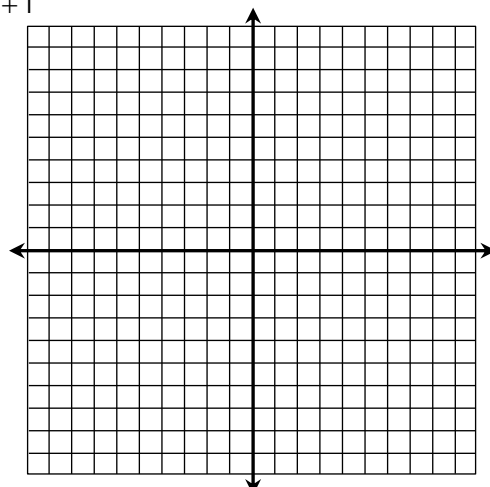
1. $f(x) = \frac{-1}{x-2}$



Domain: _____ VA: _____

Range: _____ HA: _____

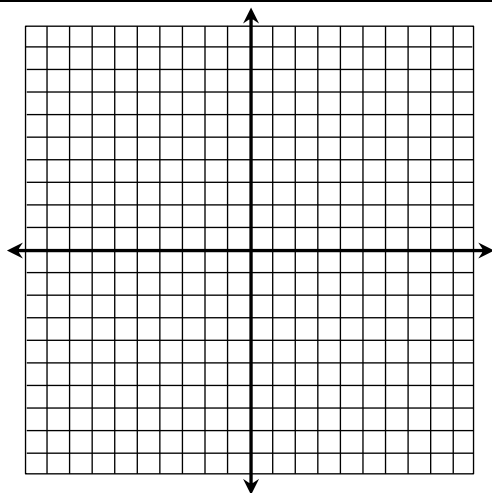
2. $f(x) = \frac{2}{x+3} + 1$



Domain: _____ VA: _____

Range: _____ HA: _____

3. $f(x) = \frac{x+4}{4x+12}$



x-int: _____

VA: _____

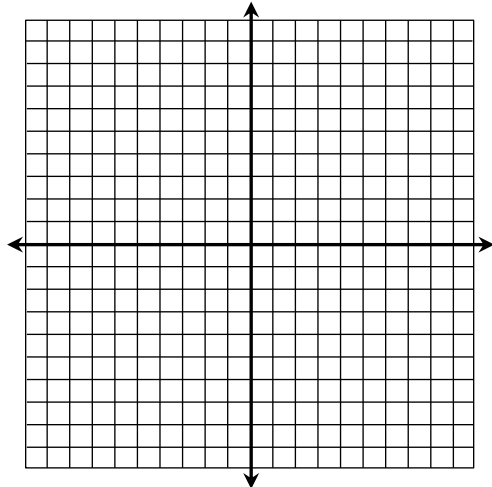
HA: _____

Hole: _____

Domain: _____

Range: _____

4. $f(x) = \frac{x^2 - 7x + 6}{x-6}$



x-int: _____

VA: _____

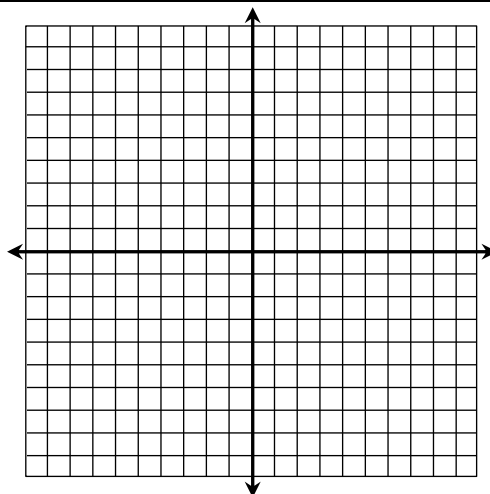
HA: _____

Hole: _____

Domain: _____

Range: _____

5. $f(x) = \frac{x^2 - 16}{x^2 + 10x + 24}$



x-int: _____

VA: _____

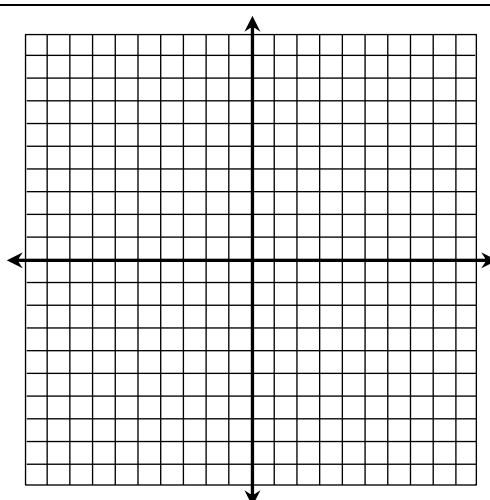
HA: _____

Hole: _____

Domain: _____

Range: _____

6. $f(x) = \frac{x - 5}{3x^2 - 16x + 5}$



x-int: _____

VA: _____

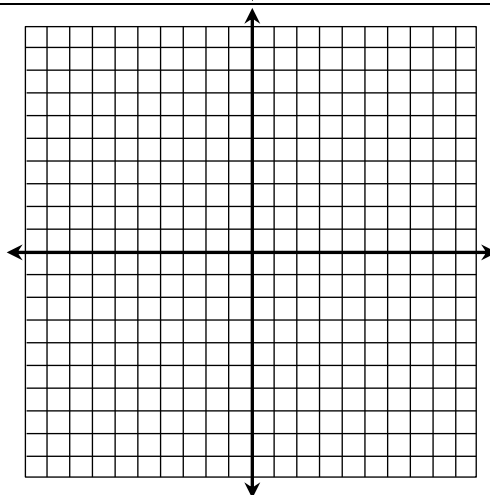
HA: _____

Hole: _____

Domain: _____

Range: _____

7. $f(x) = \frac{x^3 - 2x^2 - 8x}{3x^2 - 27}$



x-int: _____

VA: _____

HA: _____

Hole: _____

Domain: _____

Range: _____

8. If $f(x) = \frac{1}{x}$ is reflected across the x-axis, then translated 5 units right and 7 units down. Write an equation that could represent this new function, then identify the asymptotes.

9. Write a **rational function** with an x-intercept at $(-9, 0)$, a vertical asymptote at $x = 3$, and a hole located at $(1, -5)$. Then, identify the horizontal asymptote.

Name:		Date:	
Topic:		Class:	
Main Ideas/Questions	Notes/Examples		
	METHOD 1: Proportions		METHOD 2: Using an LCD
	Condense into a proportion. Then, cross-multiply to solve.		Multiply each side of the equation by the least common denominator. Solve the resulting equation.
	ALWAYS REMEMBER TO CHECK YOUR SOLUTIONS!		
Directions: Solve each equation. Check all solutions.			
1. $\frac{7}{x-6} = \frac{4}{x}$		2. $\frac{3}{a-8} = \frac{7}{2a+1}$	
3. $\frac{r}{r-1} = \frac{4}{r}$		4. $\frac{m+3}{3} = \frac{8}{m-2}$	
5. $\frac{y}{3} = \frac{y+8}{y+5}$		6. $\frac{2x-3}{2} = \frac{3}{x+4}$	

$$7. \frac{4}{3} - \frac{7}{n} = \frac{1}{6}$$

$$8. \frac{3}{k} - \frac{1}{2} = \frac{12}{k}$$

$$9. \frac{p}{p-2} + 2 = \frac{8}{p^2-4}$$

$$10. \frac{13}{3w-3} - \frac{1}{w-1} = \frac{w}{9}$$

$$11. \frac{x+6}{x+3} = 2 - \frac{5x+12}{x+3}$$

$$12. \frac{3c-2}{c^2-4c} = \frac{c}{c-4} - 1$$

Name: _____

Unit 8: Rational Functions

Date: _____ Per: _____

Homework 10: Solving Rational Equations

**** This is a 2-page document! ****

Directions: Solve each equation. Be sure to check all equations.

1. $\frac{5}{m} = \frac{8}{2m-6}$

2. $\frac{y}{4} = \frac{9}{y}$

3. $\frac{x+5}{5} = \frac{6}{x-2}$

4. $\frac{2}{a-2} = \frac{3a-1}{2a+11}$

5. $w + \frac{16}{w} = 10$

6. $\frac{1}{3} + \frac{2}{k} = \frac{13}{3k}$

7. $\frac{4}{3v} - \frac{1}{v} = \frac{v+2}{2v^2}$

8. $\frac{3}{n} - \frac{1}{2} = \frac{n-5}{n}$

$$9. \frac{4}{x} = \frac{1}{x^2 + 4x} + \frac{5}{x}$$

$$10. \frac{1}{p-2} - \frac{5}{p^2 + 4p - 12} = \frac{6}{p-2}$$

$$11. x + \frac{10}{x-9} = 2$$

$$12. \frac{5r}{r+1} = 4 - \frac{5}{r+1}$$

$$13. \frac{2}{c^2 - 4c} = \frac{1}{3c} + \frac{1}{3}$$

$$14. \frac{y+6}{y-1} - \frac{y-4}{y^2 - y} = \frac{1}{y-1}$$

Name: _____

Algebra 2

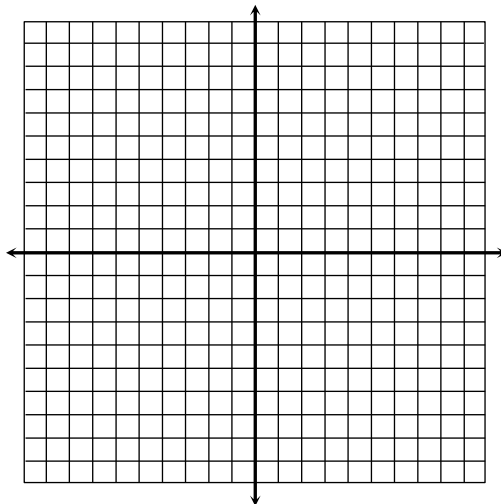
Date: _____ Per: _____

Unit 8: Rational Functions

Quiz 8-3: Graphing Rational Functions & Solving Rational Equations

Part I: Graphing – Graph each function and identify its key characteristics.

1. $f(x) = \frac{1}{x-4} + 1$



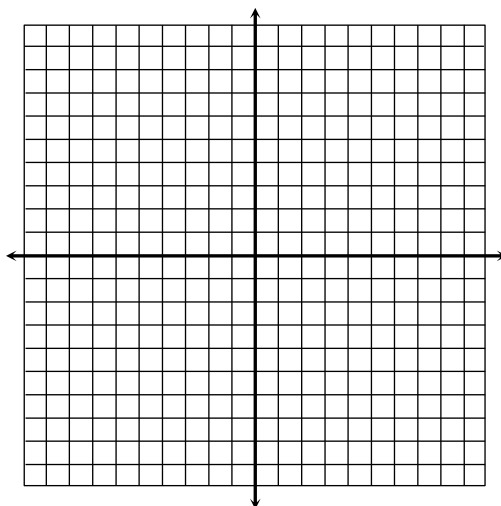
VA: _____

HA: _____

Domain: _____

Range: _____

2. $f(x) = \frac{x+2}{x+6}$



x-int: _____

VA: _____

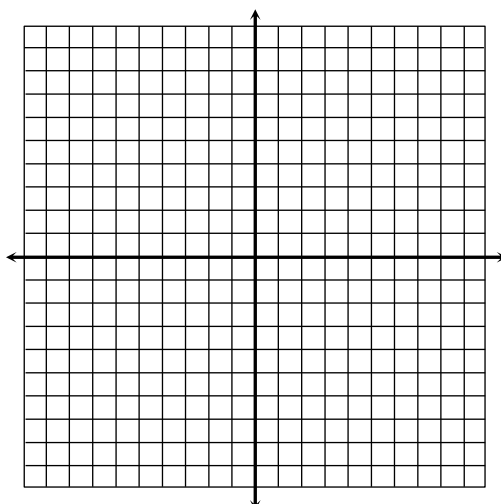
HA: _____

Hole: _____

Domain: _____

Range: _____

3. $f(x) = \frac{5x-10}{x^2-4}$



x-int: _____

VA: _____

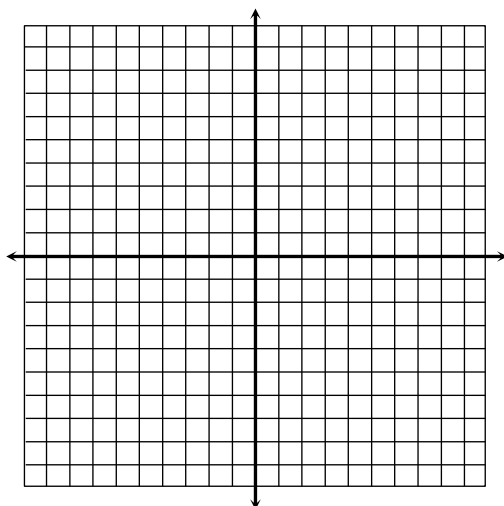
HA: _____

Hole: _____

Domain: _____

Range: _____

4. $f(x) = \frac{x^2 - 7x}{x - 7}$



x-int: _____

VA: _____

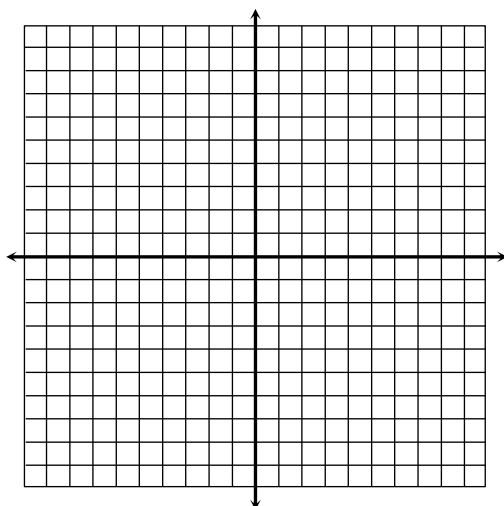
HA: _____

Hole: _____

Domain: _____

Range: _____

5. $f(x) = \frac{x^2 + 4x - 32}{x^2 - 5x + 4}$



x-int: _____

VA: _____

HA: _____

Hole: _____

Domain: _____

Range: _____

Part II: Equations – Solve each equation. Be sure to check all solutions.

6. $\frac{x+4}{2} = \frac{4x}{x-6}$

7. $\frac{1}{3} + \frac{m}{2} = \frac{m+4}{4}$

6. _____

7. _____

8. _____

9. _____

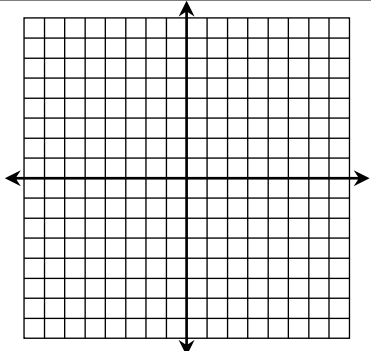
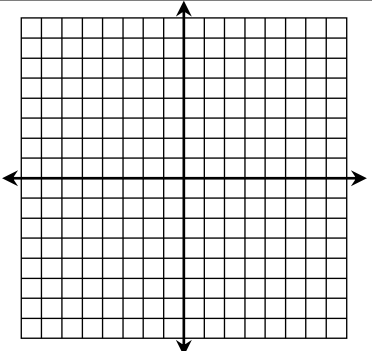
8. $\frac{a}{a-5} - \frac{1}{a+2} = \frac{47}{a^2 - 3a - 10}$

9. $\frac{3}{w^2 - w - 12} = \frac{w-1}{w-4} - 1$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
VARIATION <i>Models</i>	Variation is a special relationship between variables.		
	DIRECT VARIATION "y is directly proportional to x"	JOINT VARIATION "y varies jointly as x and z"	INVERSE VARIATION "y is inversely proportional to x"
	k is the CONSTANT OF VARIATION		
<i>Identifying</i> EQUATIONS	Directions: Determine whether the equation represents a direct, joint, or inverse variation. Identify the constant of variation.		
	1. $y = 2x$	2. $y = \frac{24}{x}$	
	3. $y = 5xz$	4. $3y = x$	
	5. $xy = 36$	6. $\frac{4}{3}y = xz$	
	7. $\frac{y}{x} = \frac{2}{5}$	8. $2A = bh$	
DIRECT VARIATION <i>Examples</i>	9. If y is directly proportional to x and y = 28 when x = 7, find x when y = 52.		10. If r is directly proportional to the square of s and r = 72 when s = 12, find r when s = 8.
	11. The dollar amount d that Megan earns varies directly with the number of hours h that she works. In her last paycheck, she earned \$148.50 working 18 hours. If her next paycheck is \$90.75, how many hours did she work?		

JOINT VARIATION <i>Examples</i>	12. If y varies jointly with x and z and $y = 48$ when $x = 8$ and $z = 15$, find y when $x = 14$ and $z = 20$.	13. If p varies jointly as q and the cube of r , and $p = 270$ when $q = 6$ and $r = 3$, find p when $q = 3$ and $r = 5$.
	14. For Tom's savings account, the interest earned varies jointly with the principal and amount of time (in years) since opening the account. If Tom opened the account 3 years ago with \$1,200 and earned \$144 in interest, how much interest will he have earned 8 years after opening the account?	
INVERSE VARIATION <i>Examples</i>	15. If y is inversely proportional to x and $y = 7.5$ when $x = 8$, find x when $y = 5$.	16. If m is inversely proportional to the square root of n and $m = 27$ when $n = 16$, find m when $n = 64$.
	17. The time it takes Adam to drive to Disney World varies inversely with his average rate of speed. If it takes Adam 10.8 hours at an average speed of 50 miles per hour, how long would it take him at an average speed of 60 miles per hour?	
COMBINED VARIATION	A combined variation is a relationship that contains both _____ and _____ variation.	
<i>Examples</i>	18. " y varies directly as x and inversely as z ."	19. " p varies inversely with q squared and directly with r cubed."
	20. a varies directly as b and inversely as c . If $a = 16$ when $b = 12$ and $c = 6$, find a when $b = 28$ and $c = 4$.	

	DIRECT <i>Variation</i>	INVERSE <i>Variation</i>	JOINT <i>Variation</i>
EQUATION FORM			
EXAMPLE			
GRAPH			X
EXAMPLE	If y varies directly as x and $y = 10$ when $x = 14$, find the constant and write an equation to represent the relationship.	If y varies inversely as x and $y = 4$ when $x = 12$, find the constant and write an equation to represent the relationship.	If y varies jointly as x and z , and $y = 42$ when $x = 9$ and $z = 7$, find the constant of variation and write an equation to represent the relationship.
COMBINED <i>Variation</i>	What is it?	Example: If p varies directly with q squared and inversely with r , and $p = 38.4$ when $q = 8$ and $r = 5$, find the constant of variation and write an equation to represent the relationship.	

Name: _____

Unit 8: Rational Functions

Date: _____ Per: _____

Homework 11: Direct, Joint, & Inverse Variation

**** This is a 2-page document! ******Directions:** Identify the equation for each variation model.

DIRECT VARIATION	INVERSE VARIATION	JOINT VARIATION

Directions: Translate into an equation to represent the relationship.

1. "w is directly proportional to v"	2. "a varies inversely with b"
3. "f varies jointly with g and h"	4. "r is inversely proportional to s cubed"
5. "y varies jointly with x and the square root of z."	6. "m varies directly with n squared and inversely with p"

Directions: Determine whether the equation represents a direct, joint, or inverse variation. Then, identify the constant.

7. $t = \frac{100}{s}$	8. $\frac{y}{5} = x$	9. $\frac{p}{qr} = 4$
10. $C = 2\pi r$	11. $V = \frac{1}{3}\pi r^2 h$	12. $xy = \frac{2}{5}$

Directions: Use the variation model to determine the missing value.

13. If y is directly proportional to x and y = 17.5 when x = 21, find y when x = 39.	14. If m varies directly with n and m = 209 when n = 22, find n when m = 361.
--	---

15. If y varies jointly with x and z and $y = 90$ when $x = 2$ and $z = 9$, find x when $y = 105$ and $z = 3$.	16. If c varies jointly as a cubed and b squared, and $c = 25.6$ when $a = 2$ and $b = 4$, find c when $a = 4$ and $b = 7$.
17. If y varies inversely with x and $y = 4.75$ when $x = 38$, find y when $x = 50$.	18. If q is inversely proportional to r squared and $q = 30$ when $r = 3$, find r when $q = 1.2$.
19. r varies directly as s and inversely as t cubed. If $r = 607.5$ when $s = 12$ and $t = 2$, find r when $s = 42$ and $t = 6$.	
20. The cost of printing fliers for the opening of an art show varies directly with the number of copies. If it costs \$45 to print 750 copies, what will be the total cost for 120 more copies?	
21. The time it takes to fill a water tank varies inversely with the water rate of the hose. At 8 gallons per minute, a hose can fill the tank in 45 minutes. How long will it take to fill the same tank at 15 gallons per minute?	
22. The volume of a pyramid varies jointly with its height and the area of its base. A pyramid with a height of 16 feet and a base area of 78 square feet has a volume of 416 cubic feet. If a pyramid has a base area of 102 square feet and a volume of 238 cubic feet, find its height.	

Unit 8 Test Study Guide

(Rational Functions)

Name: _____

Date: _____ Per: _____

Topic 1: Simplifying Rational Expressions

1. $\frac{12x^2 - 30x}{20x^3 - 50x^2}$	2. $\frac{4a^2 - 36}{24 - 8a}$	3. $\frac{n^2 - 13n + 40}{3n^2 - 14n - 5}$
--	--------------------------------	--

Topic 2: Operations with Rational Expressions

4. $\frac{6p^2 - 13p + 5}{2p^2 + 17p - 9} \cdot \frac{p^2 + 16p + 63}{4p + 28}$	5. $\frac{50 - 2w^2}{3w^2 + 9w - 30} \cdot \frac{w^2 + 5w - 14}{6w - 30}$
6. $\frac{5y + 5}{2} \div \frac{25y - 20}{40y^2 - 32y}$	7. $\frac{2c^2 + 4c - 6}{4c^2 - 7c + 3} \div \frac{16c^2 + 48c}{16c^2 - 9}$
8. $\frac{6x}{x^2 - 16} - \frac{x - 20}{x^2 - 16}$	9. $\frac{16}{3} - \frac{4k + 56}{3k + 15}$

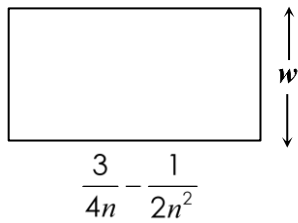
10. $\frac{m^2 - 7m - 18}{m^2 - 10m + 9} + \frac{6}{m - 1}$	11. $\frac{r}{2r + 1} + \frac{12r - 6}{4r^2 - 1}$
12. $\frac{a + 4}{8a} + \frac{5}{24}$	13. $\frac{2}{h + 2} + \frac{5}{h + 5}$
14. $\frac{3x - 1}{x - 1} - \frac{x - 1}{x - 3} + \frac{x + 1}{x^2 - 4x + 3}$	

Topic 3: Complex Fractions

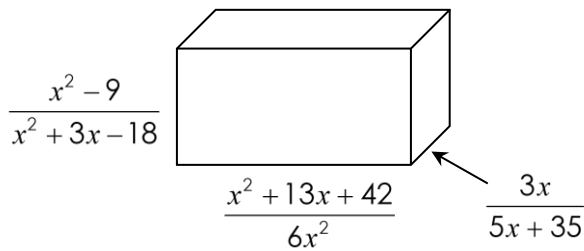
15. $\frac{\frac{12m^3}{m^2 + 14m + 45}}{\frac{3m^3 - 6m^2}{m^2 + 7m - 18}}$	16. $\frac{\frac{8k + 12}{9}}{\frac{k}{3} + \frac{1}{2}}$
17. $\frac{\frac{b}{2} - \frac{32}{b}}{1 + \frac{8}{b}}$	18. $\frac{\frac{x}{x + 1} + \frac{4}{x}}{\frac{4x + 7}{3x + 3} - \frac{1}{3}}$

Topic 4: Applications

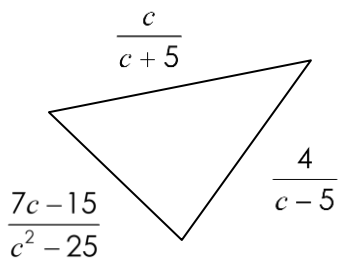
19. If the **area** of the rectangle below is $\frac{n-3}{n} + \frac{7}{2}$, find w , its **width**.



20. Find the **volume** of the rectangular prism below.



21. Find the **perimeter** of the triangle below.



Topic 5: Solving Rational Equations

22. $\frac{a-8}{a} = \frac{3}{a+5}$

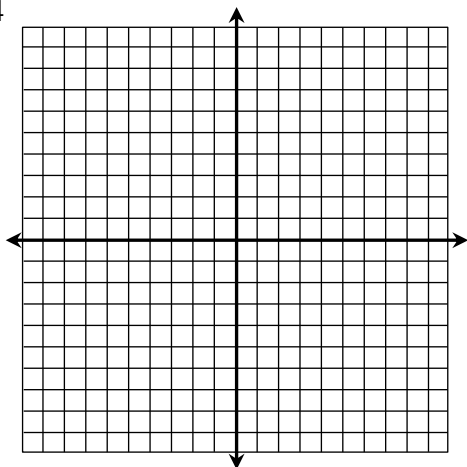
23. $\frac{3}{2n} + \frac{1}{n^2} = \frac{n-2}{2n^2}$

24. $\frac{p^2 + p + 8}{p^2 - 7p + 6} + \frac{2}{p - 1} = \frac{2}{3}$

25. $\frac{1}{3} - \frac{1}{4v + 3} = \frac{v^2 - 3v - 18}{12v + 9}$

Topic 6: Graphing Rational Functions

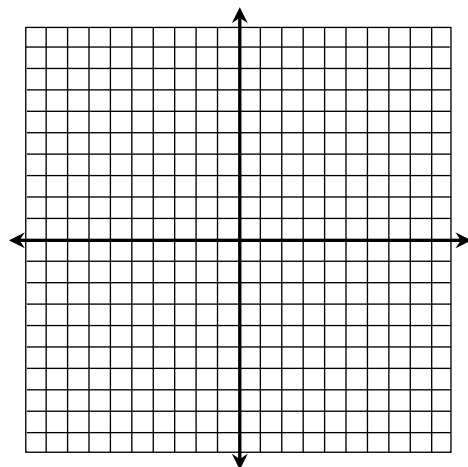
26. $f(x) = \frac{1}{x + 4} - 7$



Domain: _____ VA: _____

Range: _____ HA: _____

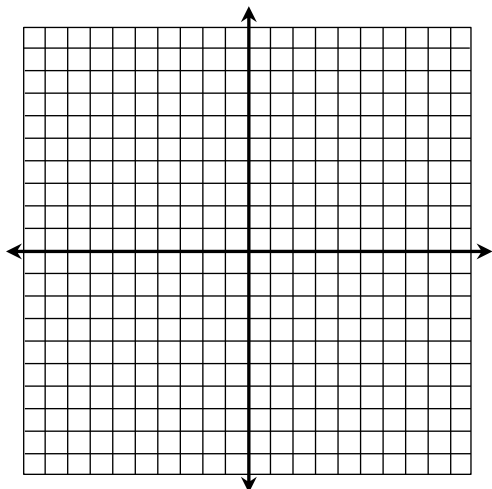
27. $f(x) = \frac{-2}{x} + 3$



Domain: _____ VA: _____

Range: _____ HA: _____

28. $f(x) = \frac{x + 5}{x + 2}$



x-int: _____

VA: _____

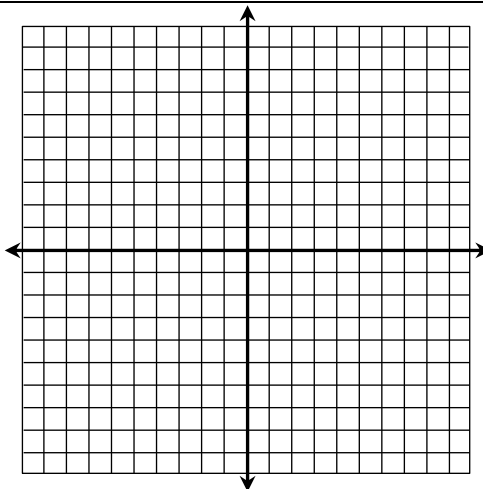
HA: _____

Hole: _____

Domain: _____

Range: _____

29. $f(x) = \frac{x^2 + 6x}{2x + 12}$



x-int: _____

VA: _____

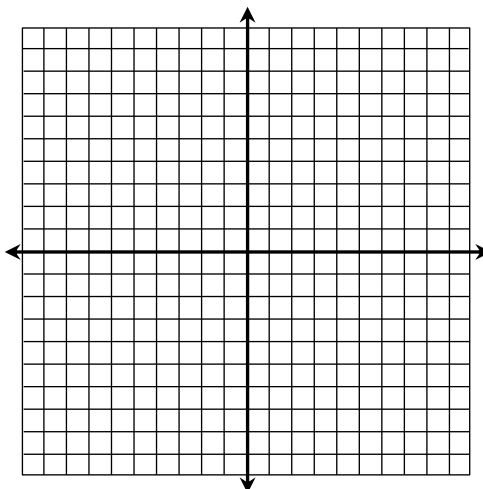
HA: _____

Hole: _____

Domain: _____

Range: _____

30. $f(x) = \frac{x^2 - 11x + 24}{x^2 - 9}$



x-int: _____

VA: _____

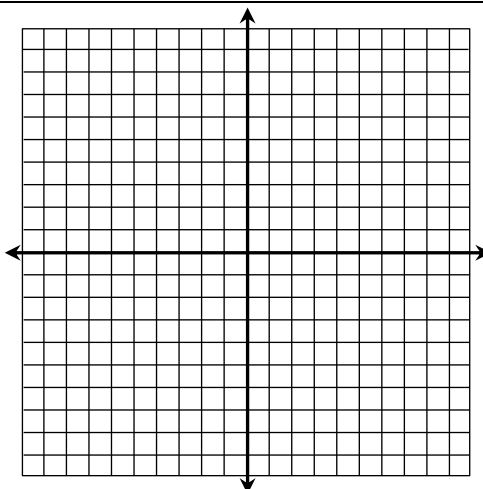
HA: _____

Hole: _____

Domain: _____

Range: _____

31. $f(x) = \frac{3x + 3}{2x^2 + 9x + 7}$



x-int: _____

VA: _____

HA: _____

Hole: _____

Domain: _____

Range: _____

32. The **reciprocal parent function** is translated 8 units left and 3 units up. Write an equation to represent this function. Then, identify the asymptotes.

33. A rational function has an x-intercept at $(-1, 0)$ and vertical asymptotes at $x = 7$ and $x = -5$. Write an equation that could represent this function, then identify the horizontal asymptote.

Topic 7: Variation

DIRECT VARIATION	JOINT VARIATION	INVERSE VARIATION
Translate into an equation to represent the relationship.		
34. "s varies jointly with t cubed and v"	35. "m varies inversely with n squared and directly with p"	
Determine if the equation represents a direct, joint, or inverse variation. Identify the constant.		
36. $3y = x^2z$	37. $\frac{1}{2}d = r$	38. $\frac{8}{x} = \frac{y}{6}$
Use the variation type to find the missing value.		
39. w varies directly as r cubed and inversely as s. If w = 24 when r = 4 and s = 8, find r when w = 81 and s = 27.	40. If b varies jointly with a and c and b = 112 when a = 12 and c = 7, find a when b = 72 and c = 2.	
41. The wind force F on a sail varies jointly as the area A of the sail and the square of the wind speed w. force on a sail with area an area of 500 ft ² is 64.8 pounds when the wind speed is 18 mph. What would be the force for a sail with an area of 250 ft ² with a wind speed of 35 mph?		
42. The volume, V, of a gas varies directly as the temperature, t, and inversely as the pressure, P. At a temperature of 200° and a pressure of 500 mmHG, the volume of the gas is 240 cm ³ . Find the volume of the gas at a temperature of 275° and a pressure of 400 mmHG.		

Name: _____

Date: _____ Per: _____

Unit 8 Test

Rational Functions

1. Simplify the expression below.

$$\frac{2a^3 - 32a}{2a^3 + 16a + 32a}$$

A. $2(a - 4)$

C. $\frac{(a - 4)}{(a + 4)}$

B. -2

D. $\frac{2(a - 4)}{(a + 4)}$

2. Simplify the expression below.

$$\frac{4yz^2}{3xy^4z} \cdot \frac{15x^2y}{8x^3yz}$$

A. $\frac{5}{2x^2y^3}$

C. $\frac{2x^2y^3}{5z}$

B. $\frac{5z}{2x^2y^3}$

D. $\frac{2x^2y^3}{5}$

3. Simplify the expression below.

$$\frac{2 - 18w^2}{3w^2 - 8w - 3} \cdot \frac{3w^2 - 7w - 6}{12w^2 + 80w - 28}$$

A. $\frac{3w + 2}{2(w + 7)}$

C. $\frac{-(3w + 2)}{2(w + 7)}$

B. $\frac{3w + 2}{w + 7}$

D. $\frac{-2(3w + 2)}{w + 7}$

4. Simplify the expression below.

$$\frac{2c^2 - 5c - 3}{c^2 + 4c - 21} \div (2c + 1)$$

A. $\frac{2c + 1}{c + 7}$

C. $\frac{(c + 3)}{(c + 7)(c - 3)}$

B. $\frac{1}{c + 7}$

D. $c + 7$

5. Simplify the expression below.

$$\frac{p^2 - 14p + 40}{16p^3} \div \frac{3p^2 - 18p + 24}{2p^2 - 4p}$$

A. $\frac{24p^2}{p - 10}$

C. $\frac{p - 10}{24p^2}$

B. $\frac{24(p - 10)}{p^2}$

D. $\frac{1}{24p^2(p - 10)}$

6. Simplify the expression below.

$$\frac{17x - 9}{3x + 6} - \frac{2x - 3}{3x + 6}$$

A. $-5x$

C. $\frac{5x - 2}{x + 2}$

B. $\frac{5x - 2}{3x}$

D. $\frac{5x - 4}{x + 2}$

7. Simplify the expression below.

$$\frac{m}{m-3} + \frac{5m-48}{m^2+5m-24}$$

A. $\frac{m}{m-3}$

C. $m+2$

B. $\frac{m+8}{m-3}$

D. $\frac{m+16}{m+8}$

8. Simplify the expression below.

$$\frac{8}{v+2} + \frac{3}{v-6}$$

A. $\frac{11}{(v+2)(v-6)}$

C. $\frac{11v-42}{2v-4}$

B. $\frac{11v-42}{(v+2)(v-6)}$

D. $\frac{11}{2v-4}$

9. Simplify the expression below.

$$\frac{\frac{9-k}{28k^3}}{\frac{k-9}{4k^6}}$$

A. $\frac{-k^3}{7}$

C. $\frac{k^3}{7}$

B. $\frac{-7}{k^2}$

D. $\frac{7}{k^2}$

10. Simplify the expression below.

$$\frac{\frac{r^2}{6} + \frac{r}{2}}{\frac{r+3}{12}}$$

A. $\frac{r+3}{6}$

C. $2r$

B. $\frac{r}{6}$

D. $\frac{1}{2r}$

11. Simplify the expression below.

$$\frac{p - \frac{9}{4p}}{1 + \frac{3}{2p}}$$

A. $\frac{2p-3}{2}$

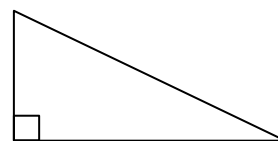
C. $2(2p-3)$

B. $\frac{p-9}{8p}$

D. $\frac{p-9}{2p}$

12. Find the area of the triangle below.

$$\frac{14x-28}{x^2-3x-4}$$



$$\frac{x^2+9x+8}{x-2}$$

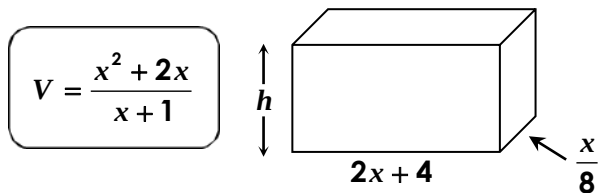
A. $7(x-2)$

C. $\frac{7(x+8)}{x-4}$

B. $14(x-2)$

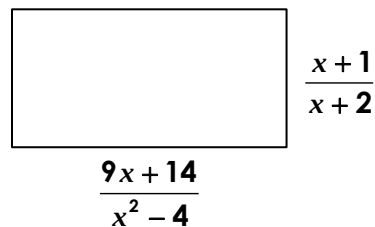
D. $\frac{14(x+8)}{x-4}$

13. Given the volume, V , of the rectangular prism below, find h , the height of the prism.



- A. $\frac{16}{x+1}$ C. $16(x+1)$
 B. $\frac{4}{x+1}$ D. $4(x+1)$

14. Find the perimeter of the rectangle below.



- A. $x - 3$ C. $\frac{x + 6}{x - 2}$
 B. $2(x - 3)$ D. $\frac{2(x + 6)}{x - 2}$

15. Solve the equation below.

$$\frac{a+2}{5-a} = \frac{6}{a-5}$$

16. Solve the equation below.

$$\frac{1}{2k} + \frac{4}{k} = \frac{k}{k+2}$$

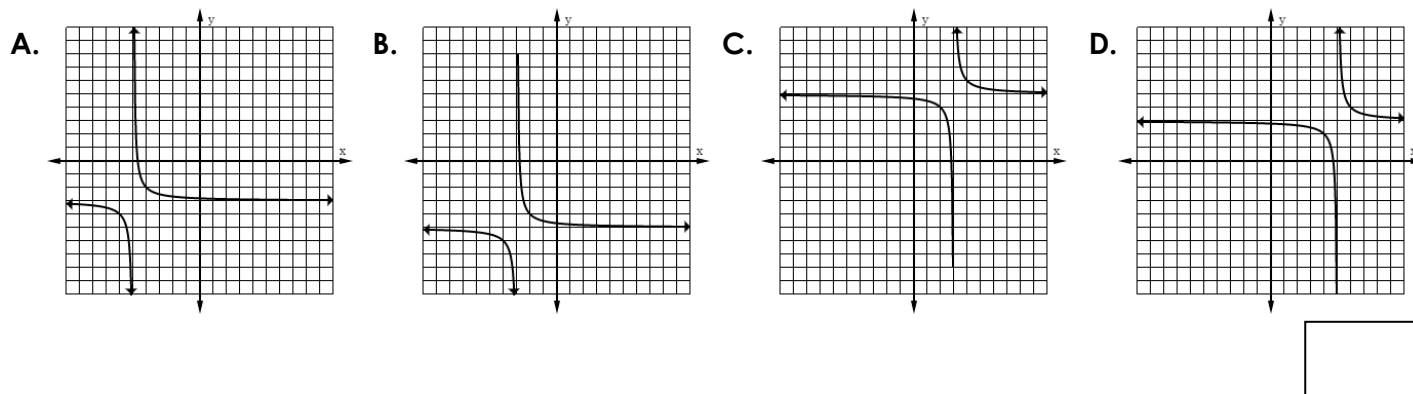
17. Solve the equation below.

$$\frac{1}{n-4} + \frac{2}{n-2} = \frac{2}{n^2 - 6n + 8}$$

18. Solve the equation below.

$$\frac{x+1}{x} + \frac{x-5}{3x} = \frac{2x+9}{4x}$$

19. Which graph represents the function $f(x) = \frac{1}{x+3} - 5$?



Use for questions 20-21:

The parent function of a reciprocal function is translated two units left and four units down.

20. Which of the following represents the horizontal asymptote of the new function?

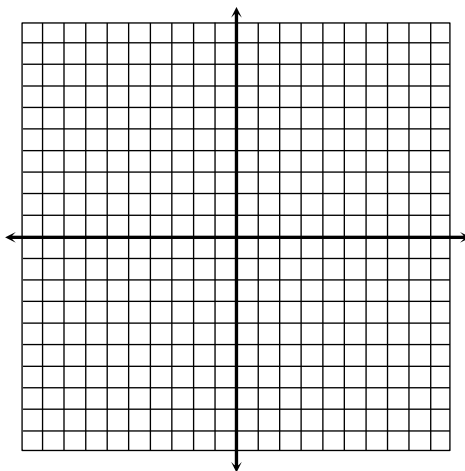
- A. $x = 2$
- B. $x = -4$
- C. $y = -2$
- D. $y = -4$

21. What is the domain of this function?

- A. $\{x|x \neq 2\}$
- B. $\{x|x \neq -2\}$
- C. $\{y|y \neq 2\}$
- D. $\{y|y \neq -4\}$

Graph each function below and identify its key characteristics.

22. $f(x) = \frac{x+4}{x-6}$



x-int: _____

VA: _____

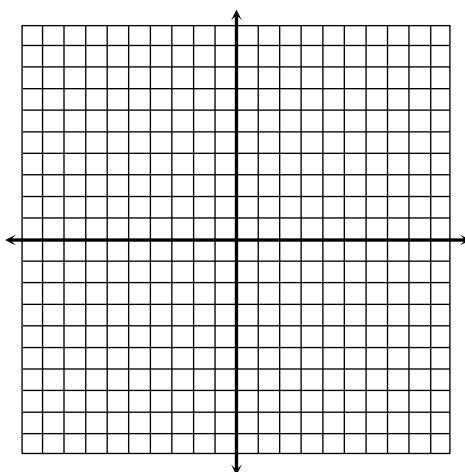
HA: _____

Hole: _____

Domain: _____

Range: _____

23. $f(x) = \frac{x+2}{x^2-3x-10}$



x-int: _____

VA: _____

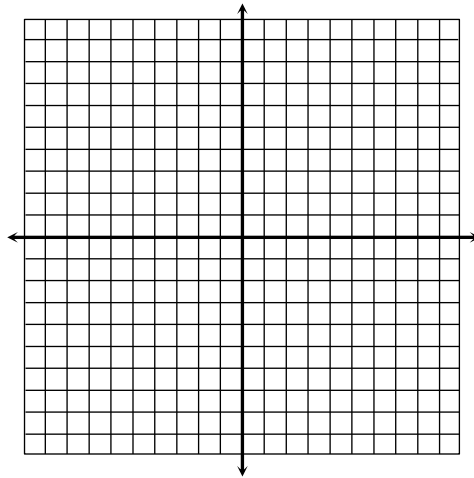
HA: _____

Hole: _____

Domain: _____

Range: _____

24. $f(x) = \frac{2x^2 - 18}{x + 3}$



x-int: _____

VA: _____

HA: _____

Hole: _____

Domain: _____

Range: _____

25. Which functions have a vertical asymptote at $x = -2$? Check all that apply.

☐ $f(x) = \frac{x^2 - 4}{x + 2}$	☐ $f(x) = \frac{1}{x + 2} - 2$	☐ $f(x) = \frac{x^2 - 8x - 9}{x^2 - 7x - 18}$
☐ $f(x) = \frac{x - 2}{x^2 - 4}$	☐ $f(x) = \frac{1}{x - 2} + 2$	☐ $f(x) = \frac{2x^2 + 4x}{x^2 + 2x}$

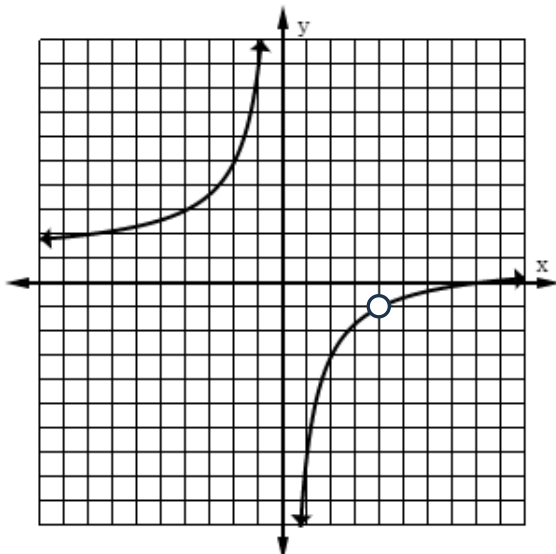
26. Which statement accurately describes the horizontal asymptotes of the functions below?

$\left\{ f(x) = \frac{1}{x - 1}, g(x) = \frac{x}{x^2 - 1} \right\}$

- A. The horizontal asymptote for $f(x)$ is $y = 0$;
The horizontal asymptote for $g(x)$ is $y = 1$ and $y = -1$.
- B. The horizontal asymptote for $f(x)$ is $y = 1$;
The horizontal asymptote for $g(x)$ is $y = 0$.
- C. The horizontal asymptotes for both functions is $y = 1$.
- D. The horizontal asymptotes for both functions is $y = 0$.

☐

27. Which equation could represent the function graphed below?



A. $f(x) = \frac{x^2 - 12x + 32}{x^2 - 4x}$

B. $f(x) = \frac{x^2 - 4x - 32}{x^2 + 4x}$

C. $f(x) = \frac{x^2 + 4x - 32}{x^2 - 4x}$

D. $f(x) = \frac{x^2 + 12x + 32}{x^2 + 4x}$

☐

28. Which statement accurately describes the relationship between variables in the equation below?

$$K = \frac{2v^2}{m}$$

- A. "K varies jointly as v squared and m"
B. "K is inversely proportional to v squared and directly proportional to m"
C. "K is directly proportional to v squared and inversely proportional to m"
D. "K varies jointly as v squared and is inversely proportional to m"

29. If p varies jointly as q and r , and $p = 150$ when $q = 18$ and $r = 5$, find q when $p = 345$ and $r = 9$.

$q =$

30. The time needed to paint a fence varies directly with the length of the fence and inversely with the number of painters. If it takes seven hours to paint 280 feet of fence with two painters, how long will it take four painters to paint 720 feet of fence?

BONUS: Solve the equation below.

$$\frac{1}{x+3} - \frac{2}{3-x} = \frac{4}{x^2-9}$$

$x =$

Name:		Date:	
Topic:		Class:	
Main Ideas/Questions		Notes/Examples	
RATIONAL EXPRESSION	A rational expression is a <u>ratio</u> of two polynomial expressions.		
	To simplify a rational expression:		
	①	Factor everything that can be factored.	
	②	Simplify the monomials (use the exponent rules!)	
③	Eliminate common binomial factors.		
SIMPLIFYING Rational Expressions	1. $\frac{20x^6}{14x^2}$	$\frac{10x^4}{7}$	2. $\frac{6k-36}{k-6} = \frac{6(k-6)}{k-6}$ $= 6$
	3. $\frac{y+8}{y^2+2y-48}$	$= \frac{y+8}{(y+8)(y-6)}$ $= \frac{1}{y-6}$	4. $\frac{2r^2-2r-40}{8r+32} = \frac{2(r^2-r-20)}{8(r+4)}$ $= \frac{2(r-5)(r+4)}{8(r+4)}$ $= \frac{r-5}{4}$
	5. $\frac{p^2-49}{7-p}$	$= \frac{(p+7)(p-7)}{7-p}$ $= -1(p+7)$ $= \frac{-p-7}{1}$	6. $\frac{12a^3-3a}{12a^3+6a^2} = \frac{3a(4a^2-1)}{6a^2(2a+1)}$ $= \frac{3a(2a+1)(2a-1)}{6a^2(2a+1)}$ $= \frac{2a-1}{2a}$
	7. $\frac{6n^2+8n+2}{2n^2-2}$	$= \frac{2(3n^2+4n+1)}{2(n^2-1)}$ $= \frac{2(3n+1)(n+1)}{2(n+1)(n-1)}$ $= \frac{3n+1}{n-1}$	8. $\frac{45-5w}{3w^2-28w+9} = \frac{5(9-w)}{(3w-1)(w-9)}$ $= \frac{-5}{3w-1}$
	Watch out! $\frac{a-b}{b-a} = -1$		

MULTIPLYING

Rational Expressions

If the problem contains-

- **Monomials Only:** Multiply together, then simplify.
- **Binomials/Trinomials:** Factor everything you can FIRST, then simplify.

$$9. \frac{6x^2y^3}{2x^2y^2} \cdot \frac{10x^3y^4}{18y^2} = \frac{60x^5y^7}{36x^2y^4} = \boxed{\frac{5x^3y^3}{3}}$$

$$10. \frac{4a^2b^2}{15ab^3} \cdot \frac{5a^3b^6}{12a^4b^7} = \frac{20a^5b^8}{180a^5b^{10}} = \boxed{\frac{1}{9b^2}}$$

$$11. \frac{4x^2-4x}{2x^2+4x-6} \cdot \frac{x^2+x-6}{4x^2+8x} = \frac{4x(x-1)}{2(x+3)(x-1)} \cdot \frac{(x+3)(x-2)}{4x(x+2)} = \boxed{\frac{x-2}{2(x+2)}}$$

$$12. \frac{10v-5v^2}{v^2-11v+18} \cdot \frac{v^2-8v-9}{15v} = \frac{5v(2-v)}{(v-9)(v-2)} \cdot \frac{(v-9)(v+1)}{15v} = \frac{-1(v+1)}{3} = \boxed{\frac{-v-1}{3}}$$

DIVIDING

Rational Expressions

To divide rational expressions, multiply by the reciprocal !

$$13. \frac{5pq}{16p^3} \div \frac{35p^2q^2}{8q^5} = \frac{5pq}{16p^3} \cdot \frac{8q^5}{35p^2q^2} = \frac{40pq^6}{560p^5q^2} = \boxed{\frac{q^4}{14p^4}}$$

$$14. \frac{10}{4x-8} \div \frac{2x^2+6x}{x^2+x-6} = \frac{10}{4(x-2)} \cdot \frac{(x+3)(x-2)}{2x(x+3)} = \boxed{\frac{5}{4x}}$$

$$15. \frac{2a^3-12a^2}{a^2-4a-12} \div \frac{24a^2-8a^3}{a^2-8a+15} = \frac{2a^2(a-6)}{(a-6)(a+2)} \cdot \frac{(a-3)(a-5)}{8a^2(3-a)} = \frac{-1(a-5)}{4(a+2)} = \boxed{\frac{-a+5}{4(a+2)}}$$

$$16. \frac{k+3}{k} \div (4k+1) \cdot \frac{16k^2-1}{k+3} = \frac{k+3}{k} \cdot \frac{1}{4k+1} \cdot \frac{(4k+1)(4k-1)}{k+3} = \boxed{\frac{4k-1}{k}}$$

Name: _____

Unit 8: Rational Functions

Date: _____ Per: _____

Homework 1: Simplify, Multiply, & Divide
Rational Expressions**** This is a 2-page document! ******Simplify each expression below.**

$$1. \frac{16m^2}{24m^7} = \boxed{\frac{2}{3m^5}}$$

$$2. \frac{n^2 + 7n}{4n^2 + 28n} = \frac{n(n+7)}{4n(n+7)} = \boxed{\frac{1}{4}}$$

$$3. \frac{x^2 - 10x - 24}{x + 2} = \frac{(x-12)(x+2)}{x+2} = \boxed{x-12}$$

$$4. \frac{1-9w^2}{12w-4} = \frac{(1-3w)(1+3w)}{4(3w-1)} = \frac{-1(1+3w)}{4} = \boxed{\frac{-1-3w}{4}}$$

$$5. \frac{4a^2 - 36a}{2a^4 - 24a^3 + 54a^2} = \frac{4a(a-9)}{2a^2(a^2 - 12a + 27)} = \frac{4a(a-9)}{2a^2(a-9)(a-3)} = \boxed{\frac{2}{a(a-3)}}$$

$$6. \frac{y^2 - 36}{5y^2 - 26y - 24} = \frac{(y+6)(y-6)}{(5y+4)(y-6)} = \boxed{\frac{y+6}{5y+4}}$$

Find the product. Give your answer in simplest form.

$$7. \frac{32x^3y}{5xy^2} \cdot \frac{15y}{8x^2y^4} = \frac{480x^3y^2}{40x^3y^6} = \boxed{\frac{12}{y^4}}$$

$$8. \frac{m^2 - 6m + 8}{2m - 2} \cdot \frac{10}{m - 4} = \frac{(m-2)(m-4)}{2(m-1)} \cdot \frac{10}{m-4} = \boxed{\frac{5(m-2)}{m-1}}$$

$$9. \frac{28n+40}{35n+50} \cdot \frac{12n+24}{8n+16} = \frac{4(7n+10)}{5(7n+10)} \cdot \frac{12(n+2)}{8(n+2)} = \frac{48}{40} = \boxed{\frac{6}{5}}$$

$$10. \frac{p+10}{9-p} \cdot \frac{p^2-5p-36}{4p^2+16p} = \frac{p+10}{9-p} \cdot \frac{(p-9)(p+4)}{4p(p+4)} = \frac{-1(p+10)}{4p} = \boxed{\frac{-p-10}{4p}}$$

$$11. \frac{v^2 - 49}{20v^3} \cdot \frac{4v^2 - 24v}{v^2 + v - 42}$$

$$= \frac{(v+7)(v-7)}{20v^3} \cdot \frac{4v(v-6)}{(v+7)(v-6)}$$

$$= \boxed{\frac{v-7}{5v^2}}$$

$$12. \frac{2n-3}{n+1} \cdot \frac{2n^2+5n+3}{9-4n^2}$$

$$= \frac{2n-3}{n+1} \cdot \frac{(2n+3)(n+1)}{(3-2n)(3+2n)}$$

$$= \boxed{-1}$$

$$13. \frac{12k^2 - 54k}{6k} \cdot \frac{6}{81 - 18k}$$

$$\frac{6k(2k-9)}{6k} \cdot \frac{6}{9(9-2k)}$$

$$= \frac{-1(6)}{9} = \boxed{-\frac{2}{3}}$$

$$14. \frac{6c^2 + 13c - 63}{6c^2 - 17c + 7} \cdot \frac{2c^2 - 9c + 4}{12c + 54}$$

$$= \frac{(2c+9)(3c-7)}{(2c-1)(3c-7)} \cdot \frac{(2c-1)(c-4)}{6(2c+9)}$$

$$= \boxed{\frac{c-4}{6}}$$

Find the quotient. Give your answer in simplest form.

$$15. \frac{14m^4}{3m} \div \frac{7m^2}{18m^5} = \frac{14m^4}{3m} \cdot \frac{18m^5}{7m^2}$$

$$= \frac{252m^9}{21m^3}$$

$$= \boxed{12m^6}$$

$$16. \frac{x^2 - 3x - 28}{6x} \div \frac{2x - 14}{2}$$

$$= \frac{(x-7)(x+4)}{6x} \cdot \frac{2}{2(x-7)}$$

$$= \boxed{\frac{x+4}{6x}}$$

$$17. \frac{2a^2 + 14a}{8a^2} \div (10a + 70)$$

$$= \frac{2a(a+7)}{8a^2} \cdot \frac{1}{10(a+7)}$$

$$= \boxed{\frac{1}{40a}}$$

$$18. \frac{1-h^2}{2h^2 - 10h - 12} \div \frac{2h-2}{6}$$

$$= \frac{(1+h)(1-h)}{2(h^2-5h-6)} \cdot \frac{6}{2(h-1)}$$

$$= \frac{(1+h)(1-h)}{2(h-6)(h+1)} \cdot \frac{6}{2(h-1)}$$

$$= \frac{-1(6)}{4(h-6)} = \boxed{\frac{-3}{2(h-6)}}$$

$$19. \frac{2r+2}{r+2} \div \frac{4r^2+8r+4}{12r+12}$$

$$= \frac{2(r+1)}{r+2} \cdot \frac{12(r+1)}{4(r^2+2r+1)}$$

$$= \frac{2(r+1)}{r+2} \cdot \frac{12(r+1)}{4(r+1)(r+1)}$$

$$= \boxed{\frac{6}{r+2}}$$

$$20. \frac{20x-4}{8x^2} \div (5x-1) \cdot \frac{6x}{5}$$

$$= \frac{4(5x-1)}{8x^2} \cdot \frac{1}{5x-1} \cdot \frac{6x}{5}$$

$$= \frac{24x}{40x^2} = \boxed{\frac{3}{5x}}$$

Name:		Date:		
Topic:		Class:		
Main Ideas/Questions		Notes/Examples		
ADDING & SUBTRACTING Rational Expressions	①	Rewrite the problem using a common denominator.		
	②	Combine the numerators and keep the common denominator.		
	③	Simplify the remaining expression.		
Examples with LIKE BASES	1.	$\frac{13a}{4a^2} + \frac{5a}{4a^2} = \frac{18a}{4a^2}$ $= \boxed{\frac{9}{2a}}$	2.	$\frac{9x+12}{16} - \frac{7x+2}{16} = \frac{2x+10}{16}$ $= \frac{2(x+5)}{16}$ $= \boxed{\frac{x+5}{8}}$
	3.	$\frac{y^2+8y}{y^2-9} + \frac{2y+21}{y^2-9}$ $= \frac{y^2+10y+21}{y^2-9}$ $= \frac{(y+7)(y+3)}{(y+3)(y-3)}$ $= \boxed{\frac{y+7}{y-3}}$	4.	$\frac{1}{12a-3} - \frac{16a^2}{12a-3}$ $= \frac{1-16a^2}{12a-3}$ $= \frac{(1-4a)(1+4a)}{3(4a-1)}$ $= \frac{-1(1+4a)}{3} = \boxed{\frac{-1-4a}{3}}$
	5.	$\frac{n^2+3n-7}{n^2+n-6} - \frac{n+1}{n^2+n-6}$ $= \frac{n^2+2n-8}{n^2+n-6}$ $= \frac{(n+4)(n-2)}{(n+3)(n-2)}$ $= \boxed{\frac{n+4}{n+3}}$	6.	$\frac{13k^2-9k}{6k^2-5k+1} + \frac{k^2+2k}{6k^2-5k+1}$ $= \frac{14k^2-7k}{6k^2-5k+1}$ $= \frac{7k(2k-1)}{(2k-1)(3k-1)} = \boxed{\frac{7k}{3k-1}}$
Examples with UNLIKE BASES	7.	$\frac{15}{2x} - \frac{16}{3x} \cdot \frac{2}{2}$ $= \frac{45}{6x} - \frac{32}{6x} = \boxed{\frac{13}{6x}}$	8.	$\frac{4a}{a^2} - \frac{7}{8a} \cdot \frac{a}{a}$ $\frac{32a}{8a^2} - \frac{7a}{8a^2} = \frac{25a}{8a^2}$ $= \boxed{\frac{25}{8a}}$

$$9. \frac{k^2+16}{5k+10} - \frac{4}{(k-2) \cdot 5}$$

$$= \frac{k^2+16-20}{5(k-2)}$$

$$= \frac{k^2-4}{5(k-2)}$$

$$= \frac{(k+2)(k-2)}{5(k-2)} = \boxed{\frac{k+2}{5}}$$

$$10. \frac{(y+3) \cdot 7}{(y+3)(y+8)} + \frac{35}{(y+8)(y+3)}$$

$$= \frac{7y+21+35}{(y+8)(y+3)}$$

$$= \frac{7y+56}{(y+8)(y+3)}$$

$$= \frac{7(y+8)}{(y+8)(y+3)} = \boxed{\frac{7}{y+3}}$$

$$11. \frac{x^2-5}{(x+3)(x+2)} + \frac{1}{x+2} \cdot \frac{(x+3)}{(x+3)}$$

$$= \frac{x^2-5+x+3}{(x+3)(x+2)}$$

$$= \frac{x^2+x-2}{(x+3)(x+2)}$$

$$= \frac{(x+2)(x-1)}{(x+3)(x+2)} = \boxed{\frac{x-1}{x+3}}$$

$$12. \frac{m^2+5m+6}{(m+2)(m-2)} + \frac{2}{(m-2)} \cdot \frac{(m+2)}{(m+2)}$$

$$= \frac{m^2+5m+6+2m+4}{(m+2)(m-2)}$$

$$= \frac{m^2+7m+10}{(m+2)(m-2)}$$

$$= \frac{(m+5)(m+2)}{(m+2)(m-2)} = \boxed{\frac{m+5}{m-2}}$$

$$13. \frac{(w-1)5}{(w-7)(w+1)} + \frac{8}{(w-7)(w+1)}$$

$$= \frac{5w-35+8w+8}{(w-7)(w+1)}$$

$$= \boxed{\frac{13w-27}{(w-7)(w+1)}}$$

$$14. \frac{b \cdot b}{b \cdot (b+3)} - \frac{4}{b} \cdot \frac{(b+3)}{(b+3)}$$

$$= \frac{b^2-4b-12}{b(b+3)}$$

$$= \boxed{\frac{(b-6)(b+2)}{b(b+3)}}$$

$$15. \frac{(p-3) \cdot 2}{4p-20} + \frac{(p-6) \cdot 4}{p^2-8p+15}$$

$$= \frac{2p-6+4p-24}{4(p-3)(p-5)}$$

$$= \frac{6p-30}{4(p-3)(p-5)}$$

$$= \frac{6(p-5)}{4(p-3)(p-5)} = \boxed{\frac{3}{2(p-3)}}$$

$$16. \frac{r \cdot r}{2r+8} - \frac{8}{r^2+4r} \cdot 2$$

$$= \frac{r^2-16}{2r(r+4)}$$

$$= \frac{(r+4)(r-4)}{2r(r+4)}$$

$$= \boxed{\frac{r-4}{2r}}$$

Name: _____

Unit 8: Rational Functions

Date: _____ Per: _____

Homework 2: Adding & Subtracting
Rational Expressions**** This is a 2-page document! ****

Find each sum/difference. Give your answer in simplest form.

$$1. \frac{9w^3}{2w^5} + \frac{3w^3}{2w^5} = \frac{12w^3}{2w^5}$$

$$= \boxed{\frac{6}{w^2}}$$

$$2. \frac{8k+9}{18} - \frac{2k+1}{18} = \frac{6k+8}{18}$$

$$= \boxed{\frac{3k+4}{9}}$$

$$3. \frac{x^2+5x}{x+7} - \frac{14}{x+7} = \frac{x^2+5x-14}{x+7}$$

$$= \frac{(x+7)(x-2)}{x+7}$$

$$= \boxed{x-2}$$

$$4. \frac{3y^2-19y}{y^2-25} + \frac{y^2-y}{y^2-25} = \frac{4y^2-20y}{y^2-25}$$

$$= \frac{4y(y-5)}{(y-5)(y+5)}$$

$$= \boxed{\frac{4y}{y+5}}$$

$$5. \frac{8m^2}{10m^3+60m^2} - \frac{6m^2-12m}{10m^3+60m^2}$$

$$= \frac{2m^2+12m}{10m^3+60m^2}$$

$$= \frac{2m(m+6)}{10m^2(m+6)} = \boxed{\frac{1}{5m}}$$

$$6. \frac{v^2-4v}{5v^2+9v-2} - \frac{2v+16}{5v^2+9v-2}$$

$$= \frac{v^2-6v-16}{5v^2+9v-2}$$

$$= \frac{(v-8)(v+2)}{(5v-1)(v+2)} = \boxed{\frac{v-8}{5v-1}}$$

$$7. \frac{5x}{4} + \frac{x}{6} \cdot \frac{2}{2} = \frac{15x}{12} + \frac{2x}{12}$$

$$= \boxed{\frac{17x}{12}}$$

$$8. \frac{2a \cdot a}{2a \cdot 12b} - \frac{b \cdot 3b}{8a \cdot 3b}$$

$$= \boxed{\frac{2a^2-3b^2}{24ab}}$$

$$9. \frac{4 \cdot 7}{4 \cdot (n+9)} + \frac{12}{4n+36}$$

$$= \frac{28+12}{4(n+9)}$$

$$= \frac{40}{4(n+9)}$$

$$= \boxed{\frac{10}{n+9}}$$

$$10. \frac{p^2-9p}{p^2-4p-45} - \frac{2 \cdot (p-9)}{p+5 \cdot (p-9)}$$

$$= \frac{p^2-9p-2p+18}{(p+5)(p-9)}$$

$$= \frac{p^2-11p+18}{(p+5)(p-9)}$$

$$= \frac{(p-9)(p-2)}{(p+5)(p-9)} = \boxed{\frac{p-2}{p+5}}$$

$$\begin{array}{l} (k-1) \\ 11. \frac{(k+5)}{(k+1)} + \frac{8}{(k-1)(k+1)} \end{array}$$

$$= \frac{k^2 + 4k - 5 + 8}{(k+1)(k-1)}$$

$$= \frac{k^2 + 4k + 3}{(k+1)(k-1)}$$

$$= \frac{(k+3)(k+1)}{(k+1)(k-1)} = \boxed{\frac{k+3}{k-1}}$$

$$\begin{array}{l} 12. \frac{22w+11}{8w^2-6w} - \frac{3}{2w} \frac{(4w-3)}{(4w-3)} \end{array}$$

$$= \frac{22w+11-12w+9}{2w(4w-3)}$$

$$= \frac{10w+20}{2w(4w-3)}$$

$$= \frac{10(w+2)}{2w(4w-3)} = \boxed{\frac{5(w+2)}{w(4w-3)}}$$

$$\begin{array}{l} (2x+5) \\ 13. \frac{x}{2x-5} - \frac{4x+15}{4x^2-25} \end{array}$$

$$= \frac{2x^2 + 5x - 4x - 15}{(2x+5)(2x-5)}$$

$$= \frac{2x^2 + x - 15}{(2x+5)(2x-5)}$$

$$= \frac{(2x-5)(x+3)}{(2x+5)(2x-5)} = \boxed{\frac{x+3}{2x+5}}$$

$$\begin{array}{l} 14. \frac{8a-18}{3a^2+14a+8} + \frac{7}{(3a+2)} \frac{(a+4)}{(a+4)} \end{array}$$

$$= \frac{8a-18+7a+28}{(3a+2)(a+4)}$$

$$= \frac{15a+10}{(3a+2)(a+4)}$$

$$= \frac{5(3a+2)}{(3a+2)(a+4)} = \boxed{\frac{5}{a+4}}$$

$$\begin{array}{l} (4m+8) \\ 15. \frac{4}{1} - \frac{5}{4m+8} \end{array}$$

$$= \frac{16m+32-5}{4m+8} = \boxed{\frac{16m+27}{4(m+2)}}$$

$$\begin{array}{l} 16. \frac{3c}{5} - \frac{2}{5} \frac{(c-4)}{(c-4)} \end{array}$$

$$= \frac{15c-2c+8}{5(c-4)}$$

$$= \boxed{\frac{13c+8}{5(c-4)}}$$

$$\begin{array}{l} (v-6) \\ 17. \frac{6}{(v+4)} + \frac{2}{(v-6)(v+4)} \end{array}$$

$$= \frac{6v-36+2v+8}{(v-6)(v+4)}$$

$$= \frac{8v-28}{(v-6)(v+4)} = \boxed{\frac{4(2v-7)}{(v-6)(v+4)}}$$

$$\begin{array}{l} (r-4) \\ 18. \frac{r}{r^2-4} + \frac{1}{r^2-6r+8} \frac{(r+2)}{(r+2)} \end{array}$$

$$= \frac{r^2-4r+r+2}{(r-4)(r+2)(r-2)}$$

$$= \frac{r^2-3r+2}{(r-4)(r+2)(r-2)}$$

$$= \frac{(r-2)(r-1)}{(r-4)(r+2)(r-2)} = \boxed{\frac{r-1}{(r-4)(r+2)}}$$

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Algebra 2

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Unit 8: Rational Functions

Quiz 8-1: Simplifying & Operations with Rational Expressions

Simplify each expression.

$$1. \frac{20x^2 + 30x}{4x^4 + 6x^3}$$

$$= \frac{10x(2x+3)}{2x^3(2x+3)}$$

$$= \boxed{\frac{5}{x^2}}$$

$$2. \frac{k^2 + 13k + 40}{k^2 - 2k - 35}$$

$$= \frac{(k+5)(k+8)}{(k-7)(k+5)}$$

$$= \boxed{\frac{k+8}{k-7}}$$

$$3. \frac{7n - 42}{36 - n^2}$$

$$= \frac{7(n-6)}{(6-n)(6+n)}$$

$$= \frac{-1(7)}{6+n} = \boxed{\frac{-7}{6+n}}$$

$$4. \frac{6a^2 + 5a - 6}{3a^2 - 5a + 2}$$

$$= \frac{(3a-2)(2a+3)}{(3a-2)(a-1)}$$

$$= \boxed{\frac{2a+3}{a-1}}$$

$$1. \frac{5}{x^2}$$

$$2. \frac{k+8}{k-7}$$

$$3. \frac{-7}{6+n}$$

$$4. \frac{2a+3}{a-1}$$

Find each product or quotient.

$$5. \frac{5w+20}{3w+15} \cdot \frac{7w+35}{10w+40}$$

$$= \frac{5(w+4)}{3(w+5)} \cdot \frac{7(w+5)}{10(w+4)}$$

$$= \frac{35}{30} = \boxed{\frac{7}{6}}$$

$$6. \frac{x-2}{x^2+4x+4} \cdot \frac{x+2}{2x-4}$$

$$= \frac{x-2}{(x+2)(x+2)} \cdot \frac{x+2}{2(x-2)}$$

$$= \boxed{\frac{1}{2(x+2)}}$$

$$5. \frac{7}{6}$$

$$6. \frac{1}{2(x+2)}$$

$$7. \frac{3m(m+1)}{m-5}$$

$$8. \frac{x^3}{6y}$$

$$7. \frac{3m^3}{m^2-25} \cdot \frac{m^2+6m+5}{m^2}$$

$$= \frac{3m^3}{(m+5)(m-5)} \cdot \frac{(m+5)(m+1)}{m^2}$$

$$= \boxed{\frac{3m(m+1)}{m-5}}$$

$$8. \frac{7x^4}{24y^5} \div \frac{21x}{12y^4}$$

$$= \frac{7x^4}{24y^5} \cdot \frac{12y^4}{21x}$$

$$= \frac{84x^4y^4}{504xy^5} = \boxed{\frac{x^3}{6y}}$$

$$9. \frac{p^2 - 2p}{p^2 + 7p - 18} \div \frac{p^2 - 11p + 18}{p^2 - 81}$$

$$= \frac{p(p-2)}{(p+9)(p-2)} \cdot \frac{(p+9)(p-9)}{(p-9)(p-2)}$$

$$= \boxed{\frac{p}{p-2}}$$

$$10. \frac{r^2 - 3r - 10}{2r^2 - 11r + 5} \div \frac{r^2 - 5r + 6}{2r^2 - 7r + 3}$$

$$\frac{(r-5)(r+2)}{(2r-1)(r-5)} \cdot \frac{(2r-1)(r-3)}{(r-2)(r-3)}$$

$$= \boxed{\frac{r+2}{r-2}}$$

$$9. \frac{p}{p-2}$$

$$10. \frac{r+2}{r-2}$$

Find each sum or difference.

$$11. \frac{4}{15v} + \frac{2}{15v}$$

$$= \frac{6}{15v} = \boxed{\frac{2}{5v}}$$

$$12. \frac{2x^2}{x^2 - 7x + 12} - \frac{x^2 + 9}{x^2 - 7x + 12}$$

$$= \frac{x^2 - 9}{x^2 - 7x + 12}$$

$$= \frac{(x+3)(x-3)}{(x-4)(x-3)}$$

$$= \boxed{\frac{x+3}{x-4}}$$

$$11. \frac{2}{5v}$$

$$12. \frac{x+3}{x-4}$$

$$13. \frac{19x}{24}$$

$$14. \frac{n+2}{4}$$

$$15. \frac{a-5}{a-2}$$

$$16. \frac{x^2 + 4x - 3}{(x+1)(x-1)}$$

$$13. \frac{5x}{3} - \frac{7x}{8} \cdot \frac{3}{3}$$

$$= \frac{40x - 21x}{24}$$

$$= \boxed{\frac{19x}{24}}$$

$$14. \frac{n^2 + 11n + 10}{4n + 36} + \frac{2}{n+9} \cdot \frac{4}{4}$$

$$= \frac{n^2 + 11n + 18}{4(n+9)}$$

$$= \frac{(n+9)(n+2)}{4(n+9)}$$

$$= \boxed{\frac{n+2}{4}}$$

$$15. \frac{a-2}{a+2} - \frac{a+10}{a^2 - 4}$$

$$= \frac{a^2 - 2a - a - 10}{(a+2)(a-2)}$$

$$= \frac{a^2 - 3a - 10}{(a+2)(a-2)}$$

$$= \frac{(a-5)(a+2)}{(a+2)(a-2)}$$

$$= \boxed{\frac{a-5}{a-2}}$$

$$16. \frac{3}{(x+1)(x-1)} + \frac{x}{(x-1)} \cdot \frac{(x+1)}{(x+1)}$$

$$= \frac{3x - 3 + x^2 + x}{(x+1)(x-1)}$$

$$= \boxed{\frac{x^2 + 4x - 3}{(x+1)(x-1)}}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
COMPLEX FRACTIONS	- A complex fraction is a fraction in which the numerator and denominator both contain fractions. - To simplify a complex fraction means to rewrite as a single fraction with no fractions in the numerator or denominator.	
<i>Steps to</i> SIMPLIFY	①	Create a single fraction in the numerator and denominator.
	②	Set up as a division problem, then multiply by the reciprocal.
	③	Simplify! (Factor first if possible.)
EXAMPLES	1.	$\frac{\frac{a}{2}}{\frac{a^2}{4}} = \frac{a}{2} \cdot \frac{4}{a^2}$ $= \frac{4a}{2a^2}$ $= \boxed{\frac{2}{a}}$
	2.	$\frac{\frac{6}{xy}}{\frac{3y}{8x^3}} = \frac{6}{xy} \cdot \frac{8x^3}{3y}$ $= \frac{48x^3}{3xy^2}$ $= \boxed{\frac{16x^2}{y^2}}$
	3.	$\frac{\frac{18p}{8p-2}}{\frac{4p^2}{4p-1}} = \frac{18p}{2(4p-1)} \cdot \frac{4p-1}{4p^2}$ $= \boxed{\frac{9}{4p}}$
	4.	$\frac{\frac{k-1}{k}}{\frac{k^2-1}{k^2}} = \frac{k-1}{k} \cdot \frac{k^2}{(k-1)(k+1)}$ $= \boxed{\frac{k}{k+1}}$
	5.	$\frac{\frac{7c+35}{c}}{\frac{c+5}{c^2}} = \frac{7(c+5)}{c} \cdot \frac{c^2}{c+5}$ $= \boxed{7c}$
	6.	$\frac{\frac{b+2}{b^2+6b+8}}{\frac{1}{b^2-b-20}} = \frac{b+2}{(b+4)(b+2)} \cdot \frac{(b-5)(b+4)}{1}$ $= \boxed{b-5}$
	7.	$\frac{\frac{15-15r}{15r^2}}{(12r-12)} = \frac{15(1-r)}{15r^2} \cdot \frac{1}{12(r-1)}$ $= \boxed{\frac{-1}{12r^2}}$
	8.	$\frac{\frac{2m^2-2m-40}{3m+12}}{\frac{m^3-5m^2}{9m^2+36m}} = \frac{2(m-5)(m+4)}{3(m+4)} \cdot \frac{9m(m+4)}{m^2(m-5)}$ $= \frac{18m(m+4)}{3m^2} = \boxed{\frac{6(m+4)}{m}}$

$$\begin{aligned}
 \frac{x \cdot \frac{x-3}{x} \cdot 3}{9 \cdot \frac{3-x}{x-3} \cdot 3} &= \frac{x^2-9}{3x} \cdot \frac{x}{x-3} \\
 &= \frac{(x+3)(x-3)}{3x} \cdot \frac{x}{x-3} \\
 &= \boxed{\frac{x+3}{3}}
 \end{aligned}$$

$$\begin{aligned}
 10. \frac{\frac{1}{pq}}{q \cdot \frac{1}{p} - \frac{1}{q} \cdot p} &= \frac{1}{pq} \cdot \frac{qp}{q-p} \\
 &= \boxed{\frac{1}{q-p}}
 \end{aligned}$$

$$\begin{aligned}
 11. \frac{\frac{5}{4}}{4 \cdot \frac{20}{(n-4)} - \frac{5}{4n-16}} \\
 &= \frac{5}{4} \cdot \frac{4(n-4)}{15} \\
 &= \boxed{\frac{n-4}{15}}
 \end{aligned}$$

$$\begin{aligned}
 12. \frac{\frac{w}{w+2}}{w+2 \cdot 1 - \frac{w}{w+2}} \\
 &= \frac{w}{w+2} \cdot \frac{w+2}{2} \\
 &= \boxed{\frac{w}{2}}
 \end{aligned}$$

$$\begin{aligned}
 13. \frac{b^2 \cdot \frac{1}{a^2} - \frac{1}{b^2} \cdot a^2}{a \cdot \frac{1}{b} + \frac{1}{a} \cdot b} \cdot a^2 \\
 &= \frac{b^2 - a^2}{a^2 b^2} \cdot \frac{ab}{a+b} \\
 &= \frac{(b+a)(b-a)}{a^2 b^2} \cdot \frac{ab}{a+b} \\
 &= \boxed{\frac{b-a}{ab}}
 \end{aligned}$$

$$\begin{aligned}
 14. \frac{\frac{m}{n} - 1 \cdot \frac{n}{n}}{\frac{m}{n} + 1 \cdot \frac{n}{n}} \\
 &= \frac{m-n}{n} \cdot \frac{n}{m+n} \\
 &= \boxed{\frac{m-n}{m+n}}
 \end{aligned}$$

$$\begin{aligned}
 15. \frac{\frac{v}{4} \cdot \frac{1}{4} + \frac{1}{4v}}{\frac{4}{4} \cdot \frac{1}{v} + \frac{1}{4} \cdot \frac{v}{v}} \\
 &= \frac{v+1}{4v} \cdot \frac{4v}{4+v} \\
 &= \boxed{\frac{v+1}{4+v}}
 \end{aligned}$$

$$\begin{aligned}
 16. \frac{\frac{5}{(a+3)} + \frac{a^2}{5a+15}}{\frac{5}{5} \cdot \frac{a}{(a+3)} + \frac{1}{5} \cdot \frac{(a+3)}{(a+3)}} \\
 &= \frac{6a^2}{5(a+3)} \cdot \frac{5(a+3)}{6a+3} \\
 &= \frac{6a^2}{3(2a+1)} = \boxed{\frac{2a^2}{2a+1}}
 \end{aligned}$$

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Unit 8: Rational Functions

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Homework 3: Complex Fractions

**** This is a 2-page document! ******Simplify each expression.**

$$1. \frac{\frac{6a^3}{5b}}{\frac{21a}{10b^2}} = \frac{6a^3}{5b} \cdot \frac{10b^2}{21a}$$

$$= \frac{60a^3b^2}{105ab}$$

$$= \boxed{\frac{4a^2b}{7}}$$

$$2. \frac{\frac{18x^2}{x-1}}{\frac{x}{2x-2}} = \frac{18x^2}{x-1} \cdot \frac{2(x-1)}{x}$$

$$= \frac{36x^2}{x}$$

$$= \boxed{36x}$$

$$3. \frac{\frac{4c^2-1}{3c}}{\frac{2c+1}{9c^2}} = \frac{(2c+1)(2c-1)}{3c} \cdot \frac{9c^2}{2c+1}$$

$$= \boxed{3c(2c-1)}$$

$$4. \frac{\frac{n^2-9n+14}{4n+28}}{\frac{n-7}{n+7}} = \frac{(n-7)(n-2)}{4(n+7)} \cdot \frac{n+7}{n-7}$$

$$= \boxed{\frac{n-2}{4}}$$

$$5. \frac{\frac{8x^2+24x}{x^2-9}}{\frac{2x}{15-5x}} = \frac{8x(x+3)}{(x+3)(x-3)} \cdot \frac{5(3-x)}{2x}$$

$$= \frac{-40x}{2x}$$

$$= \boxed{-20}$$

$$6. \frac{\frac{2v^2+16v}{2v^2-5v-7}}{\frac{4v^2+28v-32}{10v^2-10}} = \frac{2v(v+8)}{(2v-7)(v+1)} \cdot \frac{10(v+1)(v-1)}{4(v-1)(v+8)}$$

$$= \frac{20v}{4(2v-7)} = \boxed{\frac{5}{2v-7}}$$

$$7. \frac{\frac{x^2}{4}}{\frac{x}{x} \cdot \frac{x^2-4}{4} \cdot \frac{4}{x} \cdot \frac{4}{4}} = \frac{x^2}{4} \cdot \frac{4x}{x^3-16}$$

$$= \boxed{\frac{x^3}{x^3-16}}$$

$$8. \frac{\frac{a-b}{a-b} \cdot 1 - \frac{a}{a-b}}{\frac{1}{a-b}}$$

$$= \frac{-b}{a-b} \cdot \frac{a-b}{1} = \boxed{-b}$$

$$\begin{aligned}
 9. \quad & \frac{\frac{3}{p} \cdot \frac{1}{p} + \frac{2}{3} \cdot \frac{p}{p}}{\frac{1}{p}} \\
 &= \frac{3+2p}{3p} \cdot \frac{p}{1} \\
 &= \boxed{\frac{3+2p}{3}}
 \end{aligned}$$

$$\begin{aligned}
 10. \quad & \frac{5q}{q+1} \cdot \frac{1 - \frac{q}{q+1}}{q+1} \\
 &= \frac{5q}{q+1} \cdot \frac{q+1}{1} \\
 &= \boxed{5q}
 \end{aligned}$$

$$\begin{aligned}
 11. \quad & \frac{\frac{8}{y} - 4 \cdot \frac{y}{y}}{\frac{y}{y} \cdot 12 - \frac{24}{y}} \\
 &= \frac{8-4y}{y} \cdot \frac{y}{12y-24} \\
 &= \frac{4(2-y)}{y} \cdot \frac{y}{12(y-2)} = \frac{-4}{12} = \boxed{-\frac{1}{3}}
 \end{aligned}$$

$$\begin{aligned}
 12. \quad & \frac{\frac{s}{s^2} \cdot \frac{1}{r^2} + \frac{1}{s^2} \cdot \frac{r}{r^2}}{\frac{1}{s^2} \cdot \frac{1}{r^2} - \frac{1}{s^2} \cdot \frac{r^2}{r^2}} \cdot \frac{r^2}{r^2} \\
 &= \frac{s+r}{sr} \cdot \frac{r^2 s^2}{s^2 - r^2} \\
 &= \frac{s+r}{sr} \cdot \frac{r^2 s^2}{(s+r)(s-r)} \\
 &= \boxed{\frac{rs}{s-r}}
 \end{aligned}$$

$$\begin{aligned}
 13. \quad & \frac{\frac{k^2}{4} - 9 \cdot \frac{4}{4}}{\frac{k}{6} - 1 \cdot \frac{6}{6}} = \frac{k^2-36}{4} \cdot \frac{6}{k-6} \\
 &= \frac{(k+6)(k-6)}{4} \cdot \frac{6}{k-6} \\
 &= \boxed{\frac{3(k+6)}{2}}
 \end{aligned}$$

$$\begin{aligned}
 14. \quad & \frac{\frac{m}{m} \cdot \frac{m}{4} - \frac{1}{m} \cdot \frac{4}{m}}{\frac{2}{2} \cdot \frac{1}{3m} + \frac{1}{6} \cdot \frac{m}{m}} = \frac{m^2-4}{4m} \cdot \frac{6m}{2+m} \\
 &= \frac{(m+2)(m-2)}{4m} \cdot \frac{6m}{2+m} \\
 &= \boxed{\frac{3(m-2)}{2}}
 \end{aligned}$$

$$\begin{aligned}
 15. \quad & \frac{\frac{w+4}{4} + \frac{1}{2} \cdot \frac{2}{2}}{\frac{2}{2} \cdot \frac{3}{w} + \frac{1}{2} \cdot \frac{w}{w}} = \frac{w+6}{4} \cdot \frac{2w}{6+w} \\
 &= \frac{2w}{4} \\
 &= \boxed{\frac{w}{2}}
 \end{aligned}$$

$$\begin{aligned}
 16. \quad & \frac{\frac{3}{x+3} \cdot \frac{2}{3} - \frac{1}{3} \cdot \frac{x+3}{x+3}}{\frac{x^2}{3} - 3 \cdot \frac{3}{3}} \\
 &= \frac{3-x}{3(x+3)} \cdot \frac{3}{x^2-9} \\
 &= \frac{3-x}{3(x+3)} \cdot \frac{3}{(x+3)(x-3)} = \boxed{\frac{-1}{(x+3)^2}}
 \end{aligned}$$

REVIEW: Operations with Rational Expressions

SIMPLIFYING

Write each expression in simplest form.

$$1. \frac{6x^3 + 6x^2}{2x^3 - 2x} = \frac{6x^2(x+1)}{2x(x^2-1)}$$

$$= \frac{6x^2(x+1)}{2x(x+1)(x-1)}$$

$$= \boxed{\frac{3x}{x-1}}$$

$$2. \frac{4n^2 - 12n + 9}{2n^2 + 15n - 27} = \frac{(2n-3)(2n-3)}{(2n-3)(n+9)}$$

$$= \boxed{\frac{2n-3}{n+9}}$$

MULTIPLYING & DIVIDING RATIONAL EXPRESSIONS

Find each product or quotient. Write each answer in simplest form.

$$3. \frac{42a^2 - 24a}{21a - 12} \cdot \frac{3a - 9}{6a^2 + 24a}$$

$$= \frac{6a(7a-4)}{3(7a-4)} \cdot \frac{3(a-3)}{6a(a+4)}$$

$$= \boxed{\frac{a-3}{a+4}}$$

$$4. \frac{49 - y^2}{6y^2 + 42y} \cdot \frac{48y^3}{y^2 - 10y + 21}$$

$$= \frac{(7-y)(7+y)}{6y(y+7)} \cdot \frac{48y^3}{(y-7)(y-3)}$$

$$= \frac{-48y^3}{6y(y-3)} = \boxed{\frac{-8y^2}{y-3}}$$

$$5. \frac{w^2 - 81}{4w^2 + 27w - 7} \div \frac{w^2 - 5w - 36}{w^2 + 11w + 28}$$

$$= \frac{(w+9)(w-9)}{(4w-1)(w+7)} \cdot \frac{(w+7)(w+4)}{(w-9)(w+4)}$$

$$= \boxed{\frac{w+9}{4w-1}}$$

$$6. \frac{30c^3 - 60c^2}{3c} \div (2c^3 - 16c^2 + 24c)$$

$$= \frac{30c^2(c-2)}{3c} \cdot \frac{1}{2c(c-6)(c-2)}$$

$$= \frac{30c^2}{6c^2(c-6)} = \boxed{\frac{5}{c-6}}$$

ADDING & SUBTRACTING RATIONAL EXPRESSIONS

Find each sum or difference. Write each answer in simplest form.

$$7. \frac{17k+3}{8k+4} + \frac{3k+7}{8k+4} = \frac{20k+10}{8k+4}$$

$$= \frac{10(2k+1)}{4(2k+1)}$$

$$= \boxed{\frac{5}{2}}$$

$$8. \frac{9m-12}{m^2-25} - \frac{m+28}{m^2-25} = \frac{8m-40}{m^2-25}$$

$$= \frac{8(m-5)}{(m-5)(m+5)}$$

$$= \boxed{\frac{8}{m+5}}$$

$$\frac{2}{3} \cdot \frac{3}{12y} + \frac{9}{24} \cdot \frac{y}{y}$$

$$= \frac{6 + 9y}{24y}$$

$$= \frac{3(2+3y)}{24y} = \boxed{\frac{2+3y}{8y}}$$

$$10. \frac{p^2+2p-51}{p^2-6p-27} \cdot \frac{4}{p+3} \cdot \frac{(p-9)}{(p-9)(p+3)}$$

$$= \frac{p^2+2p-51-4p+36}{(p-9)(p+3)}$$

$$= \frac{p^2-2p-15}{(p-9)(p+3)} = \frac{(p-5)(p+3)}{(p-9)(p+3)} = \boxed{\frac{p-5}{p-9}}$$

$$11. \frac{r}{(r+6)} \cdot \frac{1}{r} \cdot \frac{(r+6)}{(r+6)}$$

$$= \frac{r^2-r-6}{r(r+6)}$$

$$= \boxed{\frac{(r-3)(r+2)}{r(r+6)}}$$

$$12. \frac{(h-5)}{(h-5)} \cdot \frac{3}{(h-2)} + \frac{5}{(h-5)} \cdot \frac{(h-2)}{(h-2)}$$

$$= \frac{3h-15+5h-10}{(h-5)(h-2)}$$

$$= \boxed{\frac{8h-25}{(h-5)(h-2)}}$$

COMPLEX FRACTIONS

Simplify each expression.

$$13. \frac{\frac{15x-10}{x-8}}{\frac{3x-2}{x^2-12x+32}} = \frac{5(3x-2)}{x-8} \cdot \frac{(x-8)(x-4)}{3x-2} = \boxed{5(x-4)}$$

$$14. \frac{\frac{m}{3} - \frac{1}{3}}{\frac{1-m^2}{21}} = \frac{m-1}{3} \cdot \frac{21}{(1-m)(1+m)} = \boxed{\frac{-7}{1+m}}$$

$$15. \frac{\frac{p^2}{5} - 20}{\frac{p}{10} - 1} \cdot \frac{5}{10} = \frac{p^2-100}{5} \cdot \frac{10}{p-10} = \frac{(p+10)(p-10)}{5} \cdot \frac{10}{p-10} = \boxed{2(p+10)}$$

$$16. \frac{\frac{v}{v} \cdot \frac{4v}{v} - \frac{1}{v^2}}{\frac{2v}{2v} \cdot 1 - \frac{1}{2v}} = \frac{4v^2-1}{v^2} \cdot \frac{2v}{2v-1} = \frac{(2v-1)(2v+1)}{v^2} \cdot \frac{2v}{2v-1} = \boxed{\frac{2(2v+1)}{v}}$$

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Homework 4: Rational Expressions Review

** This is a 2-page document! **

Simplify each expression.

1. $\frac{y^2 - 14y + 48}{y^2 - 64}$

$$= \frac{(y-8)(y-6)}{(y-8)(y+8)}$$

$$= \boxed{\frac{y-6}{y+8}}$$

2. $\frac{12w^4 + 30w^3}{16w^2 + 40w}$

$$= \frac{6w^3(2w+5)}{8w(2w+5)}$$

$$= \boxed{\frac{3w^2}{4}}$$

3. $\frac{16a^2b^4}{3ab^3} \cdot \frac{6a^2b^3}{8a^5b}$

$$= \frac{96a^4b^7}{24a^6b^4}$$

$$= \boxed{\frac{4b^3}{a^2}}$$

4. $\frac{x^2 + x - 6}{x^2 - x - 12} \cdot \frac{x^2 + x - 20}{x^2 - 4x + 4}$

$$= \frac{(x+3)(x-2)}{(x-4)(x+3)} \cdot \frac{(x+5)(x-4)}{(x-2)(x-2)}$$

$$= \boxed{\frac{x+5}{x-2}}$$

5. $\frac{2m^2 + 9m - 5}{3m^2 + 13m - 10} \div \frac{14m - 7}{4 - 9m^2}$

$$= \frac{(2m-1)(m+5)}{(3m-2)(m+5)} \cdot \frac{(2-3m)(2+3m)}{7(2m-1)}$$

$$= \frac{-1(2+3m)}{7} = \boxed{\frac{-2-3m}{7}}$$

6. $\frac{4p^2 - 7p - 15}{2p^2 + 3p - 27} \div (4p + 5)$

$$= \frac{(4p+5)(p-3)}{(2p+9)(p-3)} \cdot \frac{1}{4p+5}$$

$$= \boxed{\frac{1}{2p+9}}$$

7. $\frac{2k-3}{3k+24} - \frac{k-11}{3k+24}$

$$= \frac{k+8}{3(k+8)}$$

$$= \boxed{\frac{1}{3}}$$

8. $\frac{2x-13}{x^2+9x-36} + \frac{3x-2}{x^2-9x-36}$

$$= \frac{5x-15}{x^2+9x-36}$$

$$= \frac{5(x-3)}{(x+12)(x-3)} = \boxed{\frac{5}{x+12}}$$

$$5. \frac{a}{3} - \frac{4}{5} \cdot \frac{3}{3}$$

$$= \boxed{\frac{5a-12}{15}}$$

$$\frac{(m-5)}{10} \cdot \frac{2}{3m+1} + \frac{19m+17}{3m^2-14m-5}$$

$$(m-5) \cdot \frac{2}{(3m+1)(m-5)}$$

$$= \frac{2m-10 + 19m+17}{(3m+1)(m-5)}$$

$$= \frac{21m+7}{(3m+1)(m-5)} = \frac{7(3m+1)}{(3m+1)(m-5)} = \boxed{\frac{7}{m-5}}$$

$$11. \frac{x}{x-2} + \frac{5}{x} \cdot \frac{x-2}{x-2}$$

$$= \boxed{\frac{x^2+5x-10}{x(x-2)}}$$

$$\frac{(n+1)}{12} \cdot \frac{3}{(n-6)} - \frac{3}{(n+1)} \cdot \frac{(n-6)}{(n-6)}$$

$$= \frac{3n+3-3n+18}{(n+1)(n-6)}$$

$$= \boxed{\frac{21}{(n+1)(n-6)}}$$

$$13. \frac{\frac{w^2-1}{8w-8}}{\frac{w^2+6w+5}{18w}} = \frac{(w+1)(w-1)}{8(w-1)} \cdot \frac{18w}{(w+5)(w+1)}$$

$$= \boxed{\frac{9w}{4(w+5)}}$$

$$\frac{r+2}{r+2} \cdot \frac{1-\frac{r}{r+2}}{\frac{r^2}{r+2}} = \frac{2}{r+2} \cdot \frac{r+2}{r^2}$$

$$= \boxed{\frac{2}{r^2}}$$

$$15. \frac{\frac{12-6}{c} \cdot \frac{c}{c}}{\frac{c}{c} \cdot 8 - \frac{16}{c}} = \frac{12-6c}{c} \cdot \frac{c}{8c-16}$$

$$= \frac{6(2-c)}{8(c-2)}$$

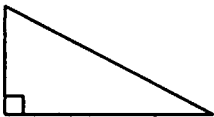
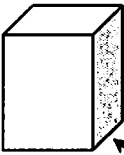
$$= \boxed{-\frac{3}{4}}$$

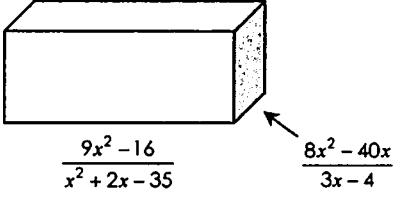
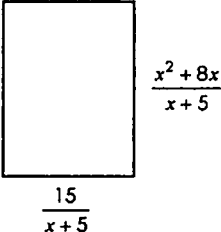
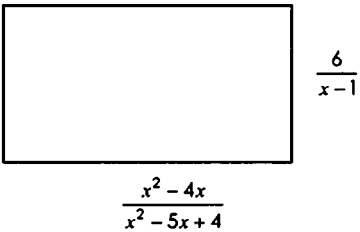
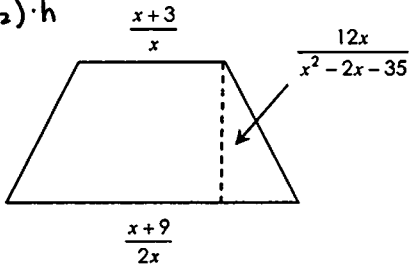
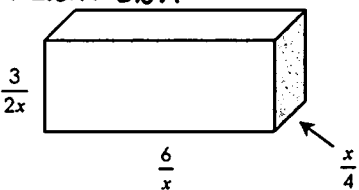
$$16. \frac{\frac{k^2-9k}{2} + 10}{\frac{2}{2} \cdot \frac{k}{8} - \frac{k+5}{16}} \cdot \frac{2}{2}$$

$$= \frac{k^2-9k+20}{2} \cdot \frac{16}{k-5}$$

$$= \frac{(k-4)(k-5)}{2} \cdot \frac{16}{k-5} = 8(k-4)$$

RATIONAL EXPRESSIONS: Applications

Example	Work and Solution
1. Write an expression in simplest form to represent the area of the rectangle. $\frac{2x}{x^2 - 12x + 27}$  $\frac{x^2 - 81}{4x^3 + 36x^2}$	$A = \ell \cdot w$ $\frac{2x}{(x-3)(x-9)} \cdot \frac{(x+9)(x-9)}{4x^2(x+9)}$ $= \frac{2x}{4x^2(x-3)} = \boxed{\frac{1}{2x(x-3)}}$
2. Write an expression in simplest form to represent the area of the rectangle.  $\frac{x^2 + 3x - 10}{7x - 14}$ $\frac{3x^2 - 11x - 20}{25 - x^2}$	$A = \ell \cdot w$ $\frac{(x+5)(x-2)}{7(x-2)} \cdot \frac{(3x+4)(x-5)}{(5-x)(5+x)}$ $= \frac{-1(3x+4)}{7}$ $= \boxed{\frac{-3x-4}{7}}$
3. If the area of a rectangular garden is $\frac{a^2}{b} - b$ and the length is $\frac{a}{b} + 1$, write an expression to represent the width of the garden in simplest form. $A = \ell \cdot w$ $w = \frac{A}{\ell}$	$\frac{\frac{a^2}{b} - b}{\frac{a}{b} + 1} = \frac{\frac{a^2 - b^2}{b}}{\frac{a+b}{b}} = \frac{a^2 - b^2}{b} \div \frac{a+b}{b}$ $= \frac{(a+b)(a-b)}{b} \cdot \frac{b}{a+b}$ $= \boxed{a-b}$
4. Write an expression to represent the area of the triangle below in simplest form. $\frac{9x+18}{6x^3-24x}$  $\frac{x^3 + 11x^2 - 26x}{3x}$	$A = \frac{1}{2} \ell \cdot w$ $\frac{1}{2} \cdot \frac{9(x+2)}{6x(x^2-4)} \cdot \frac{x(x^2+11x-26)}{3x}$ $= \frac{9x(x+2)(x+13)(x-2)}{36x^2(x+2)(x-2)}$ $= \boxed{\frac{x+13}{4x}}$
5. Write an expression to represent the volume of the rectangular prism shown below in simplest form. $\frac{5x^2 - 32x + 12}{10x^3}$  $\frac{x^2 + 14x + 48}{36 - x^2}$ $\frac{18x^2}{45x^2 - 18x}$	$V = \ell \cdot w \cdot h$ $\frac{(5x-2)(x-6)}{10x^3} \cdot \frac{18x^2}{9x(5x-2)} \cdot \frac{(x+6)(x+8)}{(6+x)(6-x)}$ $= \frac{-18x^2(x+8)}{90x^4} = \boxed{\frac{-x-8}{5x^2}}$

6. The volume of a rectangular prism below is $12x + 16$. Find the height of the prism. $V = l \cdot w \cdot h$ $h = \frac{V}{l \cdot w}$ 	$\frac{12x+16}{(3x+4)(3x-4)} \cdot \frac{8x(x-5)}{3x-4}$ $= \frac{12x+16}{8x(3x+4)} = 4(3x+4) \cdot \frac{x+7}{8x(3x+4)}$ $= \frac{x+7}{2x}$
7. Write an expression to represent the perimeter of the rectangle in simplest form. $P = 2l + 2w$ 	$2\left(\frac{x^2+8x}{x+5}\right) + 2\left(\frac{15}{x+5}\right)$ $= \frac{2x^2+16x+30}{x+5}$ $= \frac{2(x^2+8x+15)}{x+5} = \frac{2(x+3)(x+5)}{x+5} = 2x+6$
8. Write an expression to represent the perimeter of the rectangle in simplest form. $P = 2(l+w)$ 	$2\left(\frac{6}{x-1} \cdot \frac{x+4}{x-4}\right) + 2\left(\frac{x^2-4x}{(x-4)(x-1)}\right)$ $= 2\left(\frac{6x-24+x^2-4x}{(x-1)(x-4)}\right)$ $= 2\left(\frac{x^2+2x-24}{(x-1)(x-4)}\right)$ $= 2\left(\frac{(x+6)(x-4)}{(x-1)(x-4)}\right) = \frac{2(x+6)}{x-1} = \frac{2x+12}{x-1}$
9. Write an expression to represent the area of the trapezoid below in simplest form. $A = \frac{1}{2}(b_1+b_2) \cdot h$ 	$\frac{1}{2}\left(\frac{x+3}{x} + \frac{x+9}{2x}\right) \cdot \frac{12x}{x^2-2x-35}$ $= \frac{1}{2}\left(\frac{3x+15}{2x}\right) \left(\frac{12x}{(x-7)(x+5)}\right)$ $= \frac{12x \cdot 3(x+5)}{4x \cdot (x-7)(x+5)} = \frac{9}{x-7}$
10. Write an expression to represent the surface area of the rectangular prism below. $SA = 2lw + 2lh + 2wh$ 	$2\left(\frac{3}{2x} \cdot \frac{6}{x}\right) + 2\left(\frac{3}{2x} \cdot \frac{x}{4}\right) + 2\left(\frac{6}{x} \cdot \frac{x}{4}\right)$ $= 2\left(\frac{18}{2x^2}\right) + 2\left(\frac{3x}{8x}\right) + 2\left(\frac{6x}{4x}\right)$ $= \frac{18}{x^2} + \frac{3}{4} + \frac{3}{1} = \frac{18}{x^2} + \frac{3}{4} + \frac{3}{1} = \frac{72}{4x^2} + \frac{3x^2}{4x^2} + \frac{12x^2}{4x^2}$ $= \frac{15x^2+72}{4x^2}$

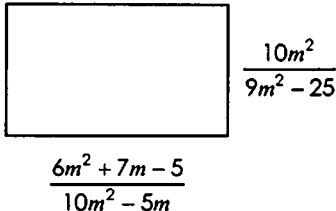
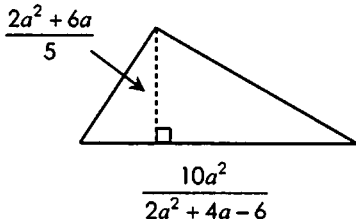
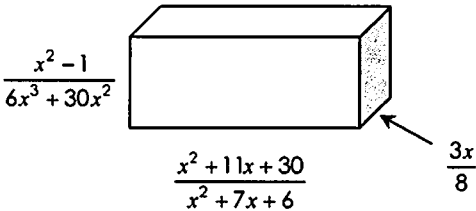
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Unit 8: Rational Functions

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Homework 5: Rational Expressions Applications

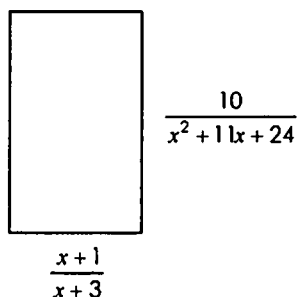
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Example	Work and Solution
1. Find an expression to represent the area of the rectangle below. 	$\frac{(2m-1)(3m+5)}{5m(2m-1)} \cdot \frac{10m^2}{(3m+5)(3m-5)}$ $= \frac{10m^2}{5m(3m-5)}$ $= \boxed{\frac{2m}{3m-5}}$
2. The area of a rectangle is $x - \frac{16}{x}$. If the width of rectangle is $\frac{x}{4} + 1$, find an expression to represent the length.	$\frac{\frac{x}{x} \cdot x - \frac{16}{x}}{\frac{x}{4} + 1} = \frac{\frac{x^2-16}{x}}{\frac{x}{4} + 1} \div \frac{x+4}{4}$ $= \frac{(x+4)(x-4)}{x} \cdot \frac{4}{x+4}$ $= \boxed{\frac{4(x-4)}{x}}$
3. Find an expression to represent the area of the triangle below. 	$\frac{1}{2} \cdot \frac{2a(a+3)}{5} \cdot \frac{10a^2}{2(a+3)(a-1)}$ $= \frac{20a^3}{20(a-1)}$ $= \boxed{\frac{a^3}{a-1}}$
4. Find an expression to represent the volume of the rectangular prism below. 	$\frac{(x+1)(x-1)}{6x^2(x+5)} \cdot \frac{(x+5)(x+6)}{(x+6)(x+1)} \cdot \frac{3x}{8}$ $= \frac{3x(x-1)}{48x^2}$ $= \boxed{\frac{x-1}{16x}}$

5. The **volume** of a rectangular prism is $\frac{7}{x} - \frac{3}{5x}$. If it has a width of $\frac{2}{x}$ and a height of $\frac{4}{x}$, find the length of the prism.

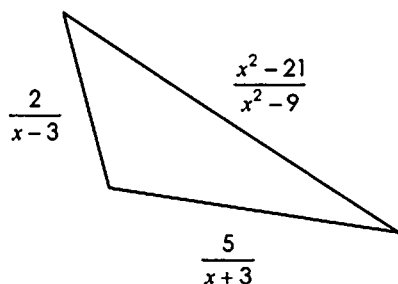
$$\begin{aligned} \frac{\frac{7}{x} - \frac{3}{5x}}{\frac{2}{x} \cdot \frac{4}{x}} &= \frac{\frac{32}{5x}}{\frac{8}{x^2}} \\ &= \frac{32}{5x} \cdot \frac{x^2}{8} \\ &= \frac{32x^2}{40x} = \boxed{\frac{4x}{5}} \end{aligned}$$

6. Find an expression to represent the **perimeter** of the rectangle below.



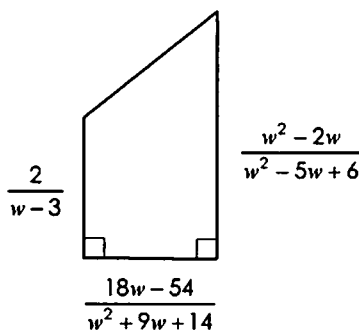
$$\begin{aligned} &2 \left(\frac{10}{(x+8)(x+3)} + \frac{(x+1)}{(x+3)} \cdot \frac{(x+8)}{(x+8)} \right) \\ &= 2 \left(\frac{10 + x^2 + 9x + 8}{(x+8)(x+3)} \right) \\ &= \frac{2(x+6)(x+3)}{(x+8)(x+3)} = \boxed{\frac{2(x+6)}{x+8}} \end{aligned}$$

7. Find an expression to represent the **perimeter** of the triangle below.



$$\begin{aligned} &\frac{(x+3)}{(x+3)} \cdot \frac{2}{(x-3)} + \frac{x^2-21}{x^2-9} + \frac{5}{(x+3)} \cdot \frac{(x-3)}{(x-3)} \\ &= \frac{2x+6 + x^2-21 + 5x-15}{(x+3)(x-3)} \\ &= \frac{x^2+7x-30}{(x+3)(x-3)} = \frac{(x+10)(x-3)}{(x+3)(x-3)} = \boxed{\frac{x+10}{x+3}} \end{aligned}$$

8. Find an expression to represent the **area** of the trapezoid below.



$$\begin{aligned} &\frac{1}{2} \left(\frac{2}{w-3} \cdot \frac{(w-2)}{(w-2)} + \frac{w^2-2w}{(w-2)(w-3)} \right) \cdot \frac{18w-54}{w^2+9w+14} \\ &= \frac{1}{2} \cdot \frac{w^2-4}{(w-3)(w-2)} \cdot \frac{18w-54}{w^2+9w+14} \\ &= \frac{1}{2} \cdot \frac{(w+2)(w-2)}{(w-3)(w-2)} \cdot \frac{18(w-3)}{(w+2)(w+7)} \\ &= \boxed{\frac{9}{w+7}} \end{aligned}$$

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Unit 8: Rational Functions

Quiz 8-2: Rational Expressions & Applications

Simplify each expression.

1. $\frac{28m^2 - 42m}{12m^3 - 18m^2}$

$$\frac{14m(2m-3)}{6m^2(2m-3)} = \boxed{\frac{7}{3m}}$$

2. $\frac{16k - 24}{6k^2 - 13k + 6}$

$$\frac{8(2k-3)}{(2k-3)(3k-2)} = \boxed{\frac{8}{3k-2}}$$

1. $\frac{7}{3m}$
2. $\frac{8}{3k-2}$

Find each product or quotient.

3. $\frac{15cd^3}{2d^2} \cdot \frac{4c^3d}{45c^4d}$

$$\frac{60c^4d^4}{90c^4d^3} = \boxed{\frac{2d}{3}}$$

4. $\frac{x^2 + 2x - 3}{2x^2 - x - 21} \cdot \frac{4x^2 - 49}{x - 1}$

$$\frac{(x+3)(x-1)}{(2x-7)(x+3)} \cdot \frac{(2x+7)(2x-7)}{x-1} = \boxed{2x+7}$$

3. $\frac{2d}{3}$
4. $2x+7$
5. $\frac{-w+7}{3w}$
6. $\frac{2}{5a-b}$

5. $\frac{5w^2 + 20w}{16 - w^2} \div \frac{15w^2}{w^2 - 11w + 28}$

$$\frac{5w(w+4)}{(4+w)(4-w)} \cdot \frac{(w-7)(w-4)}{15w^2} = \frac{-1(w-7)}{3w} = \boxed{\frac{-w+7}{3w}}$$

6. $\frac{8a^2 - 16a - 24}{5a^2 - a - 6} \div (4a - 12)$

$$\frac{8(a-3)(a+1)}{(5a-6)(a+1)} \cdot \frac{1}{4(a-3)} = \boxed{\frac{2}{5a-6}}$$

Find each sum or difference.

7. $\frac{6p}{p-1} - \frac{p+5}{p-1}$

$$\frac{5p-5}{p-1} = \frac{5(p-1)}{p-1} = \boxed{5}$$

8. $\frac{4}{4} \cdot \frac{9x}{10} + \frac{7x}{8} \cdot \frac{5}{5}$

$$\frac{36x + 35x}{40} = \boxed{\frac{71x}{40}}$$

7. 5
8. $\frac{71x}{40}$

$$9. \frac{y^2 + y - 14}{y^2 - 49} + \frac{2}{(y+7)} \cdot \frac{(y-7)}{(y-7)} \quad \frac{(n+3)}{(n+3)} \cdot \frac{n}{(n-4)} - \frac{n}{(n+3)} \cdot \frac{(n-4)}{(n-4)}$$

$$\frac{y^2 + 3y - 28}{(y+7)(y-7)} = \frac{(y+7)(y-4)}{(y+7)(y-7)} \quad \frac{n^2 + 3n - n^2 + 4n}{(n+3)(n-4)}$$

$$= \boxed{\frac{y-4}{y-7}} \quad = \boxed{\frac{7n}{(n+3)(n-4)}}$$

$$9. \frac{y-4}{y-7}$$

$$10. \frac{7n}{(n+3)(n-4)}$$

Simplify each complex fraction.

$$11. \frac{\frac{x-2}{x-2} \cdot 2 + \frac{x-8}{x-2}}{\frac{x}{4} - 1} \cdot \frac{4}{4}$$

$$\frac{2x-4+x-8}{x-2} \div \frac{x-4}{4}$$

$$= \frac{3x-12}{x-2} \cdot \frac{4}{x-4} = \frac{3(x-4)}{x-2} \cdot \frac{4}{x-4} = \frac{(r-4)(r+2)}{4r} \cdot \frac{r}{r-4}$$

$$= \boxed{\frac{12}{x-2}} \quad = \boxed{\frac{r+2}{4}}$$

$$11. \frac{12}{x-2}$$

$$12. \frac{r+2}{4}$$

13. Find the **area** of the rectangle below.

$$\frac{10x^2}{x^2 - 64} \cdot \frac{(x-8)(x-2)}{2x(x-2)}$$

$$= \frac{10x^2}{2x(x+8)} = \boxed{\frac{5x}{x+8}}$$

$$13. \frac{5x}{x+8}$$

$$14. \frac{v}{12}$$

$$15. \frac{2c-1}{c-5}$$

14. If the rectangular prism below has a **volume** of $\frac{2v^2 - 3v - 5}{8v - 20}$, find its **length**, l .

$$\frac{v+1}{2} \cdot l \cdot \frac{6}{v} = \frac{2v^2 - 3v - 5}{8v - 20}$$

$$= \frac{(2v-5)(v+1)}{4(2v-5)} \cdot \frac{2v}{6(v+1)}$$

$$= \frac{2v}{24} = \boxed{\frac{v}{12}}$$

15. Find the **perimeter** of the triangle below.

$$\frac{27c-9}{c^2 + 4c - 45} \cdot \frac{c}{c+9} \cdot \frac{c-5}{c-5} \cdot 2 \left(\frac{c}{c+9} \right) + \frac{27c-9}{(c+9)(c-5)}$$

$$= \frac{2c^2 - 10c + 27c - 9}{(c+9)(c-5)} = \frac{2c^2 + 17c - 9}{(c+9)(c-5)}$$

$$= \frac{(2c-1)(c+9)}{(c+9)(c-5)} = \boxed{\frac{2c-1}{c-5}}$$

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Main Ideas/Questions

Notes/Examples

RATIONAL

Function

A rational function is a function of the form

$$f(x) = \frac{p(x)}{q(x)}$$

where $p(x)$ and $q(x)$ are polynomials & $q(x) \neq 0$.

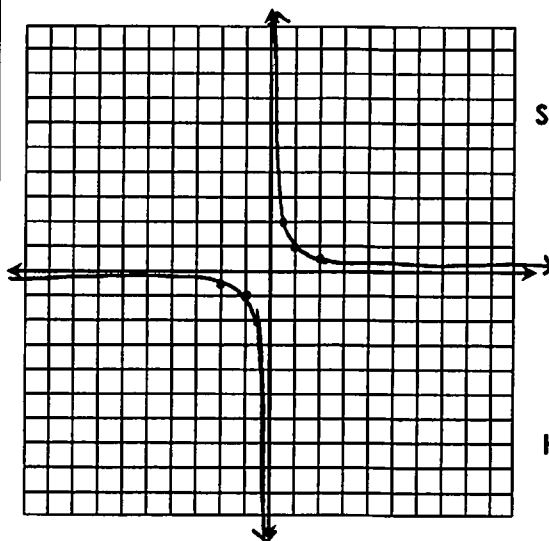
RECIPROCAL

Function

$$f(x) = \frac{1}{x}$$

One of the most basic rational functions is $f(x) = \frac{1}{x}$,
which is also known as a **reciprocal function**.

Graph this function below and identify the key characteristics.

Shape of Graph: hyperbolaDomain: $\{x | x \neq 0\}$ Range: $\{y | y \neq 0\}$ Vertical Asymptote: $x = 0$ Horizontal Asymptote: $y = 0$

TRANSFORMING

Reciprocal
Functions

$$f(x) = \frac{a}{x-h} + k$$

TRANSFORMATIONS:

- h is the horizontal shift. (+ shifts left, - shifts right)
- k is the vertical shift. (+ shifts up, - shifts down)
- If a is negative, the function is reflected across the x -axis.
- $|a| > 1$ represents a vertical stretch.
- $0 < |a| < 1$ represents a vertical compression.

VERTICAL ASYMPTOTE:

$$x = h$$

HORIZONTAL ASYMPTOTE:

$$y = k$$

DESCRIBING <i>Transformations</i>	For each function, identify the transformations from $f(x) = \frac{1}{x}$, the vertical asymptote, and the horizontal asymptote.		
	1. $f(x) = \frac{1}{x} + 4$	Transformations: up 4	
		VA: $x = 0$	HA: $y = 4$
	2. $f(x) = \frac{-2}{x+1}$	Transformations: Reflected across the x-axis, Vertical stretch by 2, Translated left 1	
		VA: $x = -1$	HA: $y = 0$
	3. $f(x) = \frac{1}{x-5} - 2$	Transformations: Translated right 5 and down 2	
		VA: $x = 5$	HA: $y = -2$
	4. $f(x) = \frac{4}{x+3} - 2$	Transformations: Vertical stretch by 4, Translated left 3 and down 2	
		VA: $x = -3$	HA: $y = -2$
WRITING <i>Functions</i>	5. The reciprocal parent function is reflected across the x-axis, then translated 4 units right and 3 units down. Write an equation to represent the new function and identify the asymptotes. $f(x) = \frac{-1}{x-4} - 3$ VA: $x = 4$ HA: $y = -3$		
	6. The reciprocal parent function is vertically stretched by a factor of 2, then translated 7 units up and 1 unit left. Write an equation to represent the new function and identify the asymptotes. $f(x) = \frac{2}{x+1} + 7$ VA: $x = -1$ HA: $y = 7$		
	7. A reciprocal function has a vertical asymptote located at $x = -2$ and a horizontal asymptote located at $y = 5$. Write an equation that could represent this function. $f(x) = \frac{1}{x+2} + 5$		

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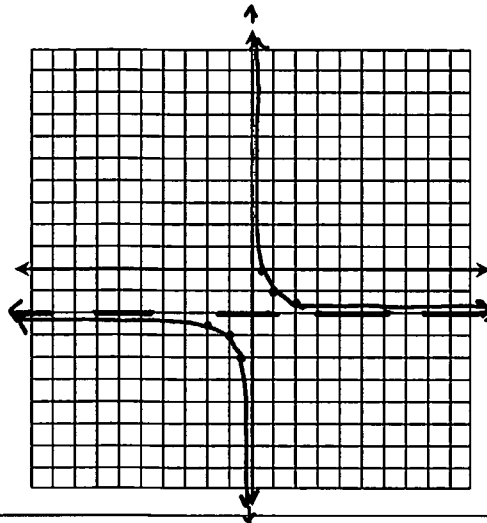
Main Ideas/Questions

Notes/Examples

Graphing RECIPROCAL FUNCTIONS

Graph each function. Identify the domain, range, and asymptotes.

1. $f(x) = \frac{1}{x} - 2$



Domain:

$\{x \mid x \neq 0\}$

Range:

$\{y \mid y \neq -2\}$

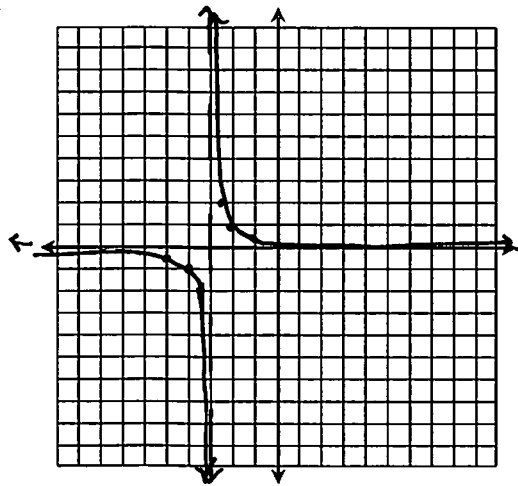
Vertical Asymptote:

$x = 0$

Horizontal Asymptote:

$y = -2$

2. $f(x) = \frac{1}{x+3}$



Domain:

$\{x \mid x \neq -3\}$

Range:

$\{y \mid y \neq 0\}$

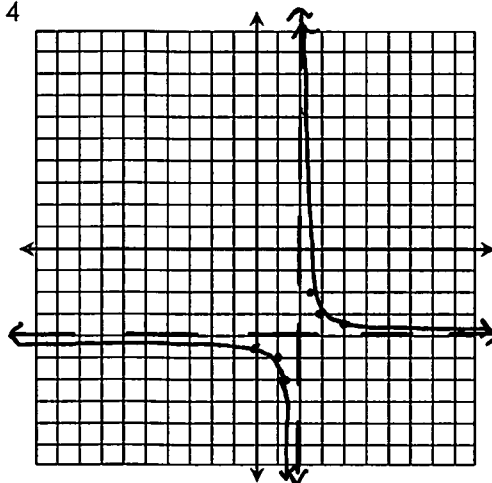
Vertical Asymptote:

$x = -3$

Horizontal Asymptote:

$y = 0$

3. $f(x) = \frac{1}{x-2} - 4$



Domain:

$\{x \mid x \neq 2\}$

Range:

$\{y \mid y \neq -4\}$

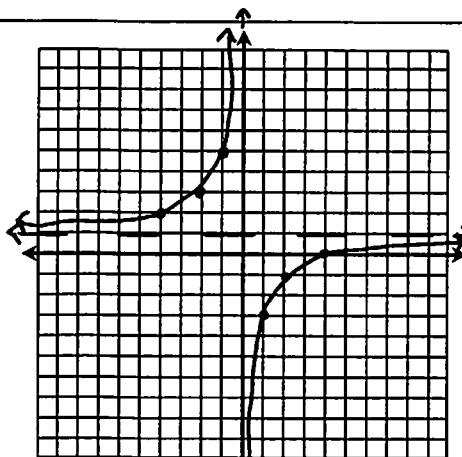
Vertical Asymptote:

$x = 2$

Horizontal Asymptote:

$y = -4$

4. $f(x) = \frac{-4}{x} + 1$



Domain:

$$\{x | x \neq 0\}$$

Range:

$$\{y | y \neq 1\}$$

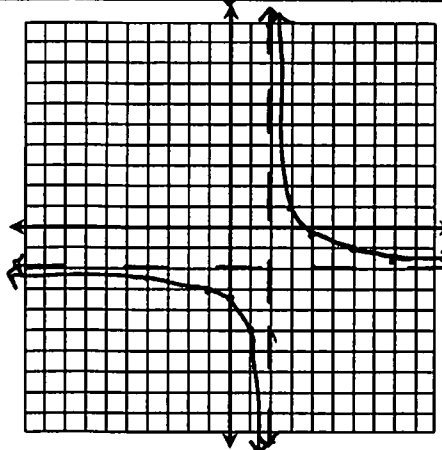
Vertical Asymptote:

$$x = 0$$

Horizontal Asymptote:

$$y = 1$$

5. $f(x) = \frac{3}{x-2} - 2$



Domain:

$$\{x | x \neq 2\}$$

Range:

$$\{y | y \neq -2\}$$

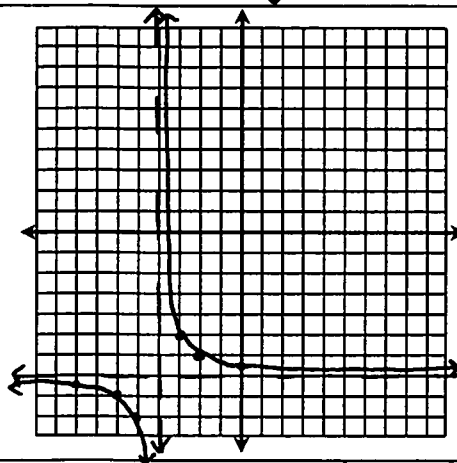
Vertical Asymptote:

$$x = 2$$

Horizontal Asymptote:

$$y = -2$$

6. $f(x) = \frac{2}{x+4} - 7$



Domain:

$$\{x | x \neq -4\}$$

Range:

$$\{y | y \neq -7\}$$

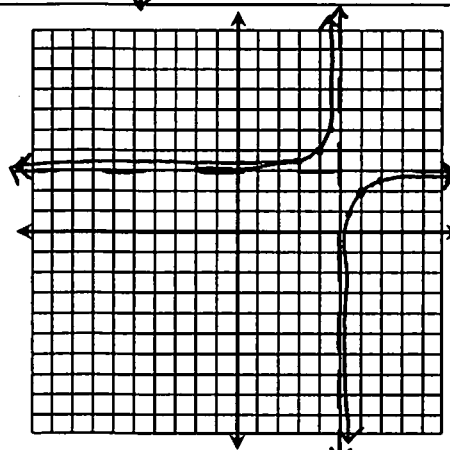
Vertical Asymptote:

$$x = -4$$

Horizontal Asymptote:

$$y = -7$$

7. $f(x) = \frac{-1}{x-5} + 3$



Domain:

$$\{x | x \neq 5\}$$

Range:

$$\{y | y \neq 3\}$$

Vertical Asymptote:

$$x = 5$$

Horizontal Asymptote:

$$y = 3$$

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Unit 8: Rational Functions

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Homework 6: Graphing Reciprocal Functions

**** This is a 2-page document! ****

Transformations from the function $f(x) = \frac{1}{x}$ are described below. Write an equation to represent the new function, then identify the vertical asymptote (VA) and horizontal asymptote (HA).

Description of Transformations	New Function	VA	HA
1. Translated 2 units left and 9 units down.	$f(x) = \frac{1}{x+2} - 9$	$x = -2$	$y = -9$
2. Reflected across the x -axis, then translated 5 units up.	$f(x) = -\frac{1}{x} + 5$	$x = 0$	$y = 5$
3. Vertically stretched by a factor of 4, then translated 3 units right.	$f(x) = \frac{4}{x-3}$	$x = 3$	$y = 0$
4. Vertically stretched by a factor of 2, reflected across the x -axis, then translated 1 unit left and 8 units up.	$f(x) = \frac{-2}{x+1} + 8$	$x = -1$	$y = 8$

The function $f(x) = \frac{1}{x}$ is transformed so that it has the given asymptotes below. Write an equation that could represent the new function.

5. Asymptotes: $x = 3$ and $y = -2$

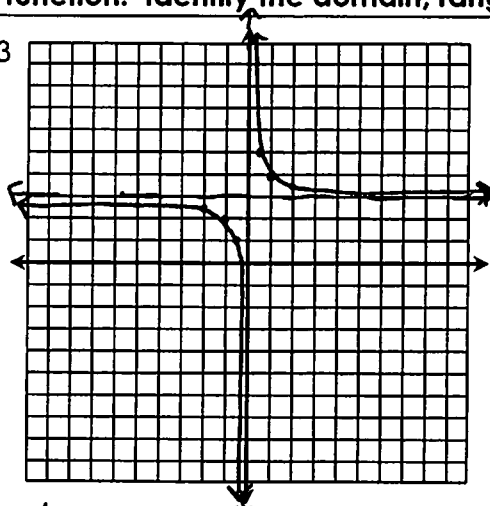
$$f(x) = \frac{1}{x-3} - 2$$

6. Asymptotes: $x = -7$ and $y = 0$

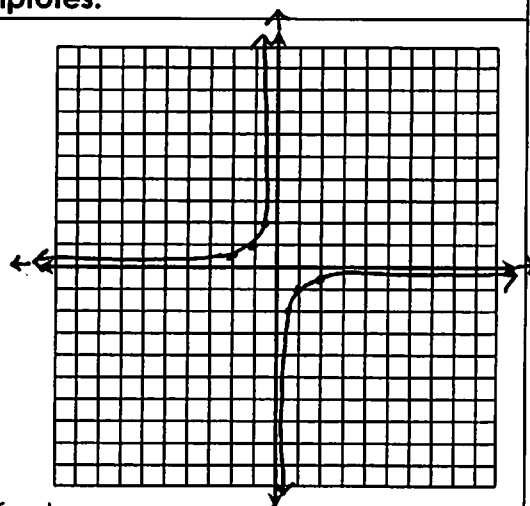
$$f(x) = \frac{1}{x+7}$$

Graph each function. Identify the domain, range, and asymptotes.

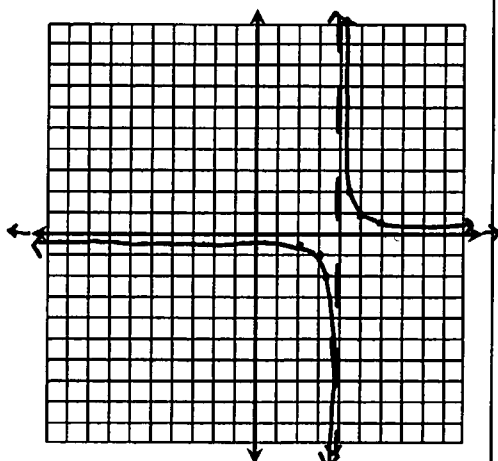
7. $f(x) = \frac{1}{x} + 3$

Domain: $\{x | x \neq 0\}$ VA: $x = 0$ Range: $\{y | y \neq 3\}$ HA: $y = 3$

8. $f(x) = \frac{-1}{x}$

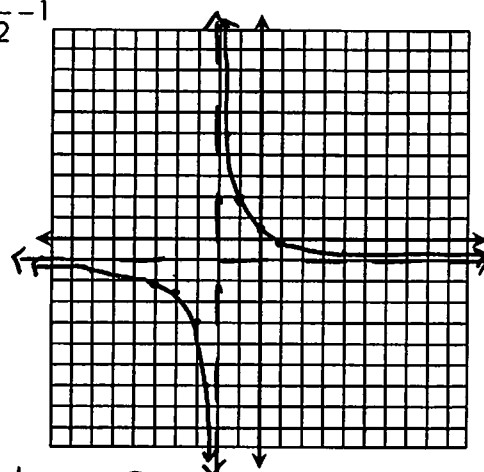
Domain: $\{x | x \neq 0\}$ VA: $x = 0$ Range: $\{y | y \neq 0\}$ HA: $y = 0$

$$9. f(x) = \frac{1}{x-4}$$



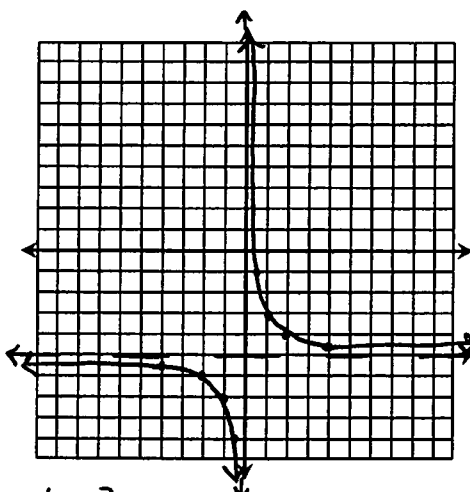
Domain: $\{x | x \neq 4\}$ VA: $x=4$
 Range: $\{y | y \neq 0\}$ HA: $y=0$

$$10. f(x) = \frac{3}{x+2} - 1$$



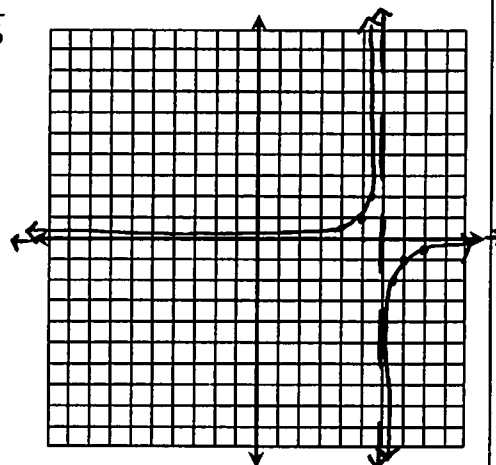
Domain: $\{x | x \neq -2\}$ VA: $x=-2$
 Range: $\{y | y \neq -1\}$ HA: $y=-1$

$$11. f(x) = \frac{2}{x} - 5$$



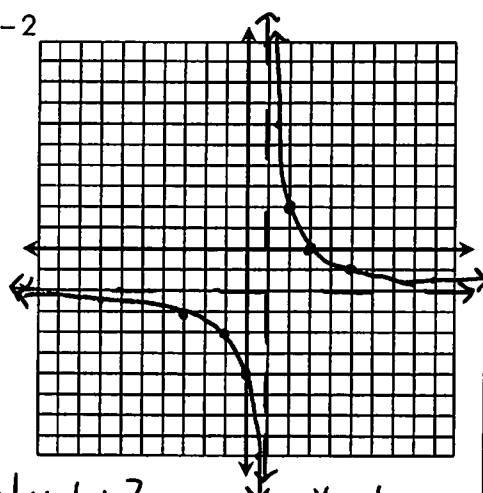
Domain: $\{x | x \neq 0\}$ VA: $x=0$
 Range: $\{y | y \neq -5\}$ HA: $y=-5$

$$12. f(x) = \frac{-1}{x-6}$$



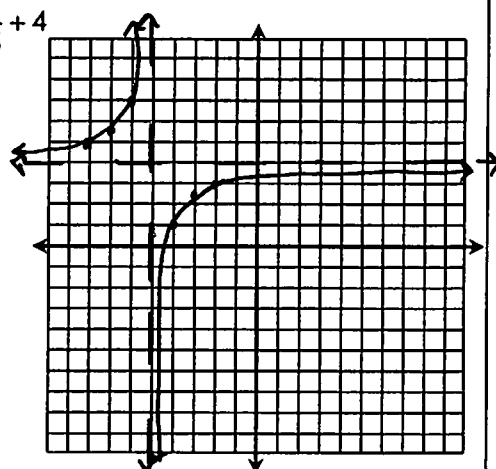
Domain: $\{x | x \neq 6\}$ VA: $x=6$
 Range: $\{y | y \neq 0\}$ HA: $y=0$

$$13. f(x) = \frac{4}{x-1} - 2$$



Domain: $\{x | x \neq 1\}$ VA: $x=1$
 Range: $\{y | y \neq -2\}$ HA: $y=-2$

$$14. f(x) = \frac{-3}{x+5} + 4$$



Domain: $\{x | x \neq -5\}$ VA: $x=-5$
 Range: $\{y | y \neq 4\}$ HA: $y=4$

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Main Ideas/Questions

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KEY FEATURES

of Rational Functions

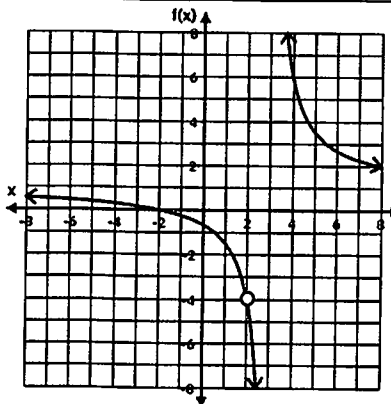
A **rational function** is a function of the form $f(x) = \frac{p(x)}{q(x)}$,

where $p(x)$ and $q(x)$ are polynomial functions. So far, we have looked at the graph of a very basic rational function (a reciprocal function).

Now, we will look at more complex rational functions and their key features: x -intercepts, vertical asymptotes, horizontal asymptotes, and holes. A hole is a point (x, y) at which there is a break in the graph.

Identify the key features of the rational function given each graph below.

1. $f(x) = \frac{x^2 - 4}{x^2 - 5x + 6}$



x-Intercept(s):

$(-2, 0)$

Vertical Asymptote(s):

$x = 3$

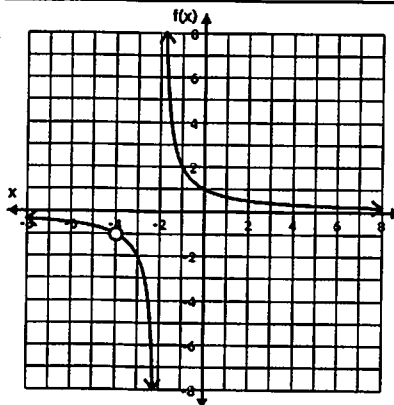
Horizontal Asymptote:

$y = 1$

Hole(s):

$(2, -4)$

2. $f(x) = \frac{2x + 8}{x^2 + 6x + 8}$



x-Intercept(s):

None

Vertical Asymptote(s):

$x = -2$

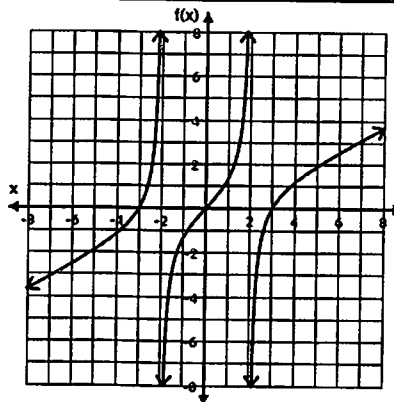
Horizontal Asymptote:

$y = 0$

Hole(s):

$(-4, -1)$

3. $f(x) = \frac{x^3 - 9x}{2x^2 - 8}$



x-Intercept(s):

$(-3, 0), (0, 0), (3, 0)$

Vertical Asymptote(s):

$x = -2, x = 2$

Horizontal Asymptote:

None

Hole(s):

None

Finding Features ALGEBRAICALLY (x-intercepts, vertical asymptotes, & holes)	Follow to find the x-intercepts, vertical asymptotes and holes of:		
	$f(x) = \frac{x^2 - 2x - 3}{x^2 + 3x + 2}$		
	Write the function in <u>factored form</u> . Then, simplify by crossing out common factors in the numerator and denominator.	$f(x) = \frac{(x-3)(x+1)}{(x+2)(x+1)} = \frac{x-3}{x+2}$	
	To find the x-intercept(s): Set the <u>numerator</u> of the simplified function equal to 0 and solve for x.	$x-3 = 0$ $x = 3 \rightarrow (3, 0)$	
	To find the vertical asymptote(s): Set the <u>denominator</u> of the simplified function equal to 0 and solve for x.	$x+2 = 0$ $x = -2$	
To find the hole(s): If there was a <u>common factor</u> , set it equal to 0 and solve for x. Substitute this x-value into the simplified function to find y.	$x+1 = 0$ $x = -1 \rightarrow \frac{-1-3}{-1+3} = \frac{-4}{2} = -2$ $(-1, -2)$		
EXAMPLES	Find the x-intercepts, vertical asymptote(s), and hole(s) of each function.		
	4. $f(x) = \frac{3x^2 - 12}{x^2 + 2x - 8}$ $f(x) = \frac{3(x+2)(x-2)}{(x+4)(x-2)}$ $f(x) = \frac{3(x+2)}{x+4}$	$3(x+2) = 0$ $x+2 = 0$ $x = -2$ $x+4 = 0$ $x = -4$ $x = -2 \rightarrow \frac{3(2+2)}{2+4} = 2$	x-Intercept(s): (-2, 0) Vertical Asymptote(s): $x = -4$ Hole(s): (2, 2)
	5. $f(x) = \frac{x+3}{x^2 + 8x + 15}$ $f(x) = \frac{x+3}{(x+3)(x+5)}$ $f(x) = \frac{1}{x+5}$	$x+5 = 0$ $x = -5$ $x = -3 \rightarrow \frac{1}{-3+5} = \frac{1}{2}$	x-Intercept(s): None Vertical Asymptote(s): $x = -5$ Hole(s): $(-3, \frac{1}{2})$
	6. $f(x) = \frac{6x^3 - 6x}{x^2 - 2x}$ $f(x) = \frac{6x(x+1)(x-1)}{x(x-2)}$ $f(x) = \frac{6(x+1)(x-1)}{x-2}$	$x+1 = 0 \quad x-1 = 0$ $x = -1 \quad x = 1$ $x-2 = 0$ $x = 2$ $x = 0 \rightarrow \frac{6(0+1)(0-1)}{0-2} = 3$	x-Intercept(s): (-1, 0), (1, 0) Vertical Asymptote(s): $x = 2$ Hole(s): (0, 3)

	$7. f(x) = \frac{x-6}{x^2+3x}$ $f(x) = \frac{x-6}{x(x+3)}$	$x-6=0$ $x=6$ $x=0;$ $x+3=0$ $x=-3$	x-Intercept(s): $(6, 0)$
			Vertical Asymptote(s): $x = -3, x = 0$
			Hole(s): None
	$8. f(x) = \frac{x^2+6x+5}{x^2+4x+3}$ $f(x) = \frac{(x+5)(x+1)}{(x+3)(x+1)}$ $f(x) = \frac{x+5}{x+3}$	$x+5=0$ $x=-5$ $x+3=0$ $x=-3$ $x=-1 \rightarrow \frac{-1+5}{-1+3} = 2$	x-Intercept(s): $(-5, 0)$
			Vertical Asymptote(s): $x = -3$
			Hole(s): $(-1, 2)$

HORIZONTAL ASYMPTOTES

To find the horizontal asymptote of a rational function, we look at the original function $f(x) = \frac{p(x)}{q(x)}$ and follow the rules below.

CASE	HORIZONTAL ASYMPTOTE
degree of $p >$ degree of q	No horizontal asymptote
degree of $p <$ degree of q	x -axis / $y = 0$
degree of $p =$ degree of q	$y = \frac{\text{leading coefficient of } p(x)}{\text{leading coefficient of } q(x)}$

Given the horizontal asymptote of each function below.

$9. f(x) = \frac{x^2-9}{x^2+2x-15}$ $y = 1$	$10. f(x) = \frac{x^2+4x}{2x-4}$ None
$11. f(x) = \frac{3x+1}{x^2-2x+1}$ $y = 0$	$12. f(x) = \frac{8x+6}{4x-2}$ $y = 2$
$13. f(x) = \frac{x^3-x^2+2x}{4x^2+4x}$ None	$14. f(x) = \frac{8}{4x-6}$ $y = 0$
$15. f(x) = \frac{-2x^2+8x}{6x^2-3x}$ $y = -\frac{1}{3}$	$16. f(x) = \frac{3x^2-6x}{x+1}$ None

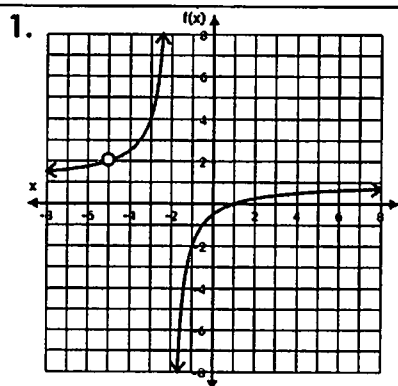
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Unit 8: Rational Functions

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Homework 7: Key Features of Rational Functions

Identify the x-intercept(s), vertical asymptote(s), horizontal asymptote, and hole(s) given the graph.

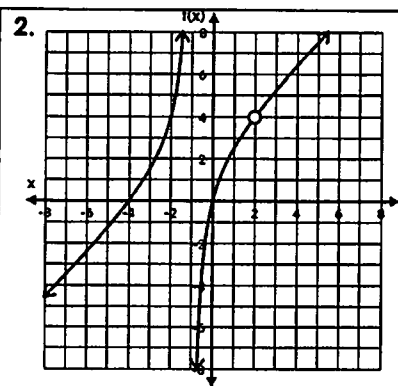


x-intercept(s):
(1, 0)

Vertical Asymptote(s):
 $x = -2$

Horizontal Asymptote:
 $y = 1$

Hole(s):
(-5, 2)



x-intercept(s):
(-4, 0), (0, 0)

Vertical Asymptote(s):
 $x = -1$

Horizontal Asymptote:
None

Hole(s):
(2, 4)

Find the x-intercept(s), vertical asymptote(s), horizontal asymptote, and hole(s) of each function.

3. $f(x) = \frac{-6x+12}{3x-12}$

$f(x) = \frac{-6(x-2)}{3(x-4)}$

x-intercept(s):
(2, 0)

Vertical Asymptote(s):
 $x = 4$

Horizontal Asymptote:
 $y = -2$

Hole(s):
None

4. $f(x) = \frac{2x-4}{x^2+4x-12}$

$f(x) = \frac{2(x-2)}{(x+6)(x-2)}$

$f(x) = \frac{2}{x+6}$

x-intercept(s):
None

Vertical Asymptote(s):
 $x = -6$

Horizontal Asymptote:
 $y = 0$

Hole(s):
(2, 1/4)

5. $f(x) = \frac{x^3+3x^2-4x}{x^2-2x-24}$

$f(x) = \frac{x(x+4)(x-1)}{(x-6)(x+4)}$

$f(x) = \frac{x(x-1)}{x-6}$

x-intercept(s):
(0, 0), (1, 0)

Vertical Asymptote(s):
 $x = 6$

Horizontal Asymptote:
None

Hole(s):
(-4, -2)

6. $f(x) = \frac{x^2-9}{x^2+5x-14}$

$f(x) = \frac{(x+3)(x-3)}{(x+7)(x-2)}$

x-intercept(s):
(-3, 0), (3, 0)

Vertical Asymptote(s):
 $x = -7, x = 2$

Horizontal Asymptote:
 $y = 1$

Hole(s):
None

7. $f(x) = \frac{2x^3+5x^2+2x}{x^2+5x+6}$

$f(x) = \frac{x(2x+1)(x+2)}{(x+3)(x+2)}$

$f(x) = \frac{x(2x+1)}{x+3}$

x-intercept(s):
(0, 0), $(-\frac{1}{2}, 0)$

Vertical Asymptote(s):
 $x = -3$

Horizontal Asymptote:
None

Hole(s):
(-2, 6)

8. $f(x) = \frac{3x^2-6x-9}{x^3-2x^2-3x}$

$f(x) = \frac{3(x-3)(x+1)}{x(x-3)(x+1)}$

$f(x) = \frac{3}{x}$

x-intercept(s):
None

Vertical Asymptote(s):
 $x = 0$

Horizontal Asymptote:
 $y = 0$

Hole(s):
(3, 1), (-1, -3)

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

GRAPHING*Rational Functions*

Use the key features of a rational function to aid in graphing:

- ① Write the function in **FACTORED FORM**, then **SIMPLIFY**.
- ② Find the **x-intercept(s)** by setting the **numerator** equal to 0.
- ③ Find the **vertical asymptote(s)** by setting the **denominator** equal to 0.
- ④ Find the **horizontal asymptote** using the rules below.

CASE

HORIZONTAL ASYMPTOTE

degree of $p >$ degree of q

No horiz. asymptote

degree of $p <$ degree of q x-axis / $y = 0$ degree of $p =$ degree of q $y = \frac{\text{leading coeff. of } p(x)}{\text{leading coeff. of } q(x)}$

- ⑤ Identify any **holes** in the function using common factors.

A hole is a point (x, y) at which there is a **break** in the graph. A hole occurs when there is a **common factor** between the numerator and denominator.

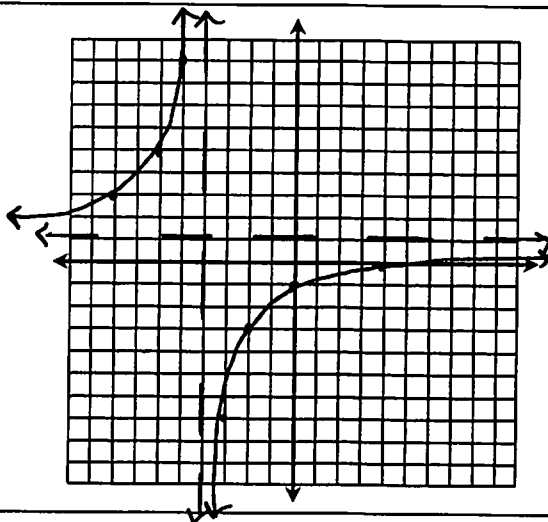
- » **To find the x-coordinate:** Set the common factor equal to 0.
- » **To find the y-coordinate:** Substitute the x-coordinate into the simplified function.

Directions: Graph each function and identify its key characteristics.

1. $f(x) = \frac{x-4}{x+4}$

$$\begin{aligned} \text{X-int: } x-4 &= 0 \\ x &= 4 \end{aligned}$$

$$\begin{aligned} \text{VA: } x+4 &= 0 \\ x &= -4 \end{aligned}$$



x-int: $(4, 0)$

VA: $x = -4$

HA: $y = 1$

Hole: None

Domain: $\{x \mid x \neq -4\}$

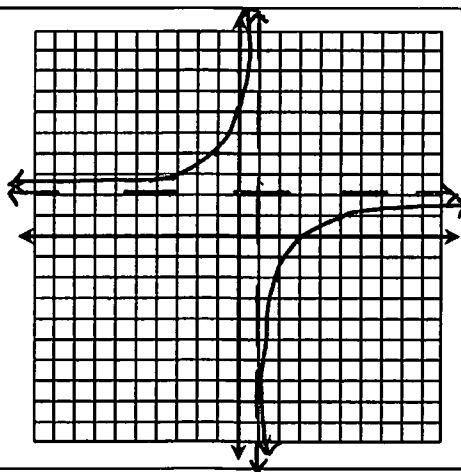
Range: $\{y \mid y \neq 1\}$

$$2. f(x) = \frac{2x-6}{x-1}$$

$$f(x) = \frac{2(x-3)}{x-1}$$

$$x\text{-int: } x-3=0 \\ x=3$$

$$VA: x-1=0 \\ x=1$$



$$x\text{-int: } (3,0)$$

$$VA: x=1$$

$$HA: y=2$$

$$\text{Hole: None}$$

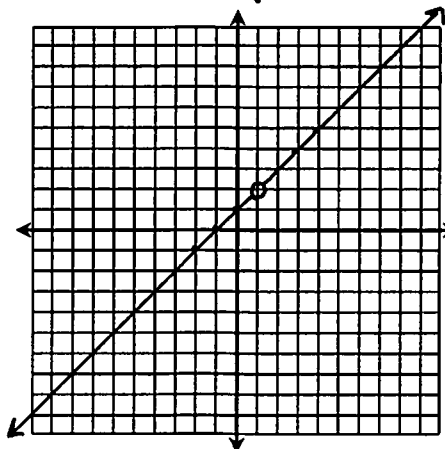
$$\text{Domain: } \{x | x \neq 1\}$$

$$\text{Range: } \{y | y \neq 2\}$$

$$3. f(x) = \frac{x^2-1}{x-1}$$

$$f(x) = \frac{(x+1)(x-1)}{x-1}$$

$$f(x) = x+1$$



$$x\text{-int: } (-1,0)$$

$$VA: \text{None}$$

$$HA: \text{None}$$

$$\text{Hole: } (1,2)$$

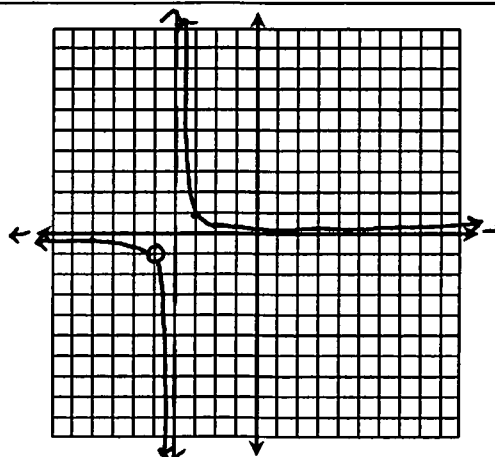
$$\text{Domain: } \{x | x \neq 1\}$$

$$\text{Range: } \{y | y \neq 2\}$$

$$4. f(x) = \frac{x+5}{x^2+9x+20}$$

$$f(x) = \frac{x+5}{(x+4)(x+5)}$$

$$f(x) = \frac{1}{x+4}$$



$$x\text{-int: None}$$

$$VA: x=-4$$

$$HA: y=0$$

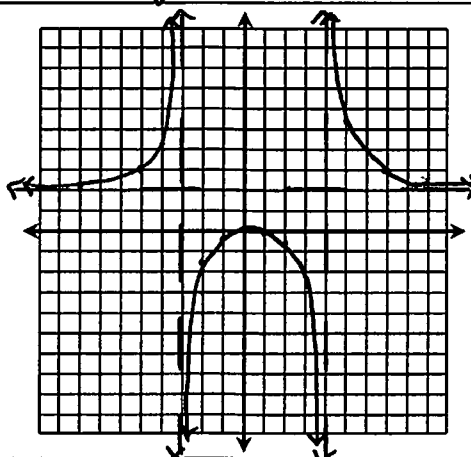
$$\text{Hole: } (-5,-1)$$

$$\text{Domain: } \{x | x \neq -5, -4\}$$

$$\text{Range: } \{y | y \neq -1\}$$

$$5. f(x) = \frac{2x^2-x-1}{x^2-x-12}$$

$$f(x) = \frac{(2x+1)(x-1)}{(x-4)(x+3)}$$



$$x\text{-int: } (-\frac{1}{2}, 0) (1, 0)$$

$$VA: x=-3, x=4$$

$$HA: y=2$$

$$\text{Hole: None}$$

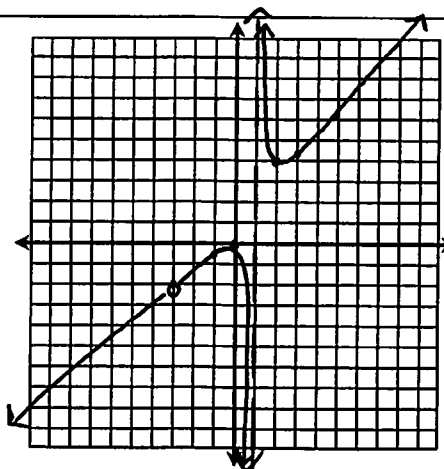
$$\text{Domain: } \{x | x \neq -3, 4\}$$

$$\text{Range: } \{y | y \neq 2\}$$

$$6. f(x) = \frac{x^3 + 3x^2}{x^2 + 2x - 3}$$

$$f(x) = \frac{x^2(x+3)}{(x+3)(x-1)}$$

$$f(x) = \frac{x^2}{x-1}$$



$$x\text{-int: } (0,0)$$

$$VA: x=1$$

$$HA: \text{None}$$

$$\text{Hole: } (-3, -9/4)$$

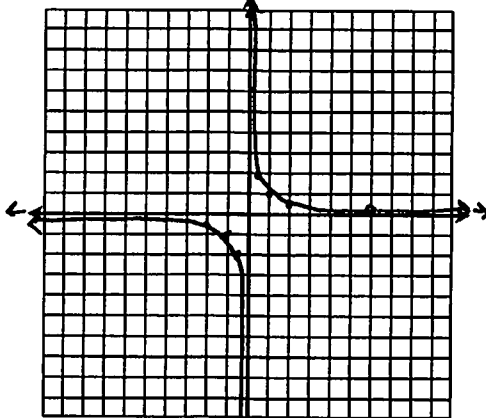
$$\text{Domain: } \{x | x \neq -3, 1\}$$

$$\text{Range: } \{y | y \neq -9/4\}$$

$$7. f(x) = \frac{x-6}{x^2-6x}$$

$$f(x) = \frac{x-6}{x(x-6)}$$

$$f(x) = \frac{1}{x}$$



$$x\text{-int: } \text{None}$$

$$VA: x=0$$

$$HA: y=0$$

$$\text{Hole: } (6, 1/6)$$

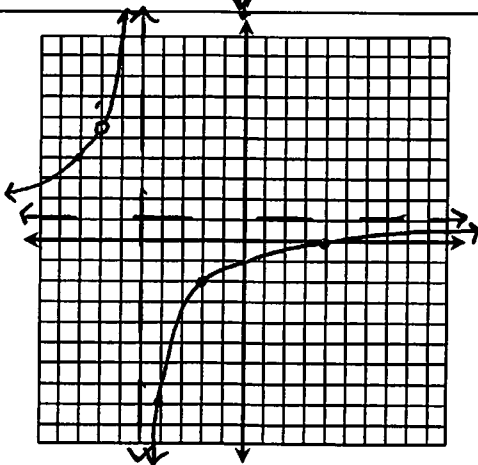
$$\text{Domain: } \{x | x \neq 0, 6\}$$

$$\text{Range: } \{y | y \neq 0, 1/6\}$$

$$8. f(x) = \frac{x^2 + 3x - 28}{x^2 + 12x + 35}$$

$$f(x) = \frac{(x+7)(x-4)}{(x+7)(x+5)}$$

$$f(x) = \frac{x-4}{x+5}$$



$$x\text{-int: } (4,0)$$

$$VA: x=-5$$

$$HA: y=1$$

$$\text{Hole: } (-7, 1/2)$$

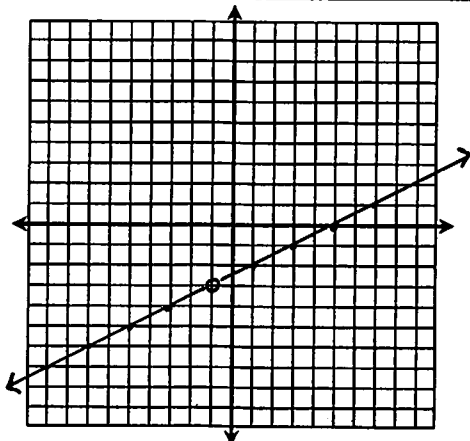
$$\text{Domain: } \{x | x \neq -7, -5\}$$

$$\text{Range: } \{y | y \neq 1, 1/2\}$$

$$9. f(x) = \frac{x^2 - 4x - 5}{2x + 2}$$

$$f(x) = \frac{(x-5)(x+1)}{2(x+1)}$$

$$f(x) = \frac{x-5}{2}$$



$$x\text{-int: } (5,0)$$

$$VA: \text{None}$$

$$HA: \text{None}$$

$$\text{Hole: } (-1, -3)$$

$$\text{Domain: } \{x | x \neq -1\}$$

$$\text{Range: } \{y | y \neq -3/2\}$$

Name: _____

Unit 8: Rational Functions

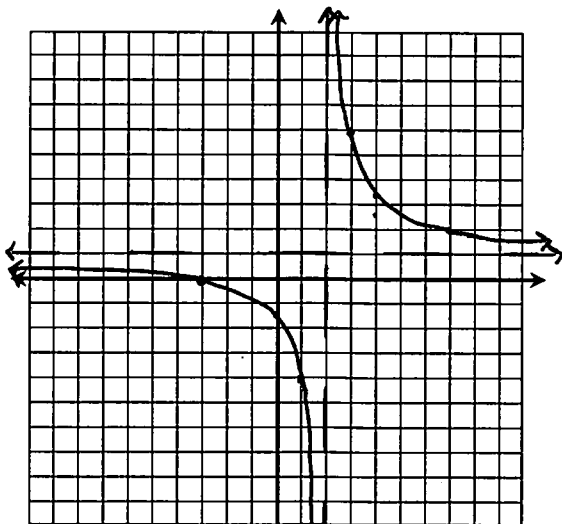
Date: _____ Per: _____

Homework 8: Graphing Rational Functions

** This is a 2-page document! **

Graph each function. Identify the domain, range, asymptotes, and holes.

1. $f(x) = \frac{x+3}{x-2}$



x-int: $(-3, 0)$

VA: $x=2$

HA: $y=1$

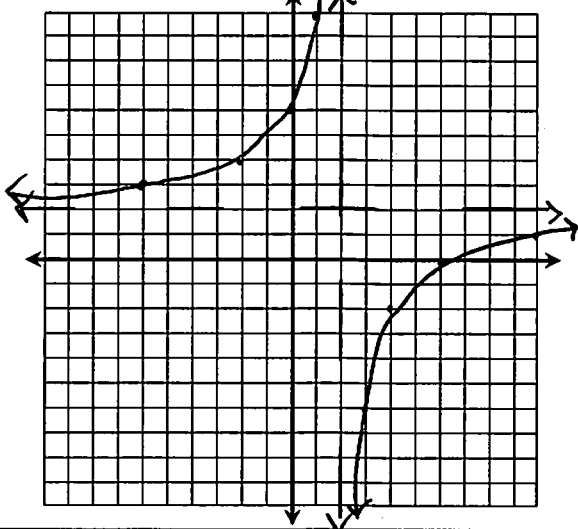
Hole: None

Domain: $\{x | x \neq 2\}$

Range: $\{y | y \neq 1\}$

2. $f(x) = \frac{4x-24}{2x-4}$

$$f(x) = \frac{4(x-6)}{2(x-2)}$$



x-int: $(6, 0)$

VA: $x=2$

HA: $y=2$

Hole: None

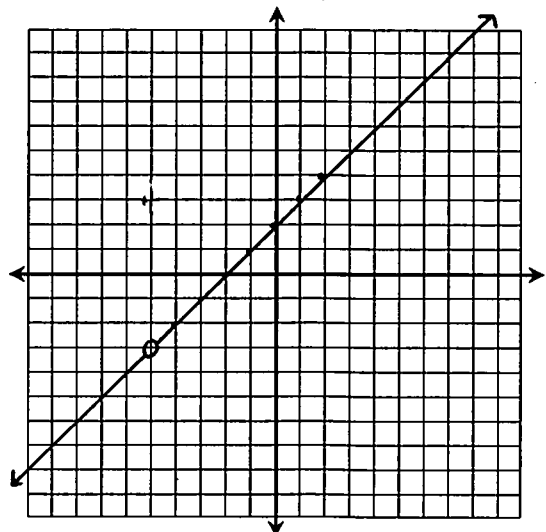
Domain: $\{x | x \neq 2\}$

Range: $\{y | y \neq 2\}$

3. $f(x) = \frac{x^2+7x+10}{x+5}$

$$f(x) = \frac{(x+5)(x+2)}{x+5}$$

$$f(x) = x+2$$



x-int: $(-2, 0)$

VA: None

HA: None

Hole: $(-5, -3)$

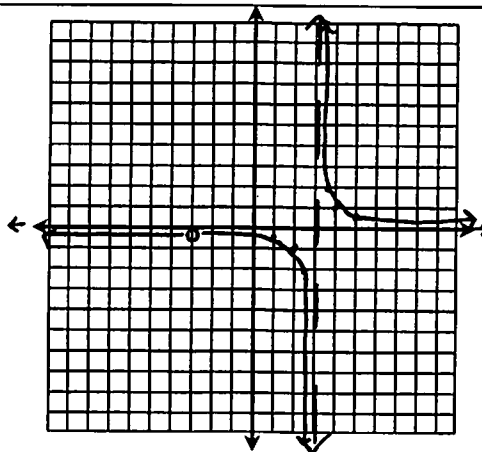
Domain: $\{x | x \neq -5\}$

Range: $\{y | y \neq -3\}$

4. $f(x) = \frac{x+3}{x^2-9}$

$f(x) = \frac{x+3}{(x+3)(x-3)}$

$f(x) = \frac{1}{x-3}$



x-int: None

VA: $x=3$

HA: $y=0$

Hole: $(-3, -1/6)$

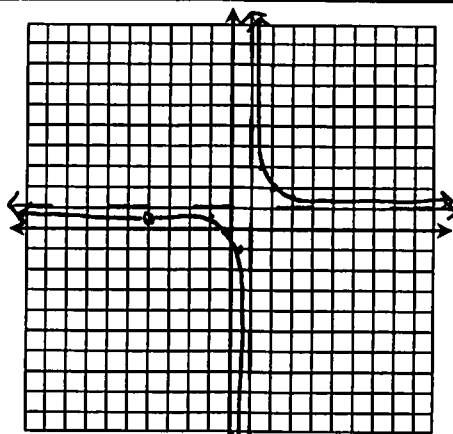
Domain: $\{x | x \neq -3, 3\}$

Range:

5. $f(x) = \frac{x^2+4x}{x^2+3x-4}$

$f(x) = \frac{x(x+4)}{(x+4)(x-1)}$

$f(x) = \frac{x}{x-1}$



x-int: $(0, 0)$

VA: $x=1$

HA: $y=1$

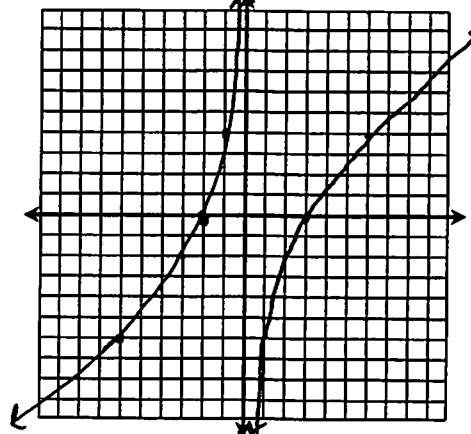
Hole: $(-4, 4/5)$

Domain: $\{x | x \neq -4, 1\}$

Range: $\{y | y \neq 4/5, 1\}$

6. $f(x) = \frac{x^2-x-6}{x}$

$f(x) = \frac{(x-3)(x+2)}{x}$



x-int: $(3, 0), (-2, 0)$

VA: $x=0$

HA: None

Hole: None

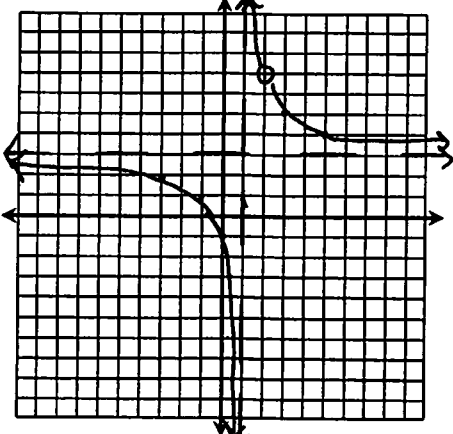
Domain: $\{x | x \neq 0\}$

Range: $\mathbb{R}$

7. $f(x) = \frac{3x^2-5x-2}{x^2-3x+2}$

$f(x) = \frac{(3x+1)(x-2)}{(x-2)(x-1)}$

$f(x) = \frac{3x+1}{x-1}$



x-int: $(-1/3, 0)$

VA: $x=1$

HA: $y=3$

Hole: $(2, 7)$

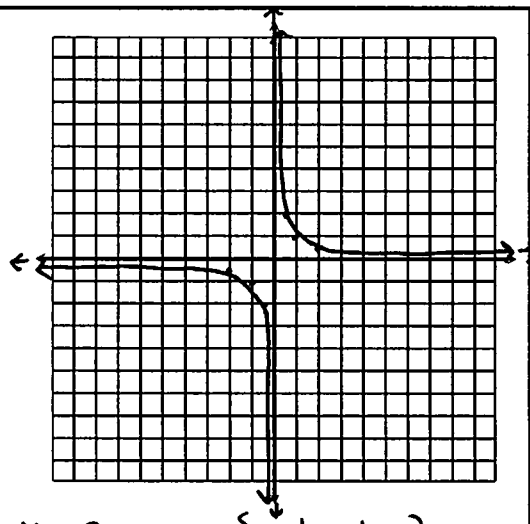
Domain: $\{x | x \neq -1/3, 2\}$

Range: $\{y | y \neq 3, 7\}$

RECIPROCAL Functions

PARENT FUNCTION:

$$f(x) = \frac{1}{x}$$



Vertical Asymptote: $x=0$ D: $\{x | x \neq 0\}$

Horizontal Asymptote: $y=0$ R: $\{y | y \neq 0\}$

..... Transformations

STANDARD FORM:

$$f(x) = \frac{a}{x-h} + k$$

- h is the horizontal shift. (+ shifts left, - shifts right)
- k is the vertical shift. (+ shifts up, - shifts down)
- If a is negative, the function is reflected across the x -axis.
- $|a| > 1$ represents a vertical stretch.
- $0 < |a| < 1$ represents a vertical compression.

VERTICAL ASYMPTOTE:

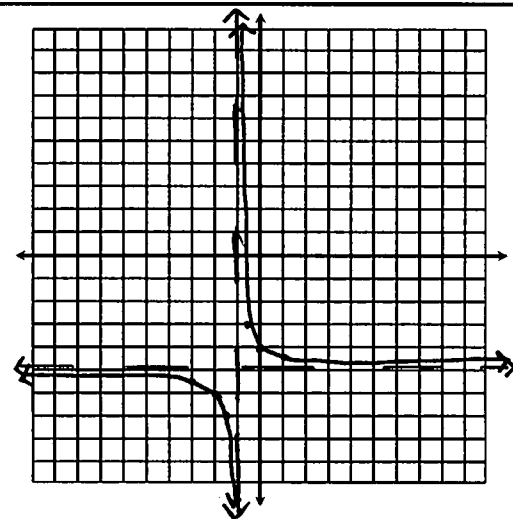
$$x=h$$

HORIZONTAL ASYMPTOTE:

$$y=k$$

EXAMPLES

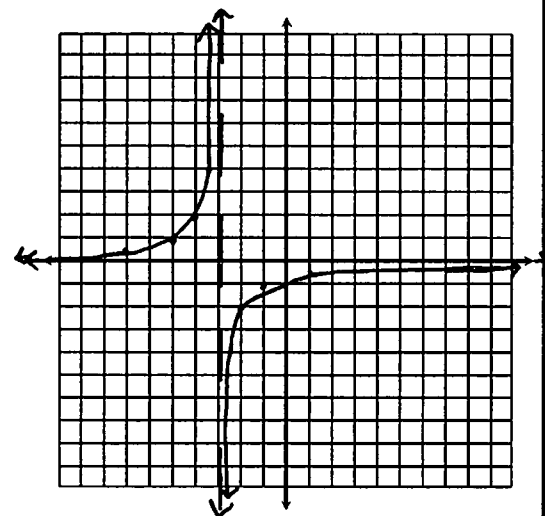
① $f(x) = \frac{1}{x+1} - 5$



Vertical Asymptote: $x=-1$ D: $\{x | x \neq -1\}$

Horizontal Asymptote: $y=-5$ R: $\{y | y \neq -5\}$

② $f(x) = \frac{-2}{x+3}$



Vertical Asymptote: $x=-3$ D: $\{x | x \neq -3\}$

Horizontal Asymptote: $y=0$ R: $\{y | y \neq 0\}$

RATIONAL Functions

EQUATION FORM:

$$f(x) = \frac{p(x)}{q(x)}$$

X-INTERCEPTS:

Set the numerator $[p(x)]$
equal to zero, then solve.

VERTICAL ASYMPTOTES:

Set the denominator $[q(x)]$
equal to zero, then solve.

HORIZONTAL ASYMPTOTES:

- If degree of $p >$ degree of q : No HA
- If degree of $p <$ degree of q : $y=0$
- If degree of $p =$ degree of q : $y = \frac{\text{lead coeff } p(x)}{\text{lead coeff } q(x)}$

HOLES: (break in graph)

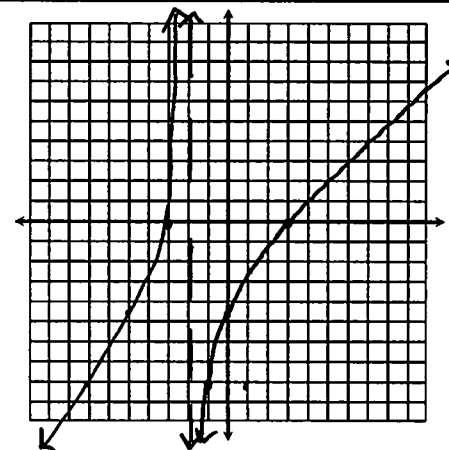
X-Coord: set common factor equal to zero, then solve.

Y-Coord: substitute X-Coord into simplified function.

EXAMPLES

① $f(x) = \frac{x^2 - 9}{x + 2}$

$$f(x) = \frac{(x+3)(x-3)}{x+2}$$



x-int: $(-3, 0), (3, 0)$ Holes: None

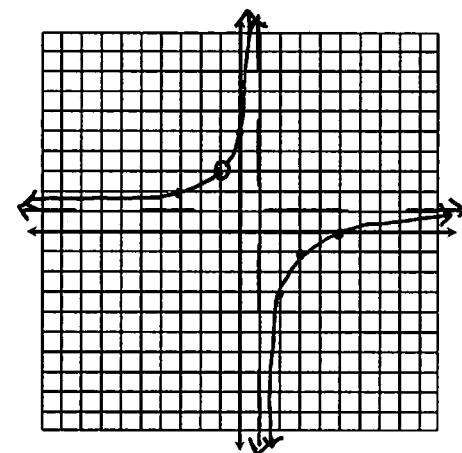
Vertical Asymptote: $x = -2$ D: $\{x | x \neq -2\}$

Horizontal Asymptote: None R: $\mathbb{R}$

② $f(x) = \frac{x^2 - 4x - 5}{x^2 - 1}$

$$f(x) = \frac{(x-5)(x+1)}{(x-1)(x+1)}$$

$$f(x) = \frac{x-5}{x-1}$$



x-int: $(5, 0)$ Holes: $(-1, 3)$

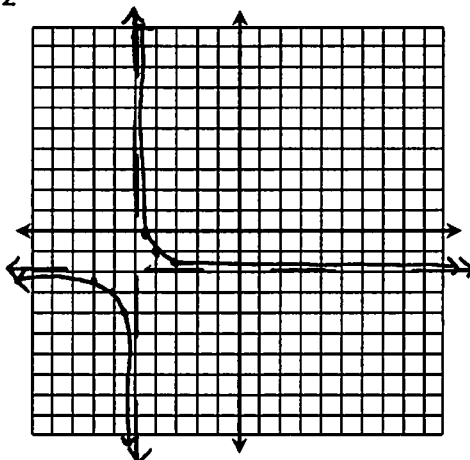
Vertical Asymptote: $x = 1$ D: $\{x | x \neq -1, 1\}$

Horizontal Asymptote: $y = 1$ R: $\{y | y \neq 1, 3\}$

GRAPHING RATIONAL FUNCTIONS Practice!

Directions: Graph each function and identify its key characteristics.

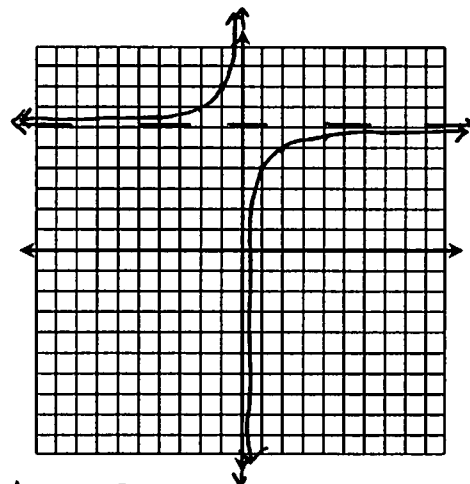
1. $f(x) = \frac{1}{x+5} - 2$



Domain: $\{x | x \neq -5\}$ VA: $x = -5$

Range: $\{y | y \neq -2\}$ HA: $y = -2$

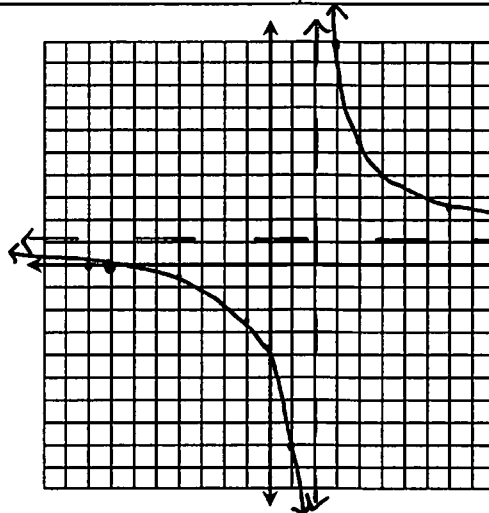
2. $f(x) = \frac{-2}{x} + 6$



Domain: $\{x | x \neq 0\}$ VA: $x = 0$

Range: $\{y | y \neq 6\}$ HA: $y = 6$

3. $f(x) = \frac{x+7}{x-2}$



x-int: $(-7, 0)$

VA: $x = 2$

HA: $y = 1$

Hole: None

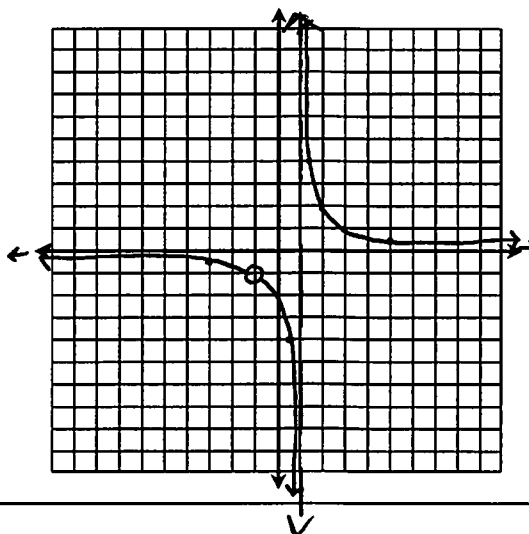
Domain: $\{x | x \neq 2\}$

Range: $\{y | y \neq 1\}$

4. $f(x) = \frac{2x+2}{x^2-1}$

$f(x) = \frac{2(x+1)}{(x+1)(x-1)}$

$f(x) = \frac{2}{x-1}$



x-int: None

VA: $x = 1$

HA: $y = 0$

Hole: $(-1, -1)$

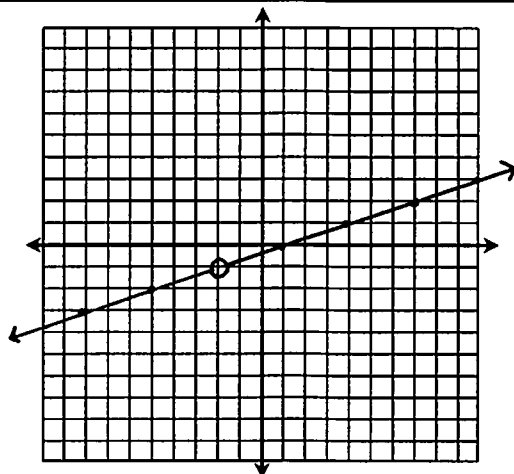
Domain: $\{x | x \neq -1, 1\}$

Range: $\{y | y \neq -1, 0\}$

$$5. f(x) = \frac{x^2 + x - 2}{3x + 6}$$

$$f(x) = \frac{(x+2)(x-1)}{3(x+2)}$$

$$f(x) = \frac{x-1}{3}$$



$$x\text{-int: } (1, 0)$$

$$VA: \text{None}$$

$$HA: \text{None}$$

$$\text{Hole: } (-2, -1)$$

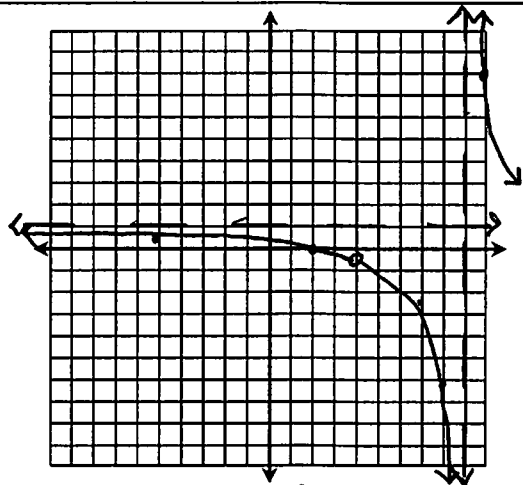
$$\text{Domain: } \{x \mid x \neq -2\}$$

$$\text{Range: } \{y \mid y \neq -1\}$$

$$6. f(x) = \frac{x^2 - 6x + 8}{x^2 - 13x + 36}$$

$$f(x) = \frac{(x-2)(x-4)}{(x-9)(x-4)}$$

$$f(x) = \frac{x-2}{x-9}$$



$$x\text{-int: } (2, 0)$$

$$VA: x = 9$$

$$HA: y = 1$$

$$\text{Hole: } (4, -2/5)$$

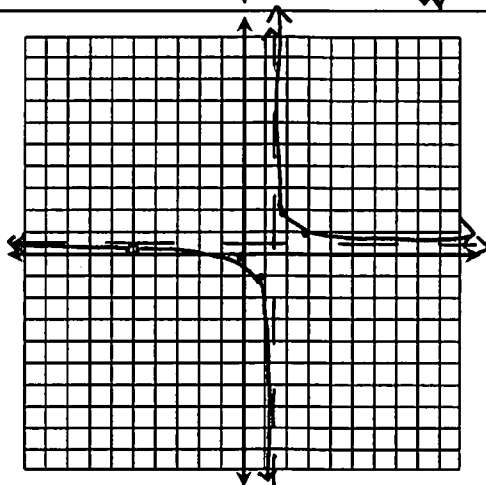
$$\text{Domain: } \{x \mid x \neq 4, 9\}$$

$$\text{Range: } \{y \mid y \neq -2/5, 1\}$$

$$7. f(x) = \frac{x^2 + 5x}{2x^2 + 7x - 15}$$

$$f(x) = \frac{x(x+5)}{(2x-3)(x+5)}$$

$$f(x) = \frac{x}{2x-3}$$



$$x\text{-int: } (0, 0)$$

$$VA: x = 1.5$$

$$HA: y = 0.5$$

$$\text{Hole: } (-5, 5/13)$$

$$\text{Domain: } \{x \mid x \neq -5, 1.5\}$$

$$\text{Range: } \{y \mid y \neq 0.5, 5/13\}$$

8. The **reciprocal parent function** is translated so that it has asymptotes at $x = 4$ and $y = -9$. Write an equation that could represent this new function.

$$f(x) = \frac{1}{x-4} - 9$$

9. Write a **rational function** with an x -intercept at $(1, 0)$ and vertical asymptotes at $x = -3$ and $x = -5$. Then give the horizontal asymptote of the function.

$$f(x) = \frac{x-1}{(x+3)(x+5)}$$

$$f(x) = \frac{x-1}{x^2+8x+15}$$

$$HA: y = 0$$

Name: _____

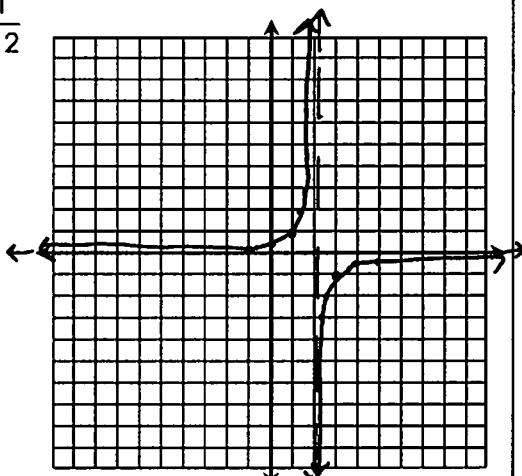
Unit 8: Rational Functions

Date: _____ Per: _____

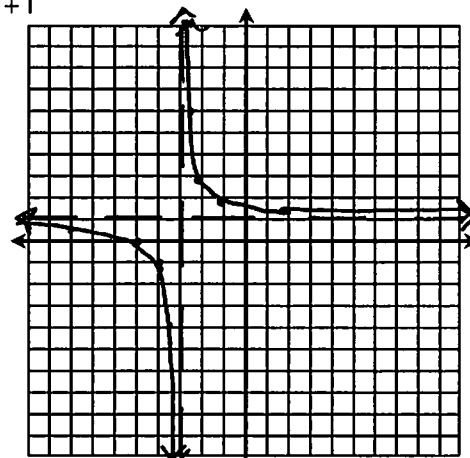
Homework 9: Graphing Review

**** This is a 2-page document! ******Graph each function and identify its key characteristics.**

1. $f(x) = \frac{-1}{x-2}$

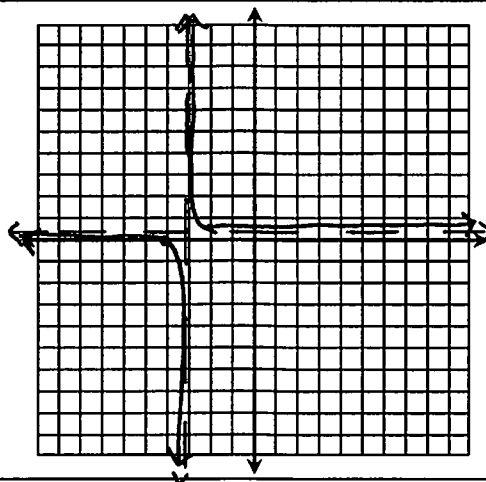
Domain: $\{x | x \neq 2\}$ VA: $x=2$ Range: $\{y | y \neq 0\}$ HA: $y=0$

2. $f(x) = \frac{2}{x+3} + 1$

Domain: $\{x | x \neq -3\}$ VA: $x=-3$ Range: $\{y | y \neq 1\}$ HA: $y=1$

3. $f(x) = \frac{x+4}{4x+12}$

$$f(x) = \frac{x+4}{4(x+3)}$$

x-int: $(-4, 0)$ VA: $x=-3$ HA: $y=0.25$

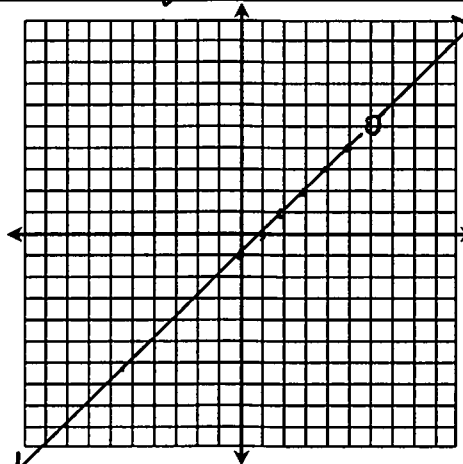
Hole: None

Domain: $\{x | x \neq -3\}$ Range: $\{y | y \neq 0.25\}$

4. $f(x) = \frac{x^2 - 7x + 6}{x - 6}$

$$f(x) = \frac{(x-6)(x-1)}{x-6}$$

$$f(x) = x-1$$

x-int: $(1, 0)$

VA: None

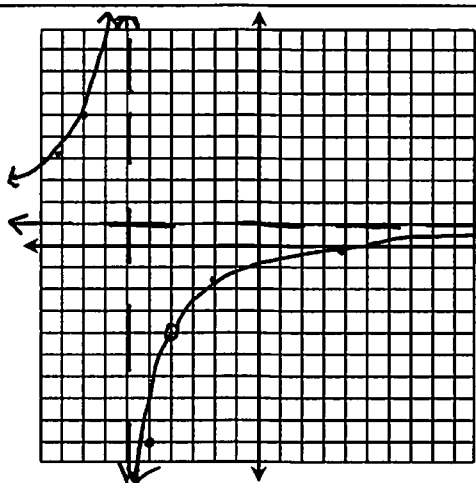
HA: None

Hole: $(6, 5)$ Domain: $\{x | x \neq 6\}$ Range: $\{y | y \neq 5\}$

5. $f(x) = \frac{x^2 - 16}{x^2 + 10x + 24}$

$$f(x) = \frac{(x+4)(x-4)}{(x+4)(x+6)}$$

$$f(x) = \frac{x-4}{x+6}$$



x-int: (4, 0)

VA: $x = -6$

HA: $y = 1$

Hole: $(-4, -4)$

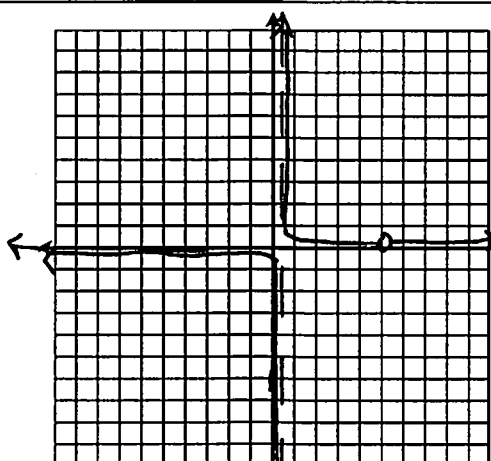
Domain: $\{x \mid x \neq -6, -4\}$

Range: $\{y \mid y \neq -4, 1\}$

6. $f(x) = \frac{x-5}{3x^2 - 16x + 5}$

$$f(x) = \frac{x-5}{(x-5)(3x-1)}$$

$$f(x) = \frac{1}{3x-1}$$



x-int: None

VA: $x = 1/3$

HA: $y = 0$

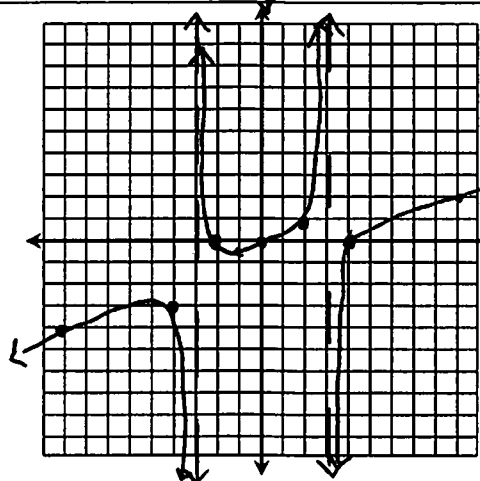
Hole: $(5, 1/4)$

Domain: $\{x \mid x \neq 1/3, 5\}$

Range: $\{y \mid y \neq 0, 1/4\}$

7. $f(x) = \frac{x^3 - 2x^2 - 8x}{3x^2 - 27}$

$$f(x) = \frac{x(x-4)(x+2)}{3(x+3)(x-3)}$$



x-int: $(0, 0), (4, 0), (-2, 0)$

VA: $x = -3, x = 3$

HA: None

Hole: None

Domain: $\{x \mid x \neq -3, 3\}$

Range: $\mathbb{R}$

8. If $f(x) = \frac{1}{x}$ is reflected across the x -axis, then translated 5 units right and 7 units down.

Write an equation that could represent this new function, then identify the asymptotes.

$$f(x) = \frac{-1}{x-5} - 7$$

HA: $y = -7$

VA: $x = 5$

9. Write a rational function with an x -intercept at $(-9, 0)$, a vertical asymptote at $x = 3$, and a hole located at $(1, -5)$. Then, identify the horizontal asymptote.

$$f(x) = \frac{(x+9)(x-1)}{(x-3)(x-1)}$$

$$f(x) = \frac{x^2 + 8x - 9}{x^2 - 4x + 3}$$

HA: $y = 1$

Name:

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Main Ideas/Questions

Notes/Examples

Solving RATIONAL EQUATIONS

METHOD 1: ProportionsCondense into a proportion.
Then, cross-multiply to solve.**METHOD 2:** Using an LCDMultiply each side of the equation by
the least common denominator.
Solve the resulting equation.**ALWAYS REMEMBER TO CHECK YOUR SOLUTIONS!****Directions:** Solve each equation. Check all solutions

1. $\frac{7}{x-6} = \frac{4}{x}$

$$4(x-6) = 7(x)$$

$$4x - 24 = 7x$$

$$-24 = 3x$$

$$x = -8$$

$$x = \{-8\}$$

2. $\frac{3}{a-8} = \frac{7}{2a+1}$

$$3(2a+1) = 7(a-8)$$

$$6a + 3 = 7a - 56$$

$$59 = a$$

$$a = \{59\}$$

3. $\frac{r}{r-1} = \frac{4}{r}$

$$r^2 = 4(r-1)$$

$$r^2 = 4r - 4$$

$$r^2 - 4r + 4 = 0$$

$$(r-2)(r-2) = 0$$

$r-2=0$	$r-2=0$
$r=2$	$r=2$

$$r = \{2\}$$

4. $\frac{m+3}{3} = \frac{8}{m-2}$

$$(m+3)(m-2) = 3(8)$$

$$m^2 - 2m + 3m - 6 = 24$$

$$m^2 + m - 30 = 0$$

$$(m+6)(m-5) = 0$$

$m+6=0$	$m-5=0$
$m=-6$	$m=5$

$$m = \{-6, 5\}$$

5. $\frac{y}{3} = \frac{y+8}{y+5}$

$$y(y+5) = 3(y+8)$$

$$y^2 + 5y = 3y + 24$$

$$y^2 + 2y - 24 = 0$$

$$(y+6)(y-4) = 0$$

$y+6=0$	$y-4=0$
$y=-6$	$y=4$

$$y = \{-6, 4\}$$

6. $\frac{2x-3}{2} = \frac{3}{x+4}$

$$(2x-3)(x+4) = 2(3)$$

$$2x^2 + 5x - 12 = 6$$

$$2x^2 + 5x - 18 = 0$$

$$(2x+9)(x-2) = 0$$

$2x+9=0$	$x-2=0$
$x=-\frac{9}{2}$	$x=2$

$$x = \{-\frac{9}{2}, 2\}$$

$$7. \left[\frac{4}{3} - \frac{7}{n} = \frac{1}{6} \right] \cdot 6n$$

$$\frac{24n}{3} - \frac{42n}{n} = \frac{6n}{6}$$

$$8n - 42 = n$$

$$-42 = -7n$$

$$6 = n$$

$$n = \{6\}$$

$$8. \left[\frac{3}{k} - \frac{1}{2} = \frac{12}{k} \right] \cdot 2k$$

$$\frac{6k}{k} - \frac{2k}{2} = \frac{24k}{k}$$

$$6 - k = 24$$

$$-k = 18$$

$$k = -18$$

$$k = \{-18\}$$

$$9. \left[\frac{p}{p-2} + 2 = \frac{8}{p^2-4} \right] p^2-4$$

$$p(p+2) + 2(p^2-4) = 8$$

$$p^2 + 2p + 2p^2 - 8 = 8$$

$$3p^2 + 2p - 16 = 0$$

$$(3p+8)(p-2) = 0$$

$$3p+8=0 \quad p-2=0$$

$$p = -\frac{8}{3}$$

$$p \neq 2$$

$$p = \{-\frac{8}{3}\}$$

$$10. \left[\frac{13}{3w-3} - \frac{1}{w-1} = \frac{w}{9} \right] \cdot 9(w-1)$$

$$13(3) - 1(9) = w(w-1)$$

$$36 = w^2 - w$$

$$0 = w^2 - w - 36$$

$$0 = (w-6)(w+5)$$

$$w-6=0$$

$$w=6$$

$$w+5=0$$

$$w=-5$$

$$w = \{-5, 6\}$$

$$11. \left[\frac{x+6}{x+3} = 2 - \frac{5x+12}{x+3} \right] \cdot x+3$$

$$x+6 = 2(x+3) - (5x+12)$$

$$x+6 = 2x+6 - 5x-12$$

$$x+6 = -3x-6$$

$$4x = -12$$

$$x \neq -3$$

No Solution

$$x = \{\}$$

$$12. \left[\frac{3c-2}{c^2-4c} = \frac{c}{c-4} - 1 \right] \cdot c^2-4c$$

$$c(c-4)$$

$$3c-2 = c^2 - c^2 + 4c$$

$$3c-2 = 4c$$

$$-2 = c$$

$$c = \{-2\}$$

Name: _____

Unit 8: Rational Functions

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Homework 10: Solving Rational Equations

**** This is a 2-page document! ******Directions:** Solve each equation. Be sure to check all equations.

1. $\frac{5}{m} = \frac{8}{2m-6}$

$5(2m-6) = 8(m)$

$10m - 30 = 8m$

$-30 = -2m$

$m = 15$

$m = \{15\}$

2. $\frac{y}{4} = \frac{9}{y}$

$y^2 = 36$

$y^2 - 36 = 0$

$(y+6)(y-6) = 0$

$$\begin{array}{l|l} y+6=0 & y-6=0 \\ y=-6 & y=6 \end{array}$$

$y = \{\pm 6\}$

3. $\frac{x+5}{5} = \frac{6}{x-2}$

$(x+5)(x-2) = 5(6)$

$x^2 + 3x - 10 = 30$

$x^2 + 3x - 40 = 0$

$(x+8)(x-5) = 0$

$$\begin{array}{l|l} x+8=0 & x-5=0 \\ x=-8 & x=5 \end{array}$$

$x = \{-8, 5\}$

4. $\frac{2}{a-2} = \frac{3a-1}{2a+11}$

$(a-2)(3a-1) = 2(2a+11)$

$3a^2 - 7a + 2 = 4a + 22$

$3a^2 - 11a - 20 = 0$

$(3a-4)(a+5) = 0$

$$\begin{array}{l|l} 3a-4=0 & a+5=0 \\ a=\frac{4}{3} & a=-5 \end{array}$$

$a = \{\frac{4}{3}, -5\}$

5. $\left[w + \frac{16}{w} = 10\right] \cdot w$

$w^2 + 16 = 10w$

$w^2 - 10w + 16 = 0$

$(w-8)(w-2) = 0$

$$\begin{array}{l|l} w-8=0 & w-2=0 \\ w=8 & w=2 \end{array}$$

$w = \{2, 8\}$

6. $\left[\frac{1}{3} + \frac{2}{k} = \frac{13}{3k}\right] \cdot 3k$

$\frac{3k}{3} + \frac{6k}{k} = \frac{39k}{3k}$

$k + 6 = 13$

$k = 7$

$k = \{7\}$

7. $\left[\frac{4}{3v} - \frac{1}{v} = \frac{v+2}{2v^2}\right] \cdot 6v^2$

$\frac{24v^2}{3v} - \frac{6v^2}{v} = \frac{6v^2(v+2)}{2v^2}$

$8v - 6v = 3v + 6$

$2v = 3v + 6$

$-6 = v$

$v = \{-6\}$

8. $\left[\frac{3}{n} - \frac{1}{2} = \frac{n-5}{n}\right] \cdot 2n$

$\frac{6n}{n} - \frac{2n}{2} = \frac{2n(n-5)}{n}$

$6 - n = 2n - 10$

$16 = 3n$

$n = \frac{16}{3}$

$n = \{\frac{16}{3}\}$

$$9. \left[\frac{4}{x} = \frac{1}{x^2+4x} + \frac{5}{x} \right] \cdot x^2+4x$$

$$\begin{aligned} 4(x+4) &= 1 + 5(x+4) \\ 4x+16 &= 1 + 5x+20 \\ 4x+16 &= 5x+21 \\ -5 &= x \end{aligned}$$

$$x = \{-5\}$$

$$10. \left[\frac{1}{p-2} - \frac{5}{p^2+4p-12} = \frac{6}{p-2} \right] \cdot (p+6)(p-2)$$

$$\begin{aligned} p+6-5 &= 6(p+6) \\ p+1 &= 6p+36 \\ -35 &= 5p \\ -7 &= p \end{aligned}$$

$$p = \{-7\}$$

$$11. \left[x + \frac{10}{x-9} = 2 \right] \cdot (x-9)$$

$$\begin{aligned} x(x-9) + 10 &= 2(x-9) \\ x^2 - 9x + 10 &= 2x - 18 \\ x^2 - 11x + 28 &= 0 \end{aligned}$$

$$\begin{array}{l|l} (x-7)(x-4) = 0 & \\ \hline x-7=0 & x-4=0 \\ x=7 & x=4 \end{array}$$

$$x = \{4, 7\}$$

$$12. \left[\frac{5r}{r+1} = 4 - \frac{5}{r+1} \right] \cdot (r+1)$$

$$\begin{aligned} 5r &= 4(r+1) - 5 \\ 5r &= 4r + 4 - 5 \\ 5r &= 4r - 1 \\ r &\neq -1 \end{aligned}$$

$$r = \{ \}$$

No solution

$$13. \left[\frac{2}{c^2-4c} = \frac{1}{3c} + \frac{1}{3} \right] \cdot 3c(c-4)$$

$$\begin{aligned} 2(3) &= c-4 + c(c-4) \\ 6 &= c-4 + c^2-4c \\ 6 &= c^2-3c-4 \\ 0 &= c^2-3c-10 \end{aligned}$$

$$\begin{array}{l|l} 0 = (c-5)(c+2) & \\ \hline c-5=0 & c+2=0 \\ c=5 & c=-2 \end{array}$$

$$c = \{-2, 5\}$$

$$14. \left[\frac{y+6}{y-1} - \frac{y-4}{y^2-y} = \frac{1}{y-1} \right] \cdot y^2-y$$

$$\begin{aligned} y(y+6) - (y-4) &= y \\ y^2 + 6y - y + 4 &= y \\ y^2 + 5y + 4 &= y \\ y^2 + 4y + 4 &= 0 \end{aligned}$$

$$\begin{array}{l|l} (y+2)(y+2) = 0 & \\ \hline y+2=0 & y+2=0 \\ y=-2 & y=-2 \end{array}$$

$$y = \{-2\}$$

Name: _____

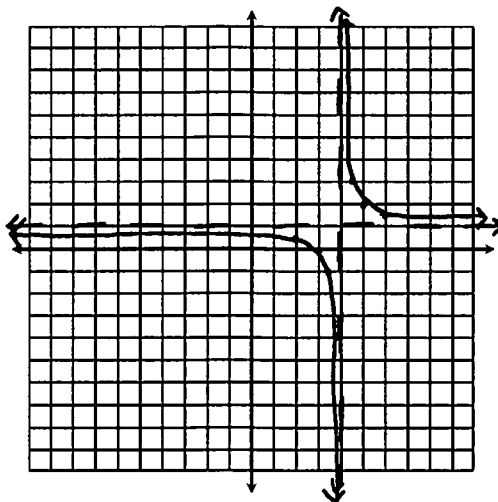
Algebra 2

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Unit 8: Rational Functions

Quiz 8-3: Graphing Rational Functions & Solving Rational Equations**Part I: Graphing** – Graph each function and identify its key characteristics.

1. $f(x) = \frac{1}{x-4} + 1$



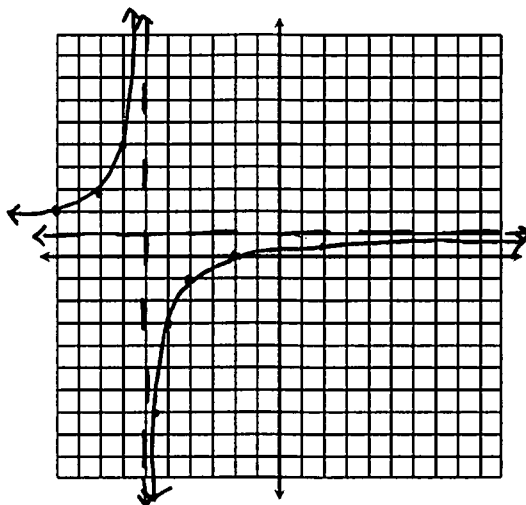
VA: $x=4$

HA: $y=1$

Domain: $\{x | x \neq 4\}$

Range: $\{y | y \neq 1\}$

2. $f(x) = \frac{x+2}{x+6}$



x-int: $(-2, 0)$

VA: $x=-6$

HA: $y=1$

Hole: None

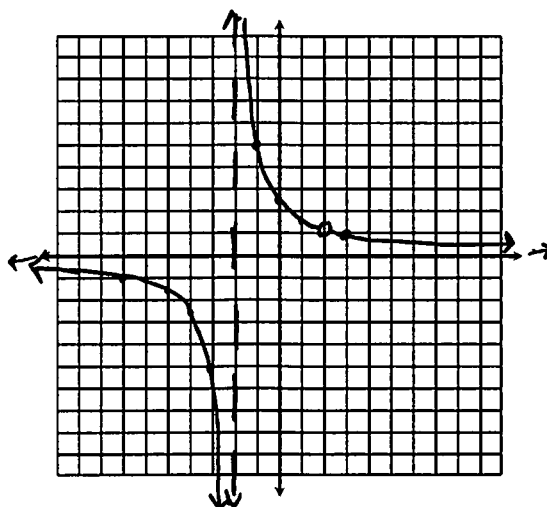
Domain: $\{x | x \neq -6\}$

Range: $\{y | y \neq 1\}$

3. $f(x) = \frac{5x-10}{x^2-4}$

$$f(x) = \frac{5(x-2)}{(x+2)(x-2)}$$

$$f(x) = \frac{5}{x+2}$$



x-int: None

VA: $x=-2$

HA: $y=0$

Hole: $(2, 5/4)$

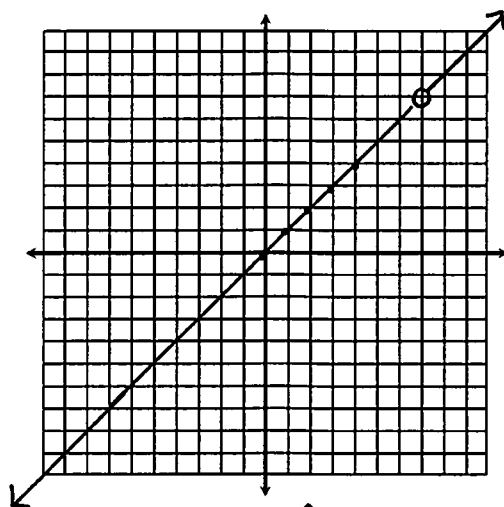
Domain: $\{x | x \neq -2, 2\}$

Range: $\{y | y \neq 0, 5/4\}$

$$4. f(x) = \frac{x^2 - 7x}{x - 7}$$

$$f(x) = \frac{x(x-7)}{x-7}$$

$$f(x) = x$$



$$x\text{-int: } (0,0)$$

$$VA: \text{None}$$

$$HA: \text{None}$$

$$\text{Hole: } (7,7)$$

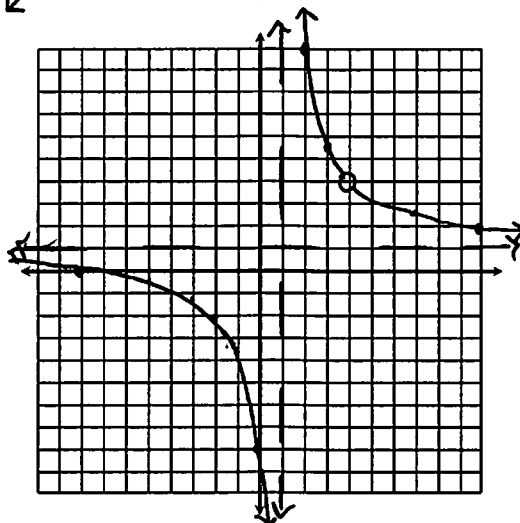
$$\text{Domain: } \{x | x \neq 7\}$$

$$\text{Range: } \{y | y \neq 7\}$$

$$5. f(x) = \frac{x^2 + 4x - 32}{x^2 - 5x + 4}$$

$$f(x) = \frac{(x+8)(x-4)}{(x-1)(x-4)}$$

$$f(x) = \frac{x+8}{x-1}$$



$$x\text{-int: } (-8,0)$$

$$VA: x=1$$

$$HA: y=1$$

$$\text{Hole: } (4,4)$$

$$\text{Domain: } \{x | x \neq 1, 4\}$$

$$\text{Range: } \{y | y \neq 1, 4\}$$

Part II: Equations – Solve each equation. Be sure to check all solutions.

$$6. \frac{x+4}{2} = \frac{4x}{x-6}$$

$$2(4x) = (x+4)(x-6)$$

$$8x = x^2 - 2x - 24$$

$$0 = x^2 - 10x - 24$$

$$0 = (x-12)(x+2)$$

$$\begin{array}{l|l} x-12=0 & x+2=0 \\ x=12 & x=-2 \end{array}$$

$$7. \left[\frac{1}{3} + \frac{m}{2} = \frac{m+4}{4} \right] \cdot 12$$

$$\frac{12}{3} + \frac{12m}{2} = \frac{12(m+4)}{4}$$

$$4 + 6m = 3m + 12$$

$$3m = 8$$

$$m = \frac{8}{3}$$

$$6. x = \{-2, 12\}$$

$$7. m = \{8/3\}$$

$$8. a = \{-7, 6\}$$

$$9. w = \{-2\}$$

$$8. \left[\frac{a}{a-5} - \frac{1}{a+2} = \frac{47}{a^2 - 3a - 10} \right] \cdot (a-5)(a+2)$$

$$a(a+2) - (a-5) = 47$$

$$a^2 + 2a - a + 5 = 47$$

$$a^2 + a - 42 = 0$$

$$(a+7)(a-6) = 0$$

$$\begin{array}{l|l} a+7=0 & a-6=0 \\ a=-7 & a=6 \end{array}$$

$$9. \left[\frac{3}{w^2 - w - 12} = \frac{w-1}{w-4} - 1 \right] \cdot (w-4)(w+3)$$

$$3 = (w-1)(w+3) - (w^2 - w - 12)$$

$$3 = w^2 + 2w - 3 - w^2 + w + 12$$

$$3 = 3w + 9$$

$$-6 = 3w$$

$$w = -2$$

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Main Ideas/Questions	Notes/Examples		
VARIATION Models	Variation is a special relationship between variables.		
	DIRECT VARIATION $y = k \cdot x$ "y is directly proportional to x"	JOINT VARIATION $y = k \cdot x \cdot z$ "y varies jointly as x and z"	INVERSE VARIATION $y = \frac{k}{x}$ "y is inversely proportional to x"
	k is the CONSTANT OF VARIATION		
Identifying EQUATIONS	Directions: Determine whether the equation represents a direct, joint, or inverse variation. Identify the constant of variation.		
	1. $y = 2x$ direct; $k = 2$	2. $y = \frac{24}{x}$ inverse; $k = 24$	
	3. $y = 5xz$ joint; $k = 5$	4. $3y = x \rightarrow y = \frac{1}{3}x$ direct; $k = \frac{1}{3}$	
	5. $xy = 36 \rightarrow y = \frac{36}{x}$ inverse; $k = 36$	6. $\frac{4}{3}y = xz \rightarrow y = \frac{3}{4} \cdot x \cdot z$ joint; $k = \frac{3}{4}$	
	7. $\frac{y}{x} = \frac{2}{5} \rightarrow 5y = 2x$ $y = \frac{2}{5}x$ direct; $k = \frac{2}{5}$	8. $2A = bh \rightarrow A = \frac{1}{2}bh$ joint; $k = \frac{1}{2}$	
DIRECT VARIATION Examples	9. If y is directly proportional to x and $y = 28$ when $x = 7$, find x when $y = 52$. $\frac{28}{7} = \frac{52}{x}$ $28x = 364$ $x = 13$		
	10. If r is directly proportional to the square of s and $r = 72$ when $s = 12$, find r when $s = 8$. $\frac{72}{144} = \frac{r}{64}$ $144r = 4608$ $r = 32$		
	11. The dollar amount d that Megan earns varies directly with the number of hours h that she works. In her last paycheck, she earned \$148.50 working 18 hours. If her next paycheck is \$90.75, how many hours did she work? $\frac{148.50}{18} = \frac{90.75}{h}$ $148.50h = 1633.50$ $h = 11$ 11 hours		

JOINT VARIATION

Examples

12. If y varies jointly with x and z and $y = 48$ when $x = 8$ and $z = 15$, find y when $x = 14$ and $z = 20$.

$$\frac{48}{8 \cdot 15} = \frac{y}{14 \cdot 20}$$

$$48(14)(20) = 8(15)(y)$$

$$13440 = 120y$$

$$112 = y$$

13. If p varies jointly as q and the cube of r , and $p = 270$ when $q = 6$ and $r = 3$, find p when $q = 3$ and $r = 5$.

$$\frac{270}{6 \cdot 27} = \frac{p}{3 \cdot 125}$$

$$162p = 101250$$

$$p = 625$$

14. For Tom's savings account, the interest earned varies jointly with the principal and amount of time (in years) since opening the account. If Tom opened the account 3 years ago with \$1,200 and earned \$144 in interest, how much interest will he have earned 8 years after opening the account?

$$\frac{144}{1200 \cdot 3} = \frac{x}{1200 \cdot 8}$$

$$3600x = 1382400$$

$$x = 384$$

$$\boxed{\$384}$$

INVERSE VARIATION

Examples

15. If y is inversely proportional to x and $y = 7.5$ when $x = 8$, find x when $y = 5$.

$$7.5(8) = 5x$$

$$60 = 5x$$

$$\boxed{x = 12}$$

16. If m is inversely proportional to the square root of n and $m = 27$ when $n = 16$, find m when $n = 64$.

$$27 \cdot 4 = m \cdot 8$$

$$108 = 8m$$

$$\boxed{m = 13.5}$$

17. The time it takes Adam to drive to Disney World varies inversely with his average rate of speed. If it takes Adam 10.8 hours at an average speed of 50 miles per hour, how long would it take him at an average speed of 60 miles per hour?

$$10.8(50) = x \cdot 60$$

$$540 = 60x$$

$$x = 9$$

$$\boxed{9 \text{ hours}}$$

COMBINED VARIATION

A **combined variation** is a relationship that contains **both**

direct and inverse variation.

Examples

18. " y varies directly as x and inversely as z ."

$$y = \frac{k \cdot x}{z}$$

19. " p varies inversely with q squared and directly with r cubed."

$$p = \frac{k \cdot r^3}{q^2}$$

20. a varies directly as b and inversely as c . If $a = 16$ when $b = 12$ and $c = 6$, find a when $b = 28$ and $c = 4$.

$$a = \frac{k \cdot b}{c}$$

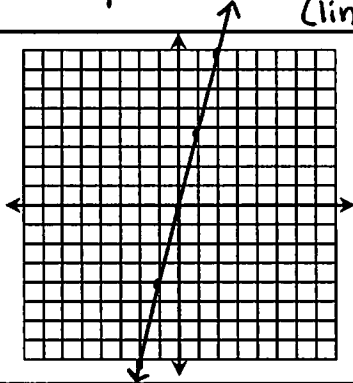
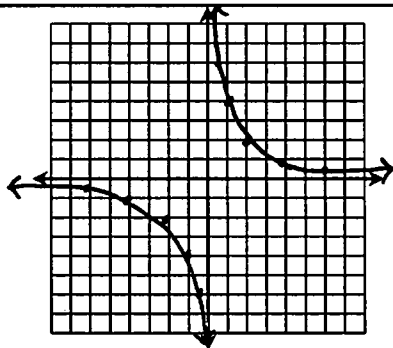
$$16 = \frac{k \cdot 12}{6}$$

$$96 = 12k$$

$$k = 8$$

$$a = \frac{8 \cdot 28}{4}$$

$$\boxed{a = 56}$$

	DIRECT Variation	INVERSE Variation	JOINT Variation
EQUATION FORM	$y = kx$	$y = \frac{k}{x}$	$y = kxz$
EXAMPLE	$y = 4x$ (linear)	$y = \frac{4}{x}$ (reciprocal)	$y = 4xz$
GRAPH			X
EXAMPLE	If y varies directly as x and $y = 10$ when $x = 14$, find the constant and write an equation to represent the relationship. $y = kx$ $10 = k \cdot 14$ $k = \frac{5}{7}; y = \frac{5}{7}x$	If y varies inversely as x and $y = 4$ when $x = 12$, find the constant and write an equation to represent the relationship. $y = \frac{k}{x}$ $4 = \frac{k}{12}$ $k = 48; y = \frac{48}{x}$	If y varies jointly as x and z, and $y = 42$ when $x = 9$ and $z = 7$, find the constant of variation and write an equation to represent the relationship. $y = k \cdot x \cdot z$ $42 = k \cdot 9 \cdot 7$ $42 = 63k$ $k = \frac{2}{3}; y = \frac{2}{3}xz$
COMBINED Variation	What is it? A combination of <u>DIRECT</u> and <u>INVERSE</u> Variation. Example: If p varies directly with q squared and inversely with r, and $p = 38.4$ when $q = 8$ and $r = 5$, find the constant of variation and write an equation to represent the relationship. $p = \frac{k \cdot q^2}{r}$ $38.4 = \frac{k \cdot 64}{5}$ $192 = 64k$ $k = 3$ $p = \frac{3q^2}{r}$		



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Unit 8: Rational Functions

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Homework 11: Direct, Joint, & Inverse Variation

**** This is a 2-page document! ****

Directions: Identify the equation for each variation model.		
DIRECT VARIATION $y = kx$	INVERSE VARIATION $y = \frac{k}{x}$	JOINT VARIATION $y = kxz$
Directions: Translate into an equation to represent the relationship.		
1. "w is directly proportional to v" $w = kv$	2. "a varies inversely with b" $a = \frac{k}{b}$	
3. "f varies jointly with g and h" $f = kgh$	4. "r is inversely proportional to s cubed" $r = \frac{k}{s^3}$	
5. "y varies jointly with x and the square root of z." $y = kx\sqrt{z}$	6. "m varies directly with n squared and inversely with p" $m = \frac{kn^2}{p}$	
Directions: Determine whether the equation represents a direct, joint, or inverse variation. Then, identify the constant.		
7. $t = \frac{100}{s}$ inverse, $k=100$	8. $\frac{y}{5} = x \rightarrow y = 5x$ direct, $k=5$	9. $\frac{p}{qr} = 4 \rightarrow p = 4qr$ joint, $k=4$
10. $C = 2\pi r$ Direct, $k=2\pi$	11. $V = \frac{1}{3}\pi r^2 h$ joint, $k=\frac{1}{3}$	12. $xy = \frac{2}{5} \rightarrow y = \frac{2}{5x}$ inverse, $k=\frac{2}{5}$
Directions: Use the variation model to determine the missing value.		
13. If y is directly proportional to x and $y = 17.5$ when $x = 21$, find y when $x = 39$. $\frac{17.5}{21} = \frac{y}{39}$ $21y = 682.5$ $y = 32.5$	14. If m varies directly with n and $m = 209$ when $n = 22$, find n when $m = 361$. $\frac{209}{22} = \frac{361}{n}$ $209n = 7942$ $n = 38$	

15. If y varies jointly with x and z and $y = 90$ when $x = 2$ and $z = 9$, find x when $y = 105$ and $z = 3$.

$$\frac{90}{2 \cdot 9} = \frac{105}{3x} \quad 1890 = 270x$$

$$\boxed{x=7}$$

16. If c varies jointly as a cubed and b squared, and $c = 25.6$ when $a = 2$ and $b = 4$, find c when $a = 4$ and $b = 7$.

$$\frac{25.6}{8 \cdot 16} = \frac{c}{64 \cdot 49}$$

$$128c = 80281.6$$

$$\boxed{c=627.2}$$

17. If y varies inversely with x and $y = 4.75$ when $x = 38$, find y when $x = 50$.

$$4.75(38) = y \cdot 50$$

$$180.5 = 50y$$

$$\boxed{y=3.61}$$

18. If q is inversely proportional to r squared and $q = 30$ when $r = 3$, find r when $q = 1.2$.

$$30 \cdot 9 = 1.2 \cdot r^2$$

$$270 = 1.2r^2$$

$$225 = r^2$$

$$\boxed{15=r}$$

19. r varies directly as s and inversely as t cubed. If $r = 607.5$ when $s = 12$ and $t = 2$, find r when $s = 42$ and $t = 6$.

$$r = \frac{ks}{t^3}$$

$$607.5 = \frac{k \cdot 12}{8}$$

$$4860 = 12k$$

$$k = 405$$

$$r = \frac{405 \cdot 42}{216}$$

$$\boxed{r=78.75}$$

20. The cost of printing fliers for the opening of an art show varies directly with the number of copies. If it costs \$45 to print 750 copies, what will be the total cost for 120 more copies?

$$C = kn$$

$$\frac{45}{750} = \frac{c}{120}$$

$$750c = 5400$$

$$c = 7.2$$

$$\$45 + \$7.20$$

$$= \$52.20$$

21. The time it takes to fill a water tank varies inversely with the water rate of the hose. At 8 gallons per minute, a hose can fill the tank in 45 minutes. How long will it take to fill the same tank at 15 gallons per minute?

$$t = \frac{k}{r}$$

$$8 \cdot 45 = x \cdot 15$$

$$360 = 15x$$

$$24 = x$$

$$\boxed{24 \text{ min}}$$

22. The volume of a pyramid varies jointly with its height and the area of its base. A pyramid with a height of 16 feet and a base area of 78 square feet has a volume of 416 cubic feet. If a pyramid has a base area of 102 square feet and a volume of 238 cubic feet, find its height.

$$V = k \cdot h \cdot B$$

$$416 = k \cdot 16 \cdot 78$$

$$416 = 1248k$$

$$k = \frac{1}{3}$$

$$238 = \frac{1}{3} \cdot 102 \cdot h$$

$$238 = 34h$$

$$7 = h$$

$$\boxed{7 \text{ ft}}$$

Unit 8 Test Study Guide

(Rational Functions)

Name: _____

Date: _____ Per: _____

Topic 1: Simplifying Rational Expressions

$1. \frac{12x^2 - 30x}{20x^3 - 50x^2}$ $= \frac{6x(2x-5)}{10x^2(2x-5)}$ $= \boxed{\frac{3}{5x}}$	$2. \frac{4a^2 - 36}{24 - 8a}$ $= \frac{4(a^2 - 9)}{8(3-a)}$ $= \frac{4(a+3)(a-3)}{8(3-a)}$ $= \frac{-(a+3)}{2} = \boxed{\frac{-a-3}{2}}$	$3. \frac{n^2 - 13n + 40}{3n^2 - 14n - 5}$ $= \frac{(n-8)(n-5)}{(3n+1)(n-5)}$ $= \boxed{\frac{n-8}{3n+1}}$
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Topic 2: Operations with Rational Expressions

$4. \frac{6p^2 - 13p + 5}{2p^2 + 17p - 9} \cdot \frac{p^2 + 16p + 63}{4p + 28}$ $\frac{(2p-1)(3p-5)}{(2p-1)(p+9)} \cdot \frac{(p+9)(p+7)}{4(p+7)}$ $= \boxed{\frac{3p-5}{4}}$	$5. \frac{50 - 2w^2}{3w^2 + 9w - 30} \cdot \frac{w^2 + 5w - 14}{6w - 30}$ $\frac{2(5+w)(5-w)}{3(w+5)(w-2)} \cdot \frac{(w+7)(w-2)}{6(w-5)}$ $= \frac{-2(w+7)}{18} = \boxed{\frac{-w-7}{9}}$
$6. \frac{5y+5}{2} \div \frac{25y-20}{40y^2-32y}$ $= \frac{5(y+1)}{2} \cdot \frac{8y(5y-4)}{5(5y-4)}$ $= \frac{40y(y+1)}{10} = \boxed{4y(y+1)}$	$7. \frac{2c^2 + 4c - 6}{4c^2 - 7c + 3} \div \frac{16c^2 + 48c}{16c^2 - 9}$ $\frac{2(c-1)(c+3)}{(4c-3)(c-1)} \cdot \frac{(4c+3)(4c-3)}{16c(c+3)}$ $= \frac{2(4c+3)}{16c} = \boxed{\frac{4c+3}{8c}}$
$8. \frac{6x}{x^2-16} - \frac{x-20}{x^2-16}$ $= \frac{6x - x + 20}{x^2 - 16}$ $= \frac{5x+20}{x^2-16} = \frac{5(x+4)}{(x+4)(x-4)} = \boxed{\frac{5}{x-4}}$	$\frac{k+5}{k+5} \cdot \frac{16}{3} - \frac{4k+56}{3k+15}$ $= \frac{16k+80-4k-56}{3(k+5)}$ $= \frac{12k+24}{3(k+5)} = \frac{12(k+2)}{3(k+5)} = \boxed{\frac{4(k+2)}{k+5}}$

$10. \frac{m^2 - 7m - 18}{m^2 - 10m + 9} + \frac{6}{m-1} \cdot \frac{m-9}{m-9}$ $= \frac{m^2 - 7m - 18 + 6m - 54}{(m-1)(m-9)}$ $= \frac{m^2 - m - 72}{(m-1)(m-9)} = \frac{(m-9)(m+8)}{(m-1)(m-9)} = \boxed{\frac{m+8}{m-1}}$	$\frac{2r-1}{2r-1} \cdot \frac{r}{2r+1} + \frac{12r-6}{4r^2-1}$ $= \frac{2r^2 - r + 12r - 6}{(2r+1)(2r-1)}$ $= \frac{2r^2 + 11r - 6}{(2r+1)(2r-1)} = \frac{(2r-1)(r+6)}{(2r+1)(2r-1)} = \boxed{\frac{r+6}{2r+1}}$
$12. \frac{a+4}{3} + \frac{5}{24} \cdot \frac{a}{a}$ $= \frac{3a+12+5a}{24a}$ $= \frac{8a+12}{24a} = \frac{4(2a+3)}{24a} = \boxed{\frac{2a+3}{6a}}$	$\frac{h+5}{h+5} \cdot \frac{2}{h+2} + \frac{5}{h+5} \cdot \frac{h+2}{h+2}$ $= \frac{2h+10+5h+10}{(h+5)(h+2)}$ $= \boxed{\frac{7h+20}{(h+5)(h+2)}}$
$\frac{x-3}{x-3} \cdot \frac{3x-1}{x-1} - \frac{x-1}{x-3} \cdot \frac{x+1}{x^2-4x+3} = \frac{(x-3)(3x-1) - (x-1)(x-1) + x+1}{(x-3)(x-1)}$ $= \frac{3x^2 - 10x + 3 - (x^2 - 2x + 1) + x + 1}{(x-3)(x-1)}$ $= \frac{2x^2 - 7x + 3}{(x-3)(x-1)} = \frac{(2x-1)(x-3)}{(x-3)(x-1)} = \boxed{\frac{2x-1}{x-1}}$	

Topic 3: Complex Fractions

$15. \frac{\frac{12m^3}{m^2+14m+45}}{\frac{3m^3-6m^2}{m^2+7m-18}} = \frac{12m^3}{(m+5)(m+9)} \cdot \frac{(m+9)(m-2)}{3m^2(m-2)}$ $= \boxed{\frac{4m}{m+5}}$	$16. \frac{\frac{8k+12}{9}}{\frac{2}{2} \cdot \frac{k}{3} + \frac{1}{2} \cdot \frac{3}{3}} = \frac{4(2k+3)}{9} \cdot \frac{6}{2k+3}$ $= \frac{24}{9}$ $= \boxed{\frac{8}{3}}$
$17. \frac{\frac{b}{b} \cdot \frac{b}{2} - \frac{32}{b} \cdot \frac{2}{2}}{\frac{b}{b} \cdot 1 + \frac{8}{b}} = \frac{\frac{b^2-64}{2b}}{\frac{b^2-8}{b}} \cdot \frac{b}{b+8}$ $= \frac{(b+8)(b-8)}{2b} \cdot \frac{b}{b+8}$ $= \boxed{\frac{b-8}{2}}$	$18. \frac{\frac{x}{x} \cdot \frac{x}{x+1} + \frac{4}{x} \cdot \frac{x+1}{x+1}}{\frac{4x+7}{3x+3} - \frac{1}{3} \cdot \frac{x+1}{x+1}} = \frac{\frac{x^2+4x+4}{x(x+1)}}{\frac{(x+2)(x+2)}{x(x+1)}} \cdot \frac{3(x+1)}{3(x+2)}$ $= \boxed{\frac{x+2}{x}}$

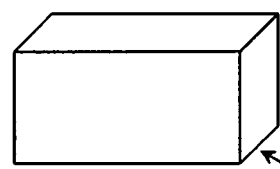
Topic 4: Applications

19. If the **area** of the rectangle below is $\frac{n-3}{n} + \frac{7}{2}$, find w , its **width**.



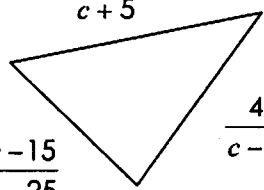
$$\begin{aligned} \frac{3}{4n} - \frac{1}{2n^2} \cdot w &= \frac{\frac{n-3}{n} + \frac{7}{2}}{\frac{n-3}{n} + \frac{7}{2}} \cdot \frac{4n^2}{3n-2} \\ &= \frac{3(3n-2)}{2n} \cdot \frac{4n^2}{3n-2} \\ &= \frac{12n^2}{2n} = \boxed{6n} \end{aligned}$$

20. Find the **volume** of the rectangular prism below.



$$\begin{aligned} \frac{x^2-9}{x^2+3x-18} \cdot \frac{x^2+13x+42}{6x^2} \cdot \frac{3x}{5x+35} &= \frac{(x+3)(x-3)}{(x+6)(x-3)} \cdot \frac{(x+7)(x+6)}{6x^2} \cdot \frac{3x}{5(x+7)} \\ &= \frac{3x(x+3)}{30x^2} \\ &= \boxed{\frac{x+3}{10x}} \end{aligned}$$

21. Find the **perimeter** of the triangle below.



$$\begin{aligned} \frac{c}{c+5} + \frac{7c-15}{c^2-25} + \frac{4}{c-5} &= \frac{c-5}{c-5} \cdot \frac{c}{c+5} + \frac{4}{c-5} \cdot \frac{c+5}{c+5} + \frac{7c-15}{(c+5)(c-5)} \\ &= \frac{c^2-5c+4c+20+7c-15}{(c+5)(c-5)} \\ &= \frac{c^2+6c+5}{(c+5)(c-5)} = \frac{(c+5)(c+1)}{(c+5)(c-5)} = \boxed{\frac{c+1}{c-5}} \end{aligned}$$

Topic 5: Solving Rational Equations

22. $\frac{a-8}{a} = \frac{3}{a+5}$

$$\begin{aligned} 3a &= (a-8)(a+5) \\ 3a &= a^2 - 3a - 40 \\ 0 &= a^2 - 6a - 40 \\ 0 &= (a-10)(a+4) \\ \hline a &= 10 \quad | \quad a = -4 \end{aligned}$$

$$\boxed{a = \{-4, 10\}}$$

23. $\frac{3}{2n} + \frac{1}{n^2} = \frac{n-2}{2n^2}$

$$\begin{aligned} 3n+2 &= n-2 \\ 2n+2 &= -2 \\ 2n &= -4 \\ n &= -2 \end{aligned}$$

$$\boxed{n = \{-2\}}$$

$$24. \left[\frac{p^2 + p + 8}{p^2 - 7p + 6} + \frac{2}{p-1} = \frac{2}{3} \right] \cdot 3(p-6)(p-1)$$

$$3p^2 + 3p + 24 + 6p - 36 = 2(p^2 - 7p + 6)$$

$$3p^2 + 9p - 12 = 2p^2 - 14p + 12$$

$$p^2 + 23p - 24 = 0$$

$$(p+24)(p-1) = 0$$

$$p = -24 \quad p = 1$$

$$p = \{-24\}$$

$$25. \left[\frac{1}{3} - \frac{1}{4v+3} = \frac{v^2 - 3v - 18}{12v+9} \right] 3(4v+3)$$

$$4v+3-3 = v^2 - 3v - 18$$

$$4v = v^2 - 3v - 18$$

$$0 = v^2 - 7v - 18$$

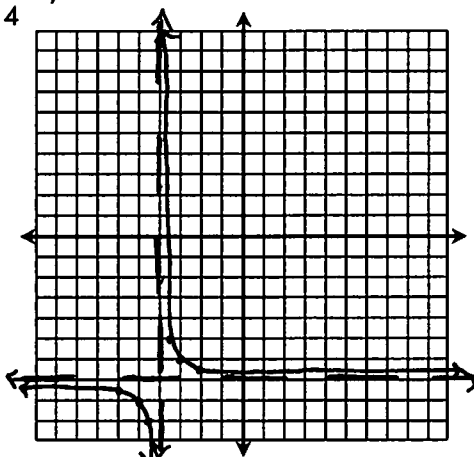
$$0 = (v-9)(v+2)$$

$$v = 9 \quad v = -2$$

$$v = \{-2, 9\}$$

Topic 6: Graphing Rational Functions

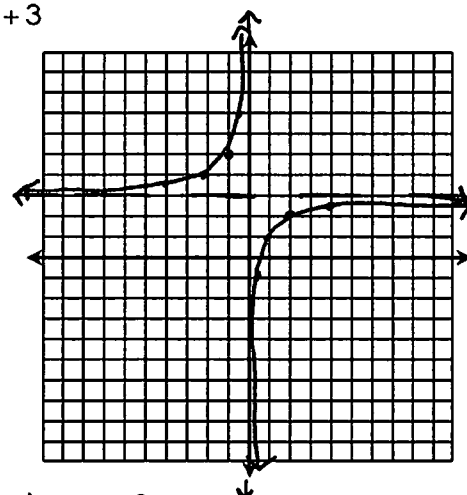
$$26. f(x) = \frac{1}{x+4} - 7$$



Domain: $\{x | x \neq -4\}$ VA: $x = -4$

Range: $\{y | y \neq -7\}$ HA: $y = -7$

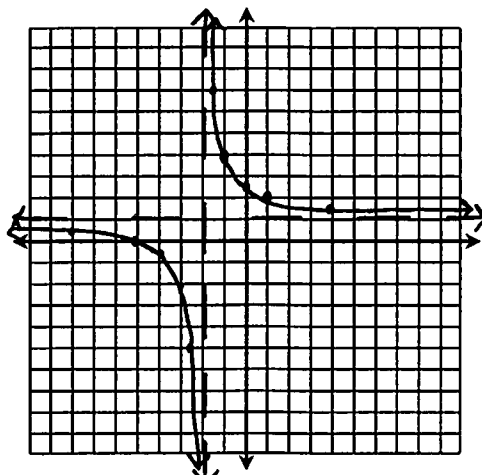
$$27. f(x) = \frac{-2}{x} + 3$$



Domain: $\{x | x \neq 0\}$ VA: $x = 0$

Range: $\{y | y \neq 3\}$ HA: $y = 3$

$$28. f(x) = \frac{x+5}{x+2}$$



x-int: $(-5, 0)$

VA: $x = -2$

HA: $y = 1$

Hole: None

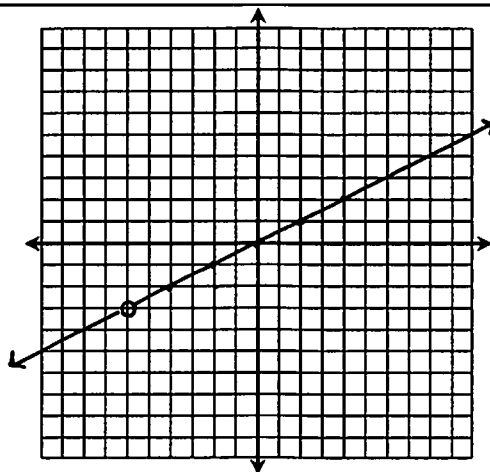
Domain: $\{x | x \neq -2\}$

Range: $\{y | y \neq 1\}$

29. $f(x) = \frac{x^2 + 6x}{2x + 12}$

$f(x) = \frac{x(x+6)}{2(x+6)}$

$f(x) = \frac{x}{2}$



x-int: (0,0)

VA: None

HA: None

Hole: (-6, -3)

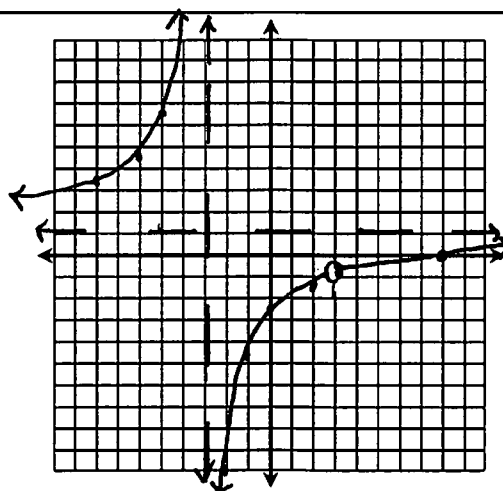
Domain: $\{x | x \neq -6\}$

Range: $\{y | y \neq -3\}$

30. $f(x) = \frac{x^2 - 11x + 24}{x^2 - 9}$

$f(x) = \frac{(x-8)(x-3)}{(x+3)(x-3)}$

$f(x) = \frac{x-8}{x+3}$



x-int: (8,0)

VA: $x = -3$

HA: $y = 1$

Hole: $(3, -\frac{5}{6})$

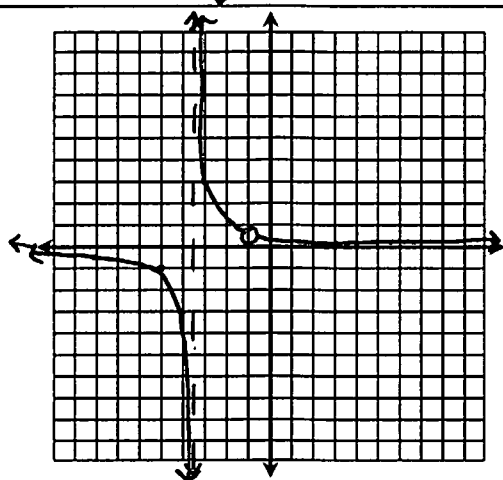
Domain: $\{x | x \neq -3, 3\}$

Range: $\{y | y \neq -\frac{5}{6}, 1\}$

31. $f(x) = \frac{3x+3}{2x^2+9x+7}$

$f(x) = \frac{3(x+1)}{(2x+7)(x+1)}$

$f(x) = \frac{3}{2x+7}$



x-int: None

VA: $x = -\frac{7}{2}$

HA: $y = 0$

Hole: $(-\frac{3}{2}, \frac{3}{5})$

Domain: $\{x | x \neq -\frac{7}{2}, -1\}$

Range: $\{y | y \neq 0, \frac{3}{5}\}$

32. The **reciprocal parent function** is translated 8 units left and 3 units up. Write an equation to represent this function. Then, identify the asymptotes.

$f(x) = \frac{1}{x+8} + 3$

VA: $x = -8$

HA: $y = 3$

33. A rational function has an x-intercept at (-1, 0) and vertical asymptotes at $x = 7$ and $x = -5$. Write an equation that could represent this function, then identify the horizontal asymptote.

$f(x) = \frac{x+1}{(x-7)(x+5)}$

$f(x) = \frac{x+1}{x^2-2x-35}$

HA: $y = 0$

Topic 7: Variation

DIRECT VARIATION	JOINT VARIATION	INVERSE VARIATION
$y = kx$; $k = \frac{y}{x}$	$y = k \cdot x \cdot z$; $k = \frac{y}{xz}$	$y = \frac{k}{x}$; $k = yx$
Translate into an equation to represent the relationship.		
34. "s varies jointly with t cubed and v" $S = kt^3v$	35. "m varies inversely with n squared and directly with p" $m = \frac{kp}{n^2}$	
Determine if the equation represents a direct, joint, or inverse variation. Identify the constant.		
36. $3y = x^2z$ $y = \frac{1}{3}x^2z$ joint ; $k = \frac{1}{3}$	37. $\frac{1}{2}d = r$ $d = 2r$ direct ; $k = 2$	38. $\frac{8}{x} = \frac{y}{6}$ $xy = 48$ $y = \frac{48}{x}$ inverse ; $k = 48$
Use the variation type to find the missing value.		
39. w varies directly as r cubed and inversely as s. If $w = 24$ when $r = 4$ and $s = 8$, find r when $w = 81$ and $s = 27$. $w = \frac{kr^3}{s}$ $24 = \frac{k \cdot 4^3}{8}$ $192 = 64k$ $k = 3$ $81 = \frac{3r^3}{27}$ $729 = r^3$ $r = 9$	40. If b varies jointly with a and c and $b = 112$ when $a = 12$ and $c = 7$, find a when $b = 72$ and $c = 2$. $b = kac$ $112 = k \cdot 12 \cdot 7$ $112 = 84k$ $\frac{4}{3} = k$ $72 = \frac{4}{3} \cdot a \cdot 2$ $72 = \frac{8}{3}a$ $27 = a$	
41. The wind force F on a sail varies jointly as the area A of the sail and the square of the wind speed w . force on a sail with area an area of 500 ft^2 is 64.8 pounds when the wind speed is 18 mph. What would be the force for a sail with an area of 250 ft^2 with a wind speed of 35 mph? $F = k \cdot A \cdot w^2$ $64.8 = k \cdot 500 \cdot 18^2$ $.0004 = k$ $F = .0004 \cdot 250 \cdot 35^2$ $F = 122.5$ 122.5 lbs		
42. The volume, V , of a gas varies directly as the temperature, t , and inversely as the pressure, P . At a temperature of 200° and a pressure of 500 mmHG , the volume of the gas is 240 cm^3 . Find the volume of the gas at a temperature of 275° and a pressure of 400 mmHG . $V = \frac{kt}{P}$ $240 = \frac{k \cdot 200}{500}$ $600 = k$ $V = \frac{600 \cdot 275}{400}$ $V = 412.5$ 412.5 cm^3		

Name: _____

Date: _____ Per: _____

Unit 8 Test

Rational Functions

1. Simplify the expression below.

$$\frac{2a^3 - 32a}{2a^3 + 16a + 32a}$$

$$\frac{2a(a+4)(a-4)}{2a(a+4)(a+4)}$$

A. $2(a-4)$

C. $\frac{(a-4)}{(a+4)}$

B. -2

D. $\frac{2(a-4)}{(a+4)}$

C

2. Simplify the expression below.

$$\frac{4yz^2}{3xy^4z} \cdot \frac{15x^2y}{8x^3yz}$$

$$\frac{60x^2y^2z^2}{24x^4y^5z^2}$$

A. $\frac{5}{2x^2y^3}$

C. $\frac{2x^2y^3}{5z}$

B. $\frac{5z}{2x^2y^3}$

D. $\frac{2x^2y^3}{5}$

A

3. Simplify the expression below.

$$\frac{2-18w^2}{3w^2-8w-3} \cdot \frac{3w^2-7w-6}{12w^2+80w-28}$$

$$\frac{2(1-3w)(1+3w)}{(3w+1)(w-3)} \cdot \frac{(3w+2)(w-3)}{4(3w-1)(w+7)}$$

A. $\frac{3w+2}{2(w+7)}$

C. $\frac{-(3w+2)}{2(w+7)}$

B. $\frac{3w+2}{w+7}$

D. $\frac{-2(3w+2)}{w+7}$

C

4. Simplify the expression below.

$$\frac{2c^2-5c-3}{c^2+4c-21} \div (2c+1)$$

$$\frac{(2c+1)(c-3)}{(c+7)(c-3)} \cdot \frac{1}{2c+1}$$

A. $\frac{2c+1}{c+7}$

C. $\frac{(c+3)}{(c+7)(c-3)}$

B. $\frac{1}{c+7}$

D. $c+7$

B

5. Simplify the expression below.

$$\frac{p^2-14p+40}{16p^3} \div \frac{3p^2-18p+24}{2p^2-4p}$$

$$\frac{(p-4)(p-10)}{16p^3} \cdot \frac{2p(p-2)}{3(p-2)(p-4)}$$

A. $\frac{24p^2}{p-10}$

C. $\frac{p-10}{24p^2}$

B. $\frac{24(p-10)}{p^2}$

D. $\frac{1}{24p^2(p-10)}$

C

6. Simplify the expression below.

$$\frac{17x-9}{3x+6} - \frac{2x-3}{3x+6}$$

$$\frac{15x-6}{3x+6} = \frac{3(5x-2)}{3(x+2)}$$

A. $-5x$

C. $\frac{5x-2}{x+2}$

B. $\frac{5x-2}{3x}$

D. $\frac{5x-4}{x+2}$

C

7. Simplify the expression below.

$$\frac{m+8}{m+8} \cdot \frac{m}{m-3} + \frac{5m-48}{m^2+5m-24}$$

$$= \frac{m^2+8m+5m-48}{(m+8)(m-3)}$$

$$= \frac{m^2+13m-48}{(m+8)(m-3)} = \frac{(m+16)(m-3)}{(m+8)(m-3)}$$

A. $\frac{m}{m-3}$

C. $m+2$

B. $\frac{m+8}{m-3}$

D. $\frac{m+16}{m+8}$

D

8. Simplify the expression below.

$$\frac{v-6}{v-6} \cdot \frac{8}{v+2} + \frac{3}{v-6} \cdot \frac{v+2}{v+2}$$

$$= \frac{8v-48+3v+6}{(v-6)(v+2)}$$

$$= \frac{11v-42}{(v-6)(v+2)}$$

A. $\frac{11}{(v+2)(v-6)}$

C. $\frac{11v-42}{2v-4}$

B. $\frac{11v-42}{(v+2)(v-6)}$

D. $\frac{11}{2v-4}$

B

9. Simplify the expression below.

$$\frac{9-k}{28k^3} \cdot \frac{4k^6}{k-9} = -\frac{4k^6}{28k^3}$$

A. $\frac{-k^3}{7}$

C. $\frac{k^3}{7}$

B. $\frac{-7}{k^2}$

D. $\frac{7}{k^2}$

A

10. Simplify the expression below.

$$\frac{\frac{r^2}{6} + \frac{r}{2} \cdot \frac{3}{3}}{\frac{r+3}{12}}$$

$$\frac{r(r+3)}{6} \cdot \frac{12}{r+3}$$

A. $\frac{r+3}{6}$

C. $2r$

B. $\frac{r}{6}$

D. $\frac{1}{2r}$

C

11. Simplify the expression below.

$$\frac{4p}{4p} \cdot \frac{p-\frac{9}{4p}}{\frac{2p}{2p} \cdot 1 + \frac{3}{2p}}$$

$$\frac{(2p+3)(2p-3)}{4p} \cdot \frac{2p}{2p+3}$$

A. $\frac{2p-3}{2}$

C. $2(2p-3)$

B. $\frac{p-9}{8p}$

D. $\frac{p-9}{2p}$

A

12. Find the area of the triangle below.

$$\frac{14x-28}{x^2-3x-4}$$



$$\frac{1}{2} \cdot \frac{14(x-2)}{(x-4)(x+1)} \cdot \frac{(x+1)(x+8)}{x-2}$$

A. $7(x-2)$

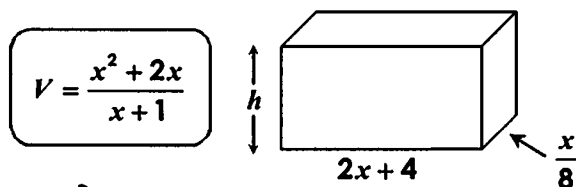
C. $\frac{7(x+8)}{x-4}$

B. $14(x-2)$

D. $\frac{14(x+8)}{x-4}$

C

13. Given the volume, V , of the rectangular prism below, find h , the height of the prism.



$$\frac{x(x+2)}{x+1} \cdot \frac{8}{2x(x+2)}$$

A. $\frac{16}{x+1}$

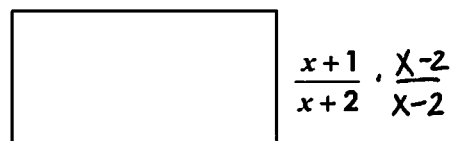
C. $16(x+1)$

B. $\frac{4}{x+1}$

D. $4(x+1)$

B

14. Find the perimeter of the rectangle below.



$$\frac{18x+28+2x^2-2x-4}{(x+2)(x-2)} = \frac{2x^2+16x+24}{(x+2)(x-2)}$$

A. $x-3$

C. $\frac{x+6}{x-2}$

B. $2(x-3)$

D. $\frac{2(x+6)}{x-2}$

D

15. Solve the equation below.

$$\frac{a+2}{5-a} = \frac{6}{a-5}$$

$$6(5-a) = (a+2)(a-5)$$

$$30-6a = a^2-3a-10$$

$$0 = a^2+3a-40$$

$$0 = (a+8)(a-5)$$

$$a = -8 \quad | \quad a = 5$$

$$a = \{-8\}$$

16. Solve the equation below.

$$\left[\frac{1}{2k} + \frac{4}{k} = \frac{k}{k+2} \right] 2k(k+2)$$

$$k+2+8k+16 = 2k^2$$

$$0 = 2k^2 - 9k - 18$$

$$0 = (k-6)(2k+3)$$

$$k = 6 \quad | \quad k = -\frac{3}{2}$$

$$k = \left\{ -\frac{3}{2}, 6 \right\}$$

17. Solve the equation below.

$$\left[\frac{1}{n-4} + \frac{2}{n-2} = \frac{2}{n^2-6n+8} \right] (n-4)(n-2)$$

$$n-2+2n-8=2$$

$$3n-10=2$$

$$3n=12$$

$$n=4$$

$$n = \emptyset$$

18. Solve the equation below.

$$\left[\frac{x+1}{x} + \frac{x-5}{3x} = \frac{2x+9}{4x} \right] \cdot 12x$$

$$12x+12+4x-20=6x+27$$

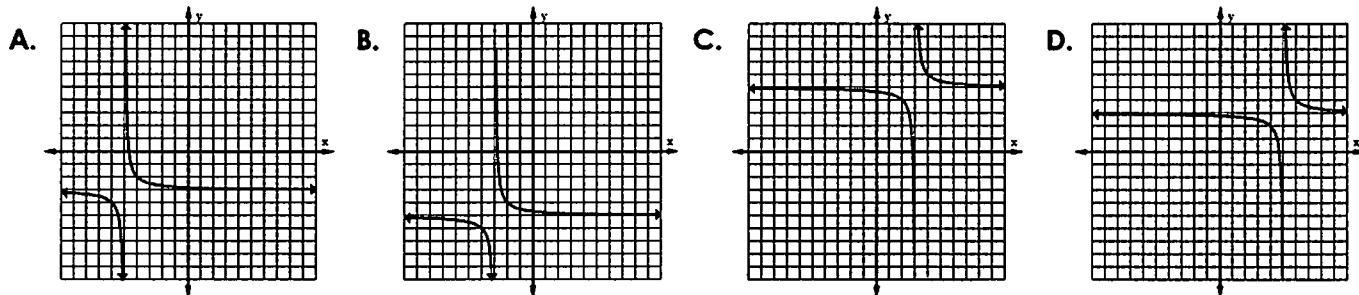
$$16x-8=6x+27$$

$$10x=35$$

$$x=3.5$$

$$x = \{3.5\}$$

19. Which graph represents the function $f(x) = \frac{1}{x+3} - 5$?



B

Use for questions 20-21:

The parent function of a reciprocal function is translated two units left and four units down.

20. Which of the following represents the horizontal asymptote of the new function?

A. $x = 2$

B. $x = -4$

C. $y = -2$

D. $y = -4$

$$f(x) = \frac{1}{x+2} - 4$$

D

21. What is the domain of this function?

A. $\{x|x \neq 2\}$

B. $\{x|x \neq -2\}$

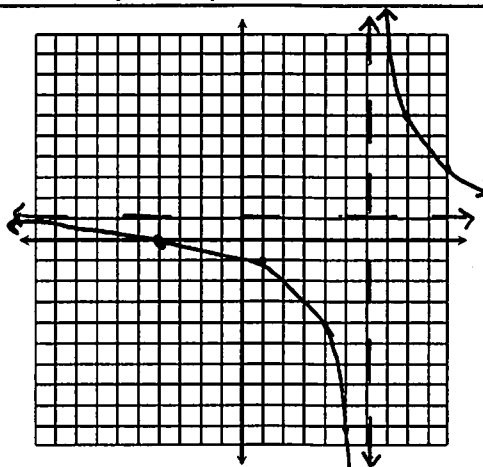
C. $\{y|y \neq 2\}$

D. $\{y|y \neq -4\}$

B

Graph each function below and identify its key characteristics.

22. $f(x) = \frac{x+4}{x-6}$



x-int: $(-4, 0)$

VA: $x = 6$

HA: $y = 1$

Hole: None

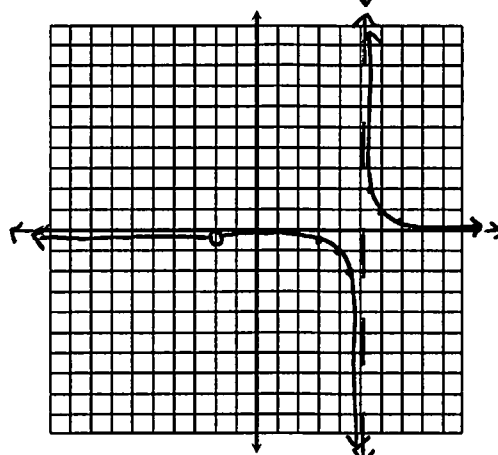
Domain: $\{x|x \neq 6\}$

Range: $\{y|y \neq 1\}$

23. $f(x) = \frac{x+2}{x^2-3x-10}$

$$f(x) = \frac{x+2}{(x-5)(x+2)}$$

$$f(x) = \frac{1}{x-5}$$



x-int: None

VA: $x = 5$

HA: $y = 0$

Hole: $(-2, -1/7)$

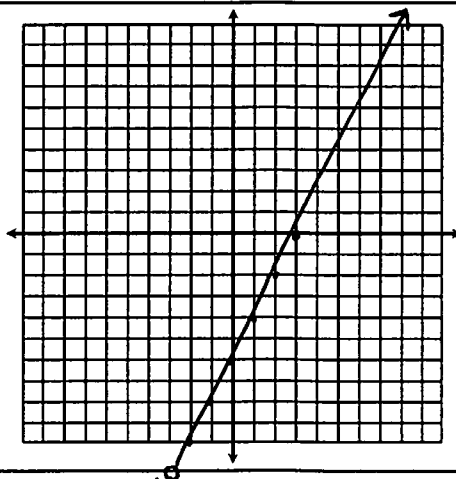
Domain: $\{x|x \neq -2, 5\}$

Range: $\{y|y \neq -1/7, 0\}$

24. $f(x) = \frac{2x^2 - 18}{x + 3}$

$f(x) = \frac{2(x+3)(x-3)}{x+3}$

$f(x) = 2(x-3)$



x-int: (3, 0)

VA: None

HA: None

Hole: (-3, -12)

Domain: $\{x | x \neq -3\}$

Range: $\{y | y \neq -12\}$

25. Which functions have a vertical asymptote at $x = -2$? Check all that apply.

☐ $f(x) = \frac{x^2 - 4}{x + 2}$	☒ $f(x) = \frac{1}{x + 2} - 2$	☒ $f(x) = \frac{x^2 - 8x - 9}{x^2 - 7x - 18}$	$\frac{(x-9)(x+1)}{(x-9)(x+2)}$
☒ $f(x) = \frac{x-2}{x^2 - 4}$	☐ $f(x) = \frac{1}{x-2} + 2$	☐ $f(x) = \frac{2x^2 + 4x}{x^2 + 2x}$	$\frac{2x(x+2)}{x(x+2)}$

$\frac{x-2}{(x+2)(x-2)}$

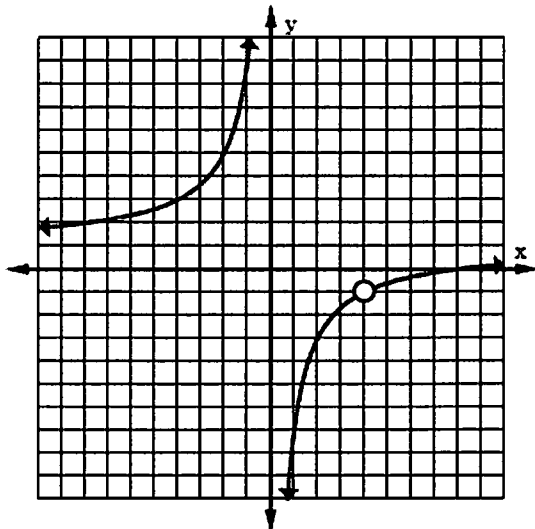
26. Which statement accurately describes the horizontal asymptotes of the functions below?

$\left\{ f(x) = \frac{1}{x-1}, g(x) = \frac{x}{x^2 - 1} \right\}$

- A. The horizontal asymptote for $f(x)$ is $y = 0$;
The horizontal asymptote for $g(x)$ is $y = 1$ and $y = -1$.
- B. The horizontal asymptote for $f(x)$ is $y = 1$;
The horizontal asymptote for $g(x)$ is $y = 0$.
- C. The horizontal asymptotes for both functions is $y = 1$.
- D. The horizontal asymptotes for both functions is $y = 0$.

D

27. Which equation could represent the function graphed below?



A. $f(x) = \frac{x^2 - 12x + 32}{x^2 - 4x} = \frac{(x-8)(x-4)}{x(x-4)}$

B. $f(x) = \frac{x^2 - 4x - 32}{x^2 + 4x}$

C. $f(x) = \frac{x^2 + 4x - 32}{x^2 - 4x}$

D. $f(x) = \frac{x^2 + 12x + 32}{x^2 + 4x}$

A

28. Which statement accurately describes the relationship between variables in the equation below?

$$K = \frac{2v^2}{m}$$

- A. "K varies jointly as v squared and m"
 B. "K is inversely proportional to v squared and directly proportional to m"
 C. "K is directly proportional to v squared and inversely proportional to m"
 D. "K varies jointly as v squared and is inversely proportional to m"

C

29. If p varies jointly as q and r, and p = 150 when q = 18 and r = 5, find q when p = 345 and r = 9.

$$p = Kqr$$

$$150 = K \cdot 18 \cdot 5$$

$$150 = 90K$$

$$\frac{5}{3} = K$$

$$345 = \frac{5}{3} \cdot 9 \cdot q$$

$$q = 23$$

$$q = 23$$

30. The time needed to paint a fence varies directly with the length of the fence and inversely with the number of painters. If it takes seven hours to paint 280 feet of fence with two painters, how long will it take four painters to paint 720 feet of fence?

$$t = \frac{L \cdot K}{n}$$

$$7 = \frac{280K}{2}$$

$$14 = 280K$$

$$0.05 = K$$

$$t = \frac{720(0.05)}{4}$$

$$t = 9$$

$$9 \text{ hrs}$$

BONUS: Solve the equation below.

$$\frac{1}{x+3} - \frac{2 \cdot \frac{-1}{-1}}{3-x} = \frac{4}{x^2-9}$$

$$\left[\frac{1}{x+3} + \frac{2}{x-3} = \frac{4}{x^2-9} \right] \cdot (x+3)(x-3)$$

$$x-3 + 2(x+3) = 4$$

$$x-3 + 2x+6 = 4$$

$$3x+3 = 4$$

$$3x = 1$$

$$x = \frac{1}{3}$$

$$x = \left\{ \frac{1}{3} \right\}$$

CREDITS

I use clipart and
fonts in my products by:



Art with Jenny K



Many thanks to these
talented artists!